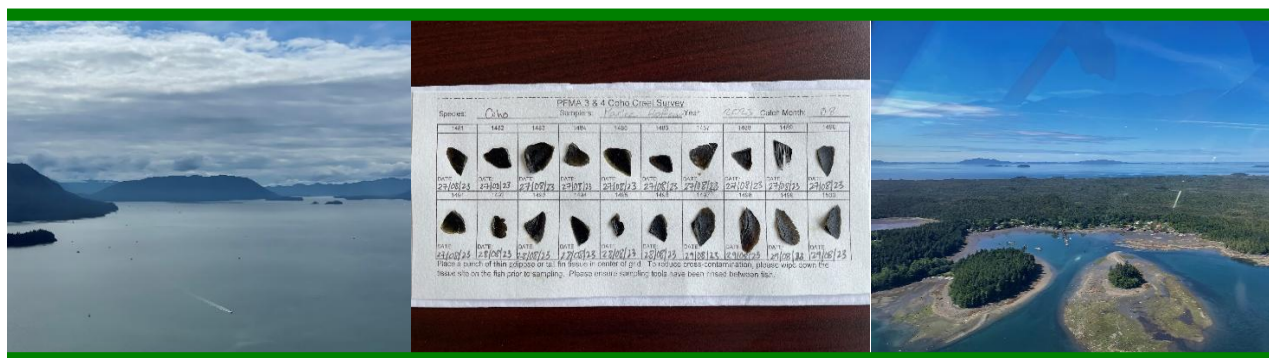


# North Coast (Areas 3 & 4) Recreational Angling Creel Survey 2023

## FINAL REPORT



*Prepared for:*

North Coast - Skeena First Nations Stewardship Society  
Prince Rupert, BC

*Prepared by:*

David Robichaud<sup>1</sup> and Angela Addison<sup>2</sup>

<sup>1</sup> LGL Limited environmental research associates  
9768 Second Street, Sidney, BC, V8L 3Y8

<sup>2</sup> North Coast - Skeena First Nations Stewardship Society  
309 2 Ave W #363, Prince Rupert, BC, V8J 3T1





## EXECUTIVE SUMMARY

Because of its potential impact on various transboundary salmon stocks, the recreational fishery in the North Coast (Areas 3 and 4) presents a high priority for monitoring. Gathering effort and harvest estimates, along with biological samples and genetic information, is crucial for effectively managing fisheries and monitoring indicator stocks. The North Coast Skeena First Nations Stewardship Society (NCSFNSS) has operated this program since 2015, with support from the Pacific Salmon Commission Northern Fund. The survey consists of access point angler interviews and aerial overflights, typically for the months of June through August but also including May since 2019. Lodge-based recreational harvest and effort are monitored by DFO via a voluntary logbook program. Survey design and implementation, and analytical methods are similar to and consistent with those used in Area 3 and 4 creel survey monitoring since 2008.

From 1 May to 31 August 2023, a total of 2,404 vessel interviews (involving 7,655 anglers) were collected by staff stationed at one of three access points (located in Prince Rupert or Port Edward) during 241 survey shifts. 32 aerial effort surveys were conducted. High priority was placed on gathering biosamples and fin-clip incidence rates from Chinook (*Oncorhynchus tshawytscha*) and Coho (*O. kisutch*) salmon. When possible, length measurements were collected from Pacific Halibut (*Hippoglossus stenolepis*). Survey effort was adequate, as relatively good precision was achieved for the harvest estimates for most of the main taxa (all precision goals were met).

Total fishing effort was estimated to be 12,086 vessel trips. In total, an estimated 15,255 Chinook Salmon were caught, with 6,443 fish kept and 8,812 released. It was estimated that 52,054 Coho Salmon were caught, with 42,525 fish kept and 9,529 released. Total Pacific Halibut catch was estimated to be 19,238 individuals, with 13,505 fish kept (biomass estimated at 74.5 net tonnes) and 5,732 released. Total harvest estimates included 9,600 Lingcod (*Ophiodon elongates*), 41 Yelloweye Rockfish (*Sebastes ruberrimus*), 5,881 other rockfishes (*Sebastes* spp.), and 1,345 groundfish other than Pacific Halibut and Lingcod.

Creel surveyors biosampled 483 Chinook Salmon, collecting length measurements (median 71 cm), adipose fin-clip incidence rates (8.1% were clipped), and scale samples. For Coho Salmon, 1,653 fish were biosampled (median fork length 61 cm; 0.74% adipose clipped). Heads were collected from 34 Chinook Salmon and six Coho Salmon for submission to the Salmon Head Recovery Program. Biosampled Pacific Halibut (n=724) had a median length of 79 cm, and average net weights of 6.0 kg and 65.3 kg in Areas 3 and 4, respectively.

During the month of May, fishing effort and Chinook Salmon harvest were estimated at 1,428 boat trips (12% of the May-August effort), and 1,191 fish (18% of the May-August harvest). The Department of Fisheries and Oceans Canada (DFO) is supportive of a May creel survey because genetic samples would help them resolve stock composition and determine exploitation rates on early-run-timing stocks. During the month of May 2023, we biosampled 170 Chinook Salmon, taking 159 scale samples, and processed 16 tagged heads.

Recreational fishing effort and salmon catch in Areas 3 & 4 have been increasing over the last 29 years. Despite the long-term trend, effort from 2020 to 2023 was below the predicted value. Despite reduced effort, Coho Salmon harvests were close to those predicted by the long-term increasing trend. Also of concern, Lingcod harvest in 2023, for the third consecutive year, was the highest observed for the available time series, despite a coast-wide reduction in the Lingcod bag limit that started in 2018. By contrast, Chinook Salmon harvests are not showing an increased trend, and may be declining, though Chinook Salmon releases are showing an increasing trend. Pacific Halibut harvest did not show a strong among-year trend, likely the result of effective management measures that reduced daily bag limits and imposed maximum size limits. While the increasing trends highlight the need for continued monitoring of this fishery into the future, there is evidence that management measures can keep harvest under control.



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## INTRODUCTION

The North Coast recreational fishery, spanning Management Areas 3 and 4 of DFO's Pacific Region, attracts resident and tourist anglers, targeting a variety of species. Predominantly focused on Chinook (*Oncorhynchus tshawytscha*) and Coho (*O. kisutch*) salmon, anglers also pursue Pacific Halibut (*Hippoglossus stenolepis*), Lingcod (*Ophiodon elongates*), and various other groundfish, rockfishes (*Sebastes* spp.), and invertebrate species. Over the last 25 years, fishing activity in this area has notably increased (Robichaud and Addison 2023), raising conservation concerns, and prompting the need to understand the harvest dynamics within the recreational fishery of Areas 3 and 4. Harvested salmon can originate from Canadian or US (enhanced and wild) stocks, hence there is a critical need for this creel program to determine BC fishery impacts, especially for stocks of concern.

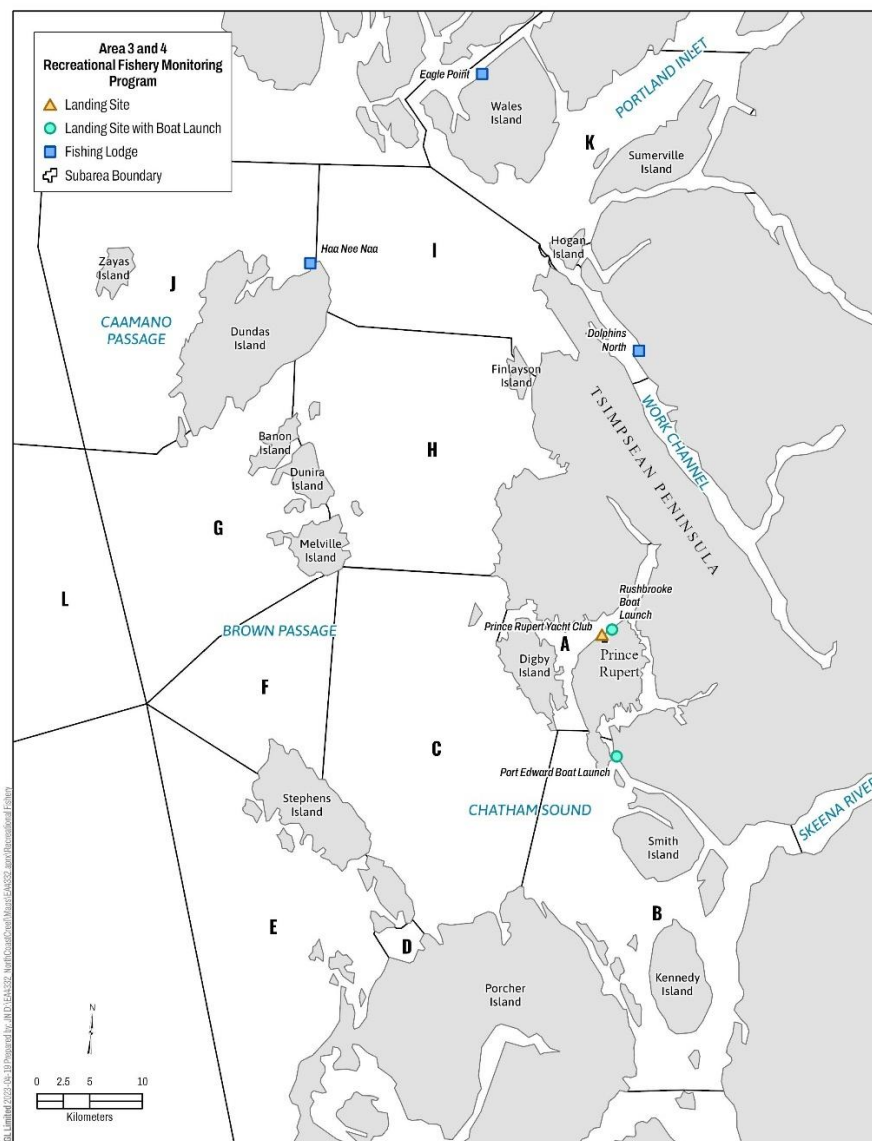
This creel survey program helps the Pacific Salmon Commission develop improved information for resource management, including better stock assessment, data acquisition, and improved scientific understanding of factors that might limit salmon production. Aside from providing precise estimates of harvest and effort, the creel program facilitates direct field observations of catches, enabling verification of species identification, quantification of harvests, and collection of biological samples from Chinook and Coho salmon. Notably, crucial data such as CWT (Coded Wire Tag) information, integral to annual stock assessments and the management of Nass and Skeena river Chinook and Coho salmon stocks, are derived from the heads of randomly sampled adipose-clipped fish collected by program observers.

Since its inception in 2008, the creel survey program in Areas 3 and 4 has been instrumental in estimating fishing effort and species-specific catch rates (Van Tongeren and Winther 2010, Robichaud and Addison 2023). In 2015, the responsibility for conducting the creel survey transitioned from Fisheries and Oceans Canada (Prince Rupert) to the North Coast Skeena First Nations Stewardship Society (NCSFNSS), with support from the Pacific Salmon Commission Northern Fund. NCSFNSS's creel survey (Robichaud et al. 2016) followed a design similar to that used in the past by Fisheries and Oceans Canada, which, in turn, was based on an adaptation of the Strait of Georgia Creel Survey design, developed in 1982 by LGL Limited (English et al. 2002).

This report documents the 2023 recreational catch and effort statistics for fishing activities in Pacific Fishery Management Areas 3 and 4, and describes the program activities, data collection methodology and the analytical approaches that were used to generate the estimates.

## SCOPE

The geographic scope of the creel survey was the same as in the previous 13 years (Van Tongeren and Winther 2010, Robichaud et al. 2016, 2017, Robichaud and Addison 2018, 2019, 2020, 2021, 2022, 2023), i.e., Management Areas 3 and 4 (from Portland Inlet, through Chatham Sound to Porcher Island (Figure 1). For the purposes of data collection and analysis, the study area was divided into 12 geographic strata ('subareas'), including three subareas in Area 3 (3I-3K) and nine subareas in Area 4 (4A-4H, and 4L).



**Figure 1.** Map of the North Coast study area, showing the major landing sites, fishing lodges, and the boundaries of the subareas that were used as spatial strata in the creel survey design.

The temporal scope of the full study was from 1 May through 31 August 2023. Survey schedules were designed to provide sufficient data to derive statistically robust catch and efforts estimates for each month and subarea.

The study included creel analysis of all major local sportfish species, with greater emphasis placed on Chinook and Coho salmon, Pacific Halibut, and Lingcod.

## GOAL AND OBJECTIVES

The goal of this study was to conduct a creel survey of the North Coast (Areas 3 and 4) recreational fishery following the design and methodology used recently by DFO (Van Tongeren

and Winther 2010), for the purpose of producing estimates of effort and catch of all species targeted by the recreational fishery. The creel survey included aerial effort counts and access point angler interviews, conducted throughout Areas 3 and 4 from May through August 2023. The precision goal for the study was to estimate total harvests of large components (such as Chinook Salmon) within 15% of the true value 19 times out of 20. Creel survey strata included temporal separation by month and day type (weekdays vs. weekend/holiday days). Spatial stratifications were based on creel survey subareas. All local fishing areas (i.e., those accessed from major boat launches in the area) were included in our survey design.

The seven objectives for this study were:

1. Maintain software for data management and analysis comparable to DFO's CREST program;
2. Provide monthly (May-August) estimates of the catch and effort of all salmon species targeted by the recreational fishery and associated estimates of precision;
3. Provide DFO and PSC with monthly in-season catch estimates for all salmon species harvested in the North Coast recreational fishery;
4. Provide communication materials to recreational harvesters to convey the usefulness and importance of their catch data in fisheries management;
5. During angler interviews, collect biological data from a representative subsample of the landed harvest, including fin clip incidence, lengths, and scale or tissue samples for Coho and Chinook salmon;
6. Produce a final written report to be available to a technical audience; and
7. Present creel survey findings at the North Coast salmon post-season review meeting to a broad audience of commercial, recreational interests, and First Nations.

Objective 1 was addressed over the course of the study period. Objectives 2 and 6 were met by completing this report. Objective 3 was met throughout the study period by providing timely and complete summaries of the monthly survey activities (with complete creel analyses done for each update report). Objective 4 will be met subsequent to the finalization of this report, as its contents will be based on the results presented herein. Objective 5 was met by biosampling fish at access points throughout the study period: fish were measured for length and sexed, scale or tissue samples and adipose fin clip incidence data were collected for Coho and Chinook salmon, heads were collected from adipose-clipped fish, and all biological data were submitted to DFO at the end of study period. Objective 7 was met on 2 December 2023 at the DFO IHPC Post-Season Review meeting.

## METHODS

A complete census of catch and effort was available for lodge-based anglers. The census data were collected by lodge staff and shared voluntarily with Fisheries and Oceans Canada stock assessment personnel. The remainder of the recreational fishery occurred over too large an area to completely census catch and effort. Instead, statistically robust methods were used to estimate catch by multiplying estimates of *angler effort* by estimates of *catch per unit of effort*. Separate estimates were generated for each creel monitoring subarea, month, day type (weekday vs. weekend), and species.

For each creel subarea during each month and day type, fishing effort was estimated by counting boats that were actively fishing during overflight surveys (sub-area counts were reduced by the

number of lodge-based boats thought to have been visible to the aircraft). Catch per unit of effort for non-lodge boats was estimated from interviews (see data forms in Appendix 1) conducted at known access points. During interviews, anglers were asked about their catch, effort, and fishing locations. They were also asked about their hourly fishing activity patterns.

Data collected during interviews included:

- Date;
- Number of anglers in party;
- Whether or not the trip was guided by a professional;
- Subarea angler activity – for each subarea, the hours during which angling activity was conducted;
- Subarea catch – the number of fish kept and the number released by species and subarea;
- Subarea gear – the amount and type of fishing gears used to catch each species in each subarea;
- Subarea targets – the number of hours spent targeting each species in each subarea (anglers that targeted only shellfish were noted);
- Biosampling – if allowed by the anglers, harvested Chinook and Coho salmon were measured (nose to fork length), sexed, checked for adipose clips and, if clipped and permission given, heads were collected for the Salmon Head Recovery Program. Scale samples were taken from Chinook Salmon, and tissue samples (fin clips) from Coho Salmon. Chinook Salmon were also examined for flesh colour. Pacific Halibut were measured (total length);
- Refusals – whether or not the anglers refused an interview, allowed catch to be observed, or allowed catch to be biosampled; and
- Weather conditions.

The analytical methods used, similar to those used in previous years (e.g., Robichaud and Addison 2023), provided a statistically unbiased estimate of catch per effort, assuming the anglers interviewed were representative of the entire non-lodge recreational fishery. Interviews were conducted at the three main access locations: Port Edward and Rushbrooke boat launches, and Prince Rupert Yacht Club. The locations surveyed were selected from all available access points, based on their geographical distribution and the amount of fishing activity that is typically conducted from that site, in order to maximize the number of interviews obtained. This approach was based on two important observations: 1) the variability in CPE (catch-per-effort) among fishing parties landing at a single access point tends to be as great as the variability in CPE among different access points within a geographic area; and 2) CPE and effort can vary substantially both within and between days at a single site (English et al. 2002). Under these conditions it is better to obtain a large number of interviews covering all temporal strata for a small number of sites than to sample a larger number of sites and obtain fewer interviews and less complete temporal coverage for any specific site.

During an observer shift, attempts were made to interview all anglers returning to the access location on that day, but some were missed (it is assumed that a random process dictated whether a boat was missed or interviewed), and some anglers refused to participate (it is not clear how the angling behaviour of the refusals differed from that of the participants). The geographic distribution of our samples was dictated by where the anglers fished, and it was assumed that the distribution of samples taken at the access locations was random relative to the fish harvested by the fishery.

The frequency of interview shifts was planned with the target of sampling 20% of estimated effort, while attempting to have “hands on” and direct observations of (and biological samples from) as many fish as possible meant allocating adequate time to each interview. The interview schedule was designed to capture data from representative angler parties in both day types, and over all time periods of the day. Interviewing sessions were separated into AM and PM shifts, with AM shifts occurring between 7:00 and 15:00, and PM shifts from 13:00 to 21:00. In total, 241 shifts were worked.

Fishing effort was estimated from boat-counts made during overflights. Overflights were conducted on 3 weekend days and 5 weekdays each month. Timing of the overflights was targeted for peak fishing periods (11 AM to 1 PM), and in 2023 most flights occurred at target times. Fog or mechanical issues delayed the start of four flights (flight on 30 July ran from 11:47 AM to 1:28 PM; flight on 2 August ran from 12:23 PM to 1:56 PM; flight on 5 August ran from 12:19 PM until 2:05 PM; flight on 27 July ran from 1:10 PM until 2:51 PM). Analytical methods accommodate the occasional off-peak flight but can result in increased uncertainty in the effort estimate.

### **Angler Activity Patterns**

Two weighting factors were used together with the interview-derived angling activity data to estimate the daily non-lodge fishing activity pattern (English et al. 2002). The first weighting factor,  $W1$ , expanded the numbers of days spent interviewing at each landing site to account for the total number of days available for sampling. That is, it was assumed that the daily activity pattern recorded during the interview shifts at landing site  $a$ , were consistent for landing site  $a$ , even during the days when no interviews occurred. A specific  $W1$  was calculated for each landing site during each month and day type:

$$W1_{mda} = \frac{N_{md}}{K_{mda}}, \quad (\text{Equation 1})$$

where  $N_{md}$  was the total number of type  $d$  days in month  $m$ ; and  $K_{mda}$  was the number of days during which interviews occurred at landing site  $a$ , on type  $d$  days during month  $m$ .

The second weighting factor,  $W2$ , expanded the numbers of interviews conducted, to account for the recreational fishing boats that were *not* interviewed. Some recreational fishing boats were not interviewed because of refusals, and some boats returned during periods of high traffic volume when it was not possible to interview all returning vessels. Some returning vessels were not doing recreational fishing at all (e.g., pleasure cruisers, Food-Social-Ceremonial harvesters). For the recreational fishing vessels, it was assumed that the activity pattern recorded during the interview shifts also held for those anglers that were not interviewed. A specific  $W2$  was calculated for each surveyor ( $u$ ) operating on each surveying date ( $k$ ) at each landing site ( $a$ ):

$$W2_{aku} = \frac{LF_{aku} + LM_{aku} \left( \frac{LF_{aku}}{LF_{aku} + LN_{aku}} \right)}{LF_{aku} - ri_{aku}}. \quad (\text{Equation 2})$$

where  $LF_{aku}$  was the number of interviewed boats that were fishing (a subset of these,  $ri_{aku}$ , refused an interview, or had an incomplete interview that could not be used),  $LN_{aku}$  was the number of interviewed boats that were not fishing, and  $LM_{aku}$  was the number of boats that were

**Table 1. The amount of data (number of complete interviews) available to estimate non-lodge angler activity patterns, for all levels of each factor, 2023.**

| Month       | Day Type | Subarea |     |     |     |     |     |     |    |     |     |     |    | TOTALS |        |         |
|-------------|----------|---------|-----|-----|-----|-----|-----|-----|----|-----|-----|-----|----|--------|--------|---------|
|             |          | 3I      | 3J  | 3K  | 4A  | 4B  | 4C  | 4D  | 4E | 4F  | 4G  | 4H  | 4L | Area 3 | Area 4 | Overall |
| May         | WD       | 3       | 5   | 4   | 54  | 27  | 26  | 14  | 6  | 74  | 7   | 9   | 1  | 12     | 218    | 230     |
|             | WE       | 6       | 9   | 4   | 108 | 50  | 64  | 21  | 9  | 80  | 9   | 13  | 3  | 19     | 357    | 376     |
|             | Total    | 9       | 14  | 8   | 162 | 77  | 90  | 35  | 15 | 154 | 16  | 22  | 4  | 31     | 575    | 606     |
| June        | WD       | 5       | 18  | 8   | 98  | 25  | 54  | 30  | 5  | 110 | 13  | 18  | 10 | 31     | 363    | 394     |
|             | WE       | 11      | 16  | 6   | 79  | 34  | 52  | 22  | 2  | 86  | 8   | 23  | 11 | 33     | 317    | 350     |
|             | Total    | 16      | 34  | 14  | 177 | 59  | 106 | 52  | 7  | 196 | 21  | 41  | 21 | 64     | 680    | 744     |
| July        | WD       | 17      | 38  | 26  | 102 | 41  | 74  | 76  | 6  | 152 | 19  | 34  | 16 | 81     | 520    | 601     |
|             | WE       | 22      | 22  | 13  | 105 | 43  | 113 | 57  | 7  | 142 | 23  | 50  | 10 | 57     | 550    | 607     |
|             | Total    | 39      | 60  | 39  | 207 | 84  | 187 | 133 | 13 | 294 | 42  | 84  | 26 | 138    | 1,070  | 1,208   |
| August      | WD       | 19      | 28  | 26  | 65  | 48  | 64  | 41  | 9  | 81  | 23  | 59  | 8  | 73     | 398    | 471     |
|             | WE       | 16      | 13  | 36  | 62  | 37  | 33  | 21  | 6  | 43  | 9   | 40  | 7  | 65     | 258    | 323     |
|             | Total    | 35      | 41  | 62  | 127 | 85  | 97  | 62  | 15 | 124 | 32  | 99  | 15 | 138    | 656    | 794     |
| GRAND TOTAL |          | 99      | 149 | 123 | 673 | 305 | 480 | 282 | 50 | 768 | 111 | 246 | 66 | 371    | 2,981  | 3,352   |

not interviewed by surveyor  $u$ , during surveying date  $k$ , at landing site  $a$ . This equation incorporates the assumption that interviewed boats were representative, allowing us to estimate the portion of missed boats that were fishing.

We used the term  $A_{mdsakut}$  to denote the number of non-lodge boats reporting activity during time-block  $t$ , as interviewed by surveyor  $u$  on survey date  $k$ , at landing site  $a$ , in subarea  $s$ , during month  $m$ , and on day type  $d$  ( $n_{mdsakut}$  was used to denote the total number of boats that reported activity during any time-block). The two correction factors were applied, and the data were summed over surveyors, survey dates and landing sites:

$$A'_{mdst} = \sum_k \sum_u \sum_a (W1_{mda} \cdot W2_{aku} \cdot A_{mdsakut}) \quad . \quad (\text{Equation 3})$$

Summing the adjusted number of anglers over the hourly time-blocks gave:

$$T'_{mds} = \sum_t A'_{mdst} \quad . \quad (\text{Equation 4})$$

The proportion of non-lodge boats ( $P_{mdst}$ ) that were actively fishing during each of the hourly time-blocks ( $t$ ) was calculated for each month, day type, and subarea:

$$P_{mdst} = \frac{A'_{mdst}}{\sum_k \sum_u \sum_a (W1_{mda} \cdot W2_{aku} \cdot n_{mdsakut})} \quad . \quad (\text{Equation 5})$$

Using this method, 96 unique angler activity patterns were to be estimated (i.e., 4 months  $\times$  2 day types  $\times$  12 subareas). To reliably describe angler activity, a relatively large number of boats (~60) needed to be interviewed in each of the 96 blocks. In the end, some blocks contained too few interviews (Table 1), so it was decided to pool activity data over subareas. The equation for non-lodge angler activity was thus:

$$P_{mdt} = \frac{\sum_s A'_{mdst}}{\sum_s \sum_k \sum_u \sum_a (W1_{mda} \cdot W2_{aku} \cdot n_{mdsakut})} \quad , \quad (\text{Equation 6})$$

and the associated variance was:

$$S_{P_{mdt}}^2 = \frac{(P_{mdt})(1 - P_{mdt})}{\sum_s \sum_k \sum_u \sum_a (W1_{mda} \cdot W2_{aku} \cdot n_{mdsakut})} \quad . \quad (\text{Equation 7})$$



### **Catch Per Effort Estimation**

Catch per effort (numbers of fish kept or released per boat trip) of non-lodge boats was estimated for each species from interviews of non-lodge anglers. Catch per effort was calculated separately for guided and non-guided boats, and overall; but only the overall estimates were used in subsequent calculations. For each interview ( $i$ ), the guided status ( $g$ ), month ( $m$ ), and the catch ( $C$ ) of each species ( $r$ ) in each subarea ( $s$ ) was recorded. Given that each interview represented one boat trip, then catch per effort (CPE), in terms of catch per boat trip, was calculated as:

$$CPE_{msrgi} = \frac{C_{msrgi}}{1} \quad \text{where } g \in \{\text{all, guided, not-guided}\}. \quad (\text{Equation 8})$$

Mean CPE was calculated as the average catch, over all  $n_{msg}$  boat trips:

$$C\hat{P}E_{msg} = \frac{\sum_{i=1}^{n_{msg}} C_{msrgi}}{n_{msg}}. \quad (\text{Equation 9})$$

The variance for the estimate of mean catch per effort was calculated as:

$$S_{C\hat{P}E_{msg}}^2 = \frac{\sum (CPE_{msrgi} - C\hat{P}E_{msg})^2}{(n_{msg} - 1)}. \quad (\text{Equation 10})$$

In the event that a month/subarea-specific sample size was  $< 5$ , adjacent subareas would be pooled together as follows:

- if  $n$  for 3I, 3J or 3K was too low, then it was combined with the other two subareas, along with data from ambiguous parts of Area 3 (“3?”) to give a 3?IJK regional average CPE;
- if  $n$  for 4A, 4B or 4C was too low, then it was combined with the other two to give a 4ABC regional average CPE;
- if  $n$  for 4D, 4E or 4F was too low, then it was combined with the other two to give a 4DEF regional average CPE; and
- if  $n$  for 4G, 4H or 4L was too low, then it was combined with the other two to give a 4GHL regional average CPE.

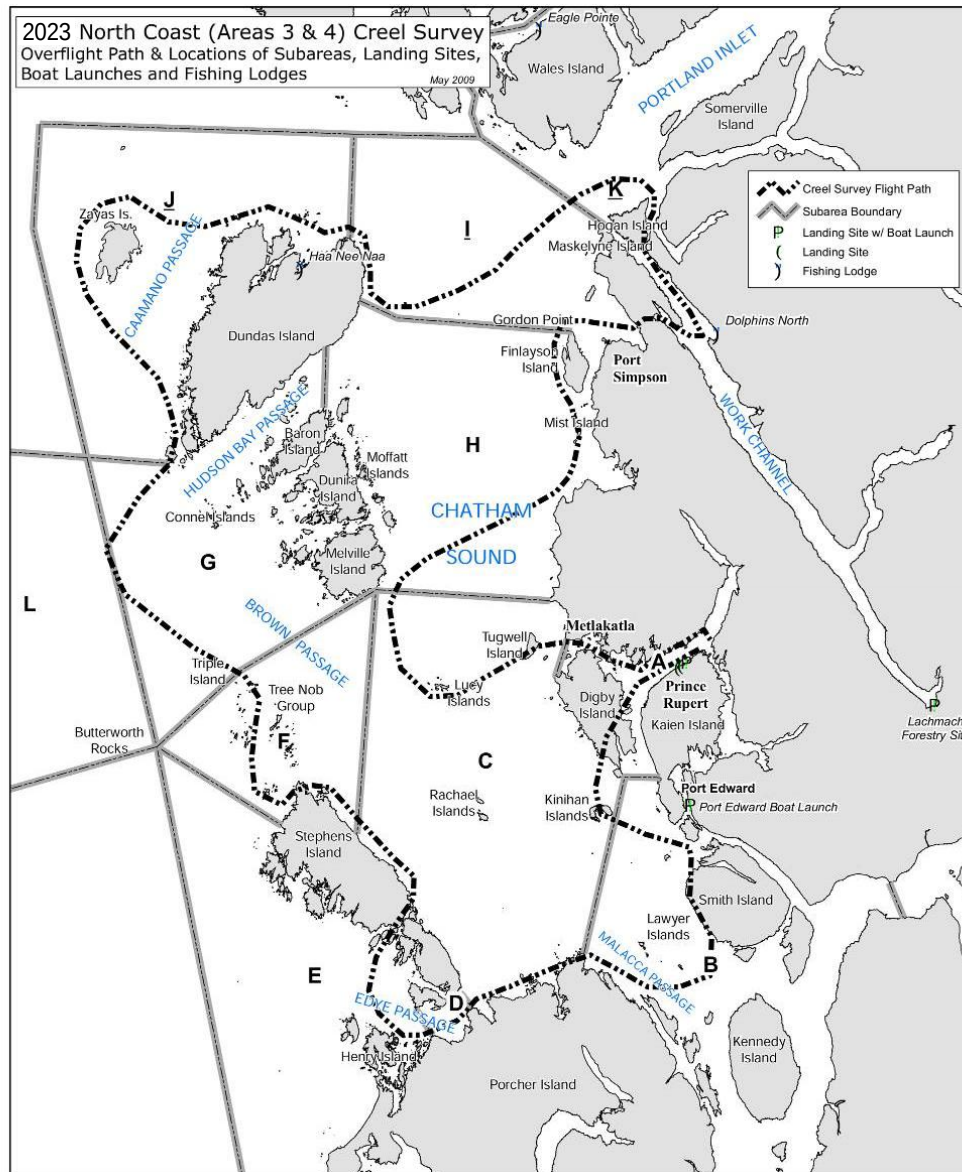
Similar methods were used to calculate the average number of fish harvested (retained) per boat trip (HPE), and the average number of fish released per boat trip (RPE).

### **Angler Effort Estimation**

To obtain statistically valid estimates of non-lodge fishing effort, boats were counted during overflight surveys conducted from a fixed-wing aircraft traveling along a set path over the study area (see Figure 2). In order to calculate the non-lodge effort, the estimated counts of active lodge boats needed to be subtracted from the overflight counts for that particular day.

#### *Lodge Effort Census*

A complete census of lodge-based angling effort was collected from lodge logbooks by DFO personnel. DFO used the following procedure to estimate the number of active lodge vessels that would have been available to be observed during the flights: 1) Lodge based vessels were considered unavailable to the flights on change-over days (days when the lodges received new



**Figure 2.** Map showing planned flight path of effort survey overflights. Letters indicate subareas. Flights originated and ended in Prince Rupert and flew in the clockwise direction. On good weather days, the flights also included a foray into subarea 4L (path not shown).

clients; on typical change-over days, departing clients leave before the flight and new clients arrive after the flight). Change-over days occurred twice a week but timing was often different for each lodge, making 29% (2/7) of the lodge effort not visible to flights due to change-over days. 2) Lodges reported their lunchtime activity to DFO, which ranged from virtually always going in for a mid-day break to rarely leaving the water during the day. Thus, approximately 60% of the remaining vessel activity was available to the flight counts on 'regular' days (i.e., not change-over). 3) Vessels on mid-day break or on change-over days were excluded from the



**Table 2. Overflight sampling effort (number of flights shifts) by day type and month, 2023.**

| Month  | Weekday | Weekend/ | Total |
|--------|---------|----------|-------|
|        |         | Holiday  |       |
| June   | 5       | 3        | 8     |
| July   | 5       | 3        | 8     |
| July   | 5       | 3        | 8     |
| August | 5       | 3        | 8     |
| Total  | 20      | 12       | 32    |

numbers of vessels visible to the flights. All of the lodge catch was reported from Area 3, without subarea-specific effort information. Thus, lodge effort was partitioned among the Area 3 subareas, see below.

### Non-lodge Effort Surveys

Table 2 shows the number of overflight surveys scheduled for each day type and month. Each survey was designed to allow visual inspection of the subareas through which the aircraft passed. Each time the aircraft entered a subarea, the start and end times were recorded, along with the numbers of small (< 30 ft) and large ( $\geq 30$  ft) recreational vessels that were observed, tallied separately depending on whether or not they were actively fishing at the time.

During survey  $o$  (conducted during month  $m$  and on day type  $d$ ), observers tallied the total number of recreational boats that were actively fishing at time  $t$  on their  $i^{\text{th}}$  pass over subarea  $s$ ,  $V_{mdsoit}$ . For each survey, the relative proportion of visible boats within Management Area 3 was calculated,  $\rho_{mdsoit}$ , and the total number of lodge boats ( $\lambda_{mdo}$ ) apportioned accordingly:

$$\lambda_{mdsoit} = \rho_{mdsoit} \cdot \lambda_{mdo} \quad . \quad (\text{Equation 11})$$

The number of non-lodge boats was hence:

$$v_{mdsoit} = \max \{0, V_{mdsoit} - \lambda_{mdsoit}\} \quad . \quad (\text{Equation 12})$$

Since angling occurs over the course of the entire day, the numbers of non-lodge boats were divided by the proportion of average daily number of boats active ( $P_{mdst}$ ) during the time block when the observations were recorded. Since the flight path (Figure 2) brought the observers into some subareas more than once during the survey (covering different ground, thus the tallies were considered additive), the adjusted tallies were summed over all visits, to calculate the total number of active non-lodge fishing boats on the day of the survey, by subarea,  $B_{mdso}$ :

$$B_{mdso} = \sum_i \frac{v_{mdsoit}}{P_{mdst}} \quad . \quad (\text{Equation 13})$$

The variance of  $B_{mdso}$  was calculated using the standard formulas for combining the variance of products and quotients of two independent random variables (Goodman 1960):

$$\begin{aligned} \text{if } z = x/y, \text{Var}(z) &= (y^{-2})\text{Var}(x) + (x^2 y^{-4})\text{Var}(y) ; \\ \text{if } z = xy, \text{Var}(z) &= (y^2)\text{Var}(x) + (x^2)\text{Var}(y) ; \text{ and} \\ \text{if } z = qx, \text{Var}(z) &= (q^2)\text{Var}(x) \quad \text{where } q \text{ is a constant.} \end{aligned} \quad (\text{Equation 14})$$

Since there was no variance associated with the boat counts, it followed that

$$S_{B_{mdso}}^2 = v_{mdsoit}^2 \cdot (P_{mdst})^{-4} \cdot S_{P_{mdst}}^2 \quad . \quad (\text{Equation 15})$$

Next, the survey-specific ‘daily effort’ estimates were averaged over the number of surveys

conducted,  $n_{mds}$ , as:

$$\hat{B}_{mds} = \frac{\sum_{o=1}^{n_{mds}} B_{mdso}}{n_{mds}} \quad (\text{Equation 16})$$

with a variance of:

$$S_{\hat{B}_{mds}}^2 = (n_{mds})^{-2} \cdot S_{B_{mdso}}^2 \quad (\text{Equation 17})$$

Total fishing effort for each month was calculated for each subarea by multiplying the month-specific average daily effort by the number days of day type  $d$  that occurred in month  $m$ :

$$E_{mds} = \hat{B}_{mds} \cdot N_{md} \quad (\text{Equation 18})$$

which was then summed over day type to give:

$$E_{ms} = \sum_d \hat{B}_{mds} \cdot N_{md} \quad (\text{Equation 19})$$

with a variance of:

$$S_{E_{ms}}^2 = \sum_d S_{\hat{B}_{mds}}^2 \cdot N_{md}^2 \quad (\text{Equation 20})$$

Because of inclement weather, not all overflights surveyed every subarea. Especially for the more exposed subareas, it was possible to have no overflight data in a given combination of month and day type. In this event, subarea-specific effort was scaled using: 1) the proportion of effort in that subarea, relative to that in a three-subarea cluster (see pooling of adjacent subareas for CPE estimation, above), measured during overflights for which all three subareas were surveyed; and 2) the effort in the cluster's other two subareas, estimated for the combination of month and day type during which the data gap occurred.

Rarely, this method produced an effort estimate of zero in a stratum for which interview data suggested otherwise. In 2023, no boats were counted in Subarea 4L (except on June weekdays) or in May or June in Subarea 4E, but interviews showed that effort had occurred. In these cases, the interview data obtained were taken as representative of total effort.

### **Catch and Harvest Estimation**

Total catch was calculated for each month, subarea, and species by multiplying month-specific angling effort by month-specific catch per effort, and then summing over day type:

$$C_{msr} = \sum_d (E_{mds} \cdot C\hat{P}E_{msrg}) \quad (\text{Equation 21})$$

where  $g$  = 'all' (see Equation 8). The standard errors for these catch estimates were:

$$S_{C_{msr}} = \sqrt{\sum_d \left( E_{mds}^2 \frac{S_{CPE_{msrg}}^2}{n_{msg}} + CPE_{msrg}^2 \frac{S_{E_{mds}}^2}{n_{mds}} + \frac{S_{CPE_{msrg}}^2 S_{E_{mds}}^2}{n_{msg} n_{mds}} \right)} \quad (\text{Equation 22})$$

where  $n_{mds}$  and  $n_{msg}$  are the sample sizes for E and CPE estimates, respectively.

Similar methods were used to estimate monthly numbers of fish harvested, and monthly numbers of fish released by replacing CPE with HPE or with RPE in Equations 21 and 22.

### **Biosampling**

When given permission by the anglers, surveyors collected biological data from Chinook and Coho salmon and Pacific Halibut. Biosampled species were sexed when they were intact. Chinook and Coho salmon were examined for clipped adipose fins (mark rate data used by DFO

for coded wire tag analyses) and when possible, heads from clipped fish were collected. The flesh color was noted for Chinook Salmon. Scale samples were collected from Chinook Salmon, and tissue samples (fin clips) were collected from Coho Salmon (used by DFO to determine age, and to carry out genetic stock identification).

Ideally, all anglers would allow their catch to be biosampled, and the examined fish would represent an unbiased sample of all harvested fish. But Robichaud et al. (2016) theorized that the interview protocol, used until the end of 2015, may have resulted in bias. The protocol had surveyors ask anglers about adipose clips before asking permission to observe the fish. Anglers were often engaged in a conversation about the purpose of the fin-clips and how the data would be used, and it appeared that anglers with clipped fish were more likely to permit handling and biosampling of catch than when no clipped fish were present. To avoid this bias, we modified our data collection protocol in 2016 so that anglers were asked if their catch could be biosampled before any other interactions took place. Also starting 2016, no fin clip data were accepted on the word of the anglers: the only fin-clip data we used were those which came from observers when they handled and inspected the catch.

### **Biomass of Harvested Pacific Halibut**

Halibut lengths were converted to weights using the International Pacific Halibut Commission (Clark 1992) regression equations:

$$\begin{aligned}\text{Weight(lb)}_{\text{net}} &= 6.921 \times 10^{-6} \times \text{FL(cm)}^{3.24} \\ \text{Weight(lb)}_{\text{round}} &= 9.166 \times 10^{-6} \times \text{FL(cm)}^{3.24}\end{aligned}\quad (\text{Equation 23})$$

Converting average length to average weight allowed harvested biomass to be calculated from harvested fish counts.

### **Statistical Analyses**

For each fish taxa, differences in CPE among months were examined using Kruskal Wallis tests (Zar 1984), whereas differences between trip-types (guided vs. non-guided) were examined using paired t-tests (i.e., the difference between guided vs unguided catch for a given subarea and month was tested against a null of zero; Sokal and Rohlf 1995). Kruskal Wallis tests were used to examine the effects of month and day type on effort, and the effects of month on catch and harvest. When statistically significant effects were observed, *post-hoc* Kruskal-Wallis tests were used to compare between each possible pair, where the false discovery rate was controlled using the Benjamini–Hochberg procedure (Benjamini and Hochberg 1995).

### **Trailer Census**

The total number of parked trailers was recorded daily at the landing locations during the peak of the activity timing window (i.e., between 11:00 AM and 1:30 PM; similar to the targeted timing of aerial overflights). Theoretically, trailer counts could be used as an index of fishing effort, are more economical than overflights, and are not weather dependant, though no sub-area-specific effort data are collected. DFO has been collecting trailer count data for years and we are maintaining the time series. Linear regression was used to determine the relationship between trailer counts and fishing effort. A data point was generated each time a trailer census occurred on the same day as an overflight. For the regression analysis, the location-specific trailer counts and non-lodge boat counts were used as the independent and dependent variables, respectively.

## RESULTS

### Angler Interviews

Over the four month study period, there were 241 observer shifts, during which 2,404 interviews were conducted, involving 7,655 non-lodge anglers (Figure 3; Table 3). July was the busiest month, with 799 interviews conducted (average 13.3 interviews per shift), which represented 37% of the total number of study period interviews (Table 3).

Non-lodge boat trip activity was unevenly distributed among the subareas, with 4F being the favourite (Table 4). Subareas 4A and 4C were the next most likely area for anglers to have reported activity, followed by 4B, 4D, and 4H. The subareas with the lowest number of trips reporting activity were 4E and 4L. Only 75 interviews were of parties that only targeted shellfish, and those were overwhelmingly in subarea 4A (Table 5).

On average, 27% of the non-lodge boat trips were guided by a professional. The guiding rate varied by landing site. Only 1 of 169 trips (0.6%) interviewed at Port Edward were guided, whereas trips departing from Rushbrooke and Prince Rupert Yacht Club were guided 25% and 62% of the time, respectively (Table 6).

In all there were 38 refusals (Table 4), or 1.6% refusal rate. The highest refusal rate (2.1%) was at the Rushbrooke boat launch, where 58% of the refusals occurred. The refusal rates for the Port Edward boat launch and the Prince Rupert Yacht Club were 1.4% and 1.0% (24% and 18% of the refusals), respectively.

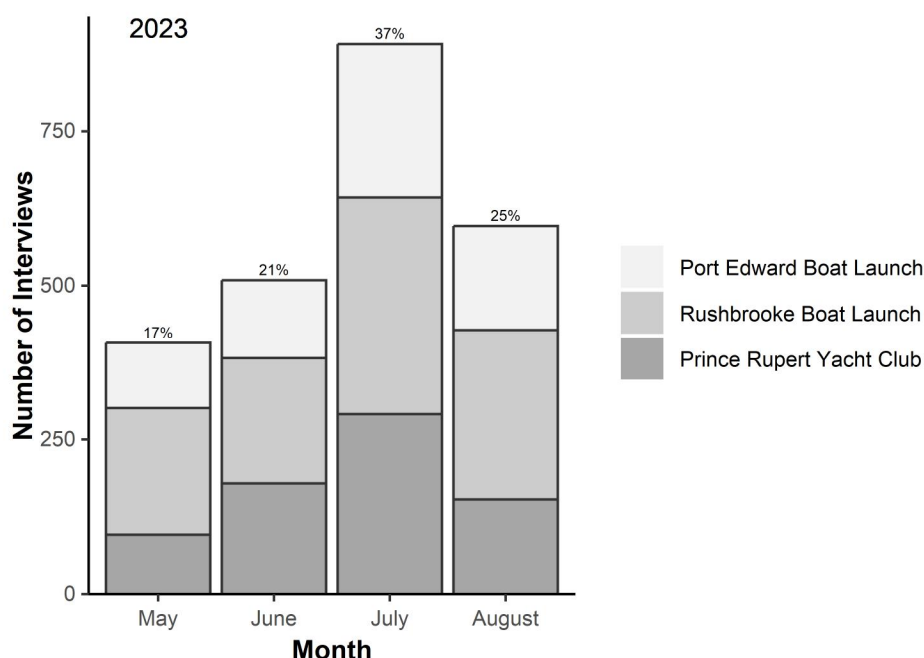


Figure 3. Number of interviews conducted at each of three landing sites, by month, 2023.

**Table 3. Monthly number of shifts, and numbers of boats interviewed per landing site, 2023. Non-lodge effort is expressed in units of boat trips. Sampling efficiency is also calculated.**

| Landing Site                     | May    |           |          |            | June   |           |          |            | Jul    |           |          |            |
|----------------------------------|--------|-----------|----------|------------|--------|-----------|----------|------------|--------|-----------|----------|------------|
|                                  | # ints | # anglers | # stints | ints/stint | # ints | # anglers | # stints | ints/stint | # ints | # anglers | # stints | ints/stint |
| Port Edward Boat Launch          | 106    | 263       | 21       | 5.0        | 127    | 335       | 21       | 6.0        | 248    | 779       | 22       | 11.3       |
| Prince Rupert Yacht Club         | 96     | 334       | 20       | 4.8        | 179    | 653       | 18       | 9.9        | 291    | 1038      | 22       | 13.2       |
| Rushbrooke Boat Launch           | 205    | 544       | 22       | 9.3        | 203    | 593       | 19       | 10.7       | 352    | 1135      | 22       | 16.0       |
| TOTAL                            | 407    | 1,141     | 63       | 6.5        | 509    | 1,581     | 58       | 8.8        | 891    | 2,952     | 66       | 13.5       |
| % of Total Interviews            |        |           | 17%      |            |        |           | 21%      |            |        |           | 37%      |            |
| Estimated Non-lodge Effort       |        |           | 1,244    |            |        |           | 2,166    |            |        |           | 3,207    |            |
| % of Cumulative Non-lodge Effort |        |           | 12%      |            |        |           | 22%      |            |        |           | 32%      |            |
| % of Effort Sampled              |        |           | 33%      |            |        |           | 23%      |            |        |           | 28%      |            |

| Landing Site                     | August |           |          |            | TOTAL  |           |          |            |
|----------------------------------|--------|-----------|----------|------------|--------|-----------|----------|------------|
|                                  | # ints | # anglers | # stints | ints/stint | # ints | # anglers | # stints | ints/stint |
| Port Edward Boat Launch          | 169    | 499       | 18       | 9.4        | 650    | 1876      | 82       | 7.9        |
| Prince Rupert Yacht Club         | 153    | 572       | 18       | 8.5        | 719    | 2597      | 78       | 9.2        |
| Rushbrooke Boat Launch           | 275    | 910       | 18       | 15.3       | 1035   | 3182      | 81       | 12.8       |
| TOTAL                            | 597    | 1,981     | 54       | 11.1       | 2,404  | 7,655     | 241      | 10.0       |
| % of Total Interviews            |        |           | 25%      |            |        |           |          |            |
| Estimated Non-lodge Effort       |        |           | 3,445    |            |        |           | 10,062   |            |
| % of Cumulative Non-lodge Effort |        |           | 34%      |            |        |           |          |            |
| % of Effort Sampled              |        |           | 17%      |            |        |           | 24%      |            |

**Table 4. Fishing locations as reported by interviewed non-lodge parties, by month and subarea, 2023. For interviews that reported activity in multiple subareas, the boat trip tally was evenly partitioned among the subareas visited.**

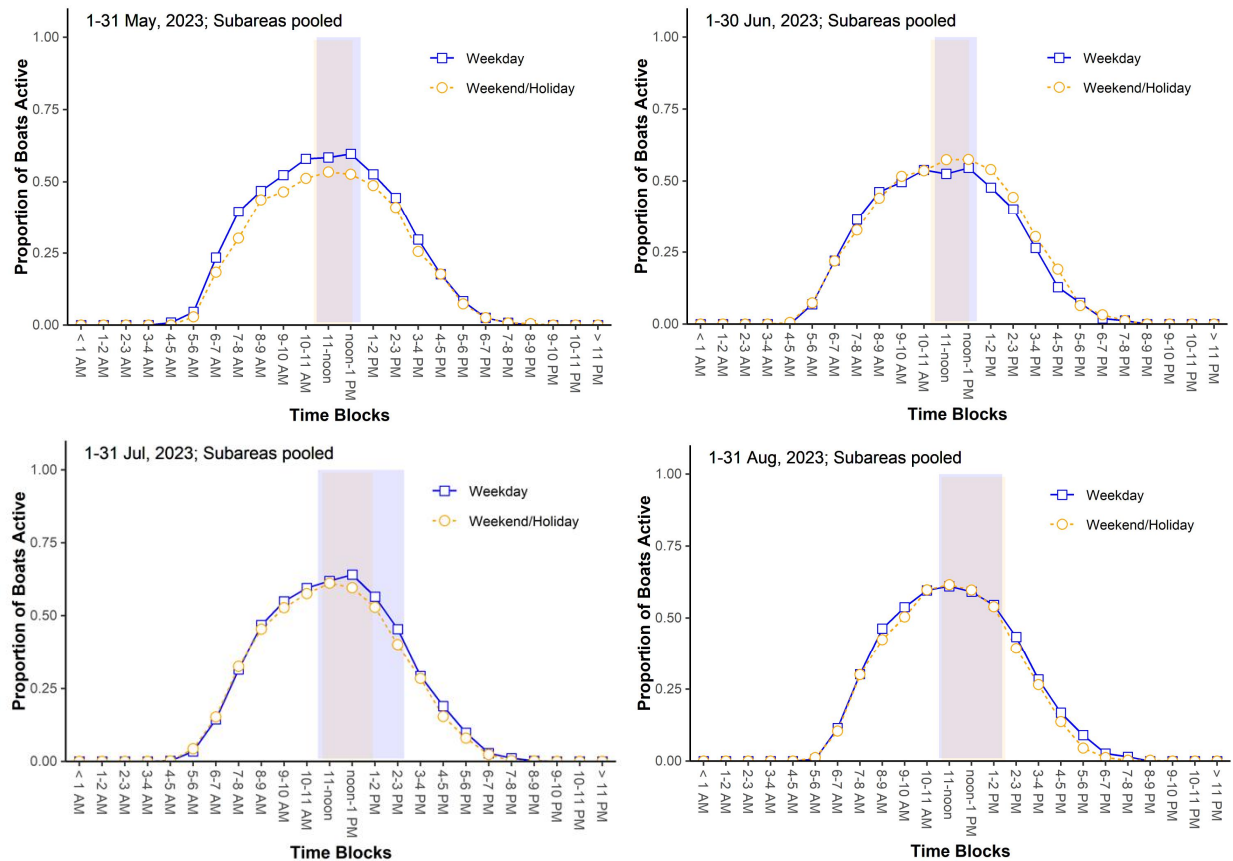
| Month  | Subarea |     |    |     |     |     |     |    |     |    |     |    |    |    |         | TOTAL |
|--------|---------|-----|----|-----|-----|-----|-----|----|-----|----|-----|----|----|----|---------|-------|
|        | 3I      | 3J  | 3K | 4A  | 4B  | 4C  | 4D  | 4E | 4F  | 4G | 4H  | 4L | 3? | 4? | Refusal |       |
| May    | 7       | 9   | 6  | 91  | 55  | 60  | 23  | 10 | 110 | 10 | 14  | 2  |    | 7  | 5       | 407   |
| June   | 13      | 25  | 10 | 93  | 43  | 77  | 40  | 5  | 140 | 13 | 30  | 13 | 1  | 6  | 4       | 509   |
| July   | 29      | 42  | 33 | 110 | 60  | 139 | 114 | 8  | 224 | 31 | 58  | 19 | 1  | 11 | 13      | 891   |
| August | 28      | 31  | 47 | 70  | 66  | 76  | 46  | 12 | 102 | 22 | 67  | 9  | 1  | 3  | 16      | 597   |
| TOTAL  | 76      | 107 | 95 | 364 | 224 | 351 | 223 | 34 | 576 | 76 | 169 | 43 | 2  | 27 | 38      | 2,404 |

**Table 5. Numbers of non-lodge boat trips that reported only fishing for shellfish, by month and subarea, 2023.**

| Month  | Subarea |    |    |    |    |    |    |    |    |    |    |    |    |    |  | TOTAL |
|--------|---------|----|----|----|----|----|----|----|----|----|----|----|----|----|--|-------|
|        | 3I      | 3J | 3K | 4A | 4B | 4C | 4D | 4E | 4F | 4G | 4H | 4L | 3? | 4? |  |       |
| May    | 0       | 0  | 0  | 17 | 1  | 1  | 0  | 0  | 0  | 0  | 2  | 0  | 0  | 0  |  | 21    |
| June   | 1       | 0  | 0  | 10 | 1  | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |  | 13    |
| July   | 0       | 0  | 1  | 18 | 4  | 0  | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  |  | 24    |
| August | 0       | 0  | 0  | 14 | 1  | 1  | 0  | 0  | 0  | 0  | 1  | 0  | 0  | 0  |  | 17    |
| TOTAL  | 1       | 0  | 1  | 59 | 7  | 3  | 0  | 0  | 0  | 1  | 3  | 0  | 0  | 0  |  | 75    |

**Table 6. Numbers of guided (i.e., led by professional guides) and unguided boat trips, by landing site, 2023.**

| Landing Site             | Trip Type |          | % guided |
|--------------------------|-----------|----------|----------|
|                          | Guided    | Unguided |          |
| Port Edward Boat Launch  | 16        | 634      | 2.5%     |
| Prince Rupert Yacht Club | 428       | 291      | 60%      |
| Rushbrooke Boat Launch   | 257       | 778      | 25%      |
| TOTAL                    | 701       | 1,703    | 29%      |



**Figure 4.** Angler activity pattern (proportion recreational angling boats that were actively fishing during each hour-long time block) by day type and month, pooled over all subareas, 2023. Shaded regions bound the earliest and latest flight times for each day type (blue = weekday; orange = weekend/holiday) and period.

### Angler Activity Patterns

Non-lodge angler activity pattern estimates varied little among periods (Figure 4). There was virtually no difference between the weekday and weekend/holiday activity patterns, except for a slightly greater tendency for weekday anglers to be active during the peak times in May, as compared to weekend anglers. Timing of our overflights lined up with peak activity levels for all but one weekday flight (27 July 2023) that occurred at an off-peak time (1:10 to 2:51 PM) as a result of fog earlier in the day.

### Catch Per Effort Estimates

Sample sizes were adequate ( $n \geq 5$ ) to produce reliable estimates of non-lodge CPE in most combinations of periods and creel study sub-areas (Table 7), except in subareas 4L in May. However, when the interview data were split between guided and non-guided groups, period-specific sample sizes did not suffice for several more combinations of period and management area (Table 7). In those cases where sample sizes were limited, data from three adjacent subareas were pooled together for CPE calculations (Table 8).



Average number of fish harvested per non-lodge boat trip (HPE) was calculated for each species by period, subarea, and by trip type (Appendix 2).

HPE varied significantly among periods (Figure 5, Table 9) for Chinook ( $\chi^2 = 22.5$ ,  $df = 3$ ,  $P < 0.0001$ ), Coho ( $\chi^2 = 29.0$ ,  $df = 3$ ,  $P < 0.0001$ ) and Pink salmon (*Oncorhynchus gorbuscha*,  $\chi^2 = 24.8$ ,  $df = 3$ ,  $P < 0.0001$ ). Chinook Salmon HPE increased from May to June and was reduced thereafter. For Coho Salmon, median HPE increased over the course of the study period (Table 9; Figure 5), with May being significantly lower than June, which was significantly lower than July and August. HPE for Pink Salmon was significantly lower in May and June as compared to July and August. HPE for Bocaccio Rockfish (*Sebastes paucispinis*) also varied significantly among periods (Table 9), but the among period differences were not statistically significant after controlling for the rate of false positives. There were no significant differences for any other species examined (including Pacific Halibut and Lingcod; Figure 5).

Guided trips typically had higher HPE than non-guided trips, and the differences were statistically significant for a variety of species (Table 10). In particular, guided trips harvested 0.5 more Chinook Salmon ( $t_{47} = 5.2$ ,  $P < 0.0001$ ), and 2.1 more Coho Salmon ( $t_{47} = 5.7$ ,  $P < 0.0001$ ) than unguided trips. In addition, guided trips harvested 0.6 more Lingcod ( $t_{47} = 4.4$ ,  $P < 0.0001$ ), 0.1 more Pacific Cod (*Gadus macrocephalus*,  $t_{47} = 2.1$ ,  $P = 0.039$ ), and 0.8 more Pacific Halibut per trip ( $t_{47} = 6.5$ ,  $P < 0.0001$ ) than unguided trips. Guided trips also harvested significantly more Quillback (*Sebastes maliger*), Tiger (*S. nigrocinctus*), and Widow rockfish than unguided trips.

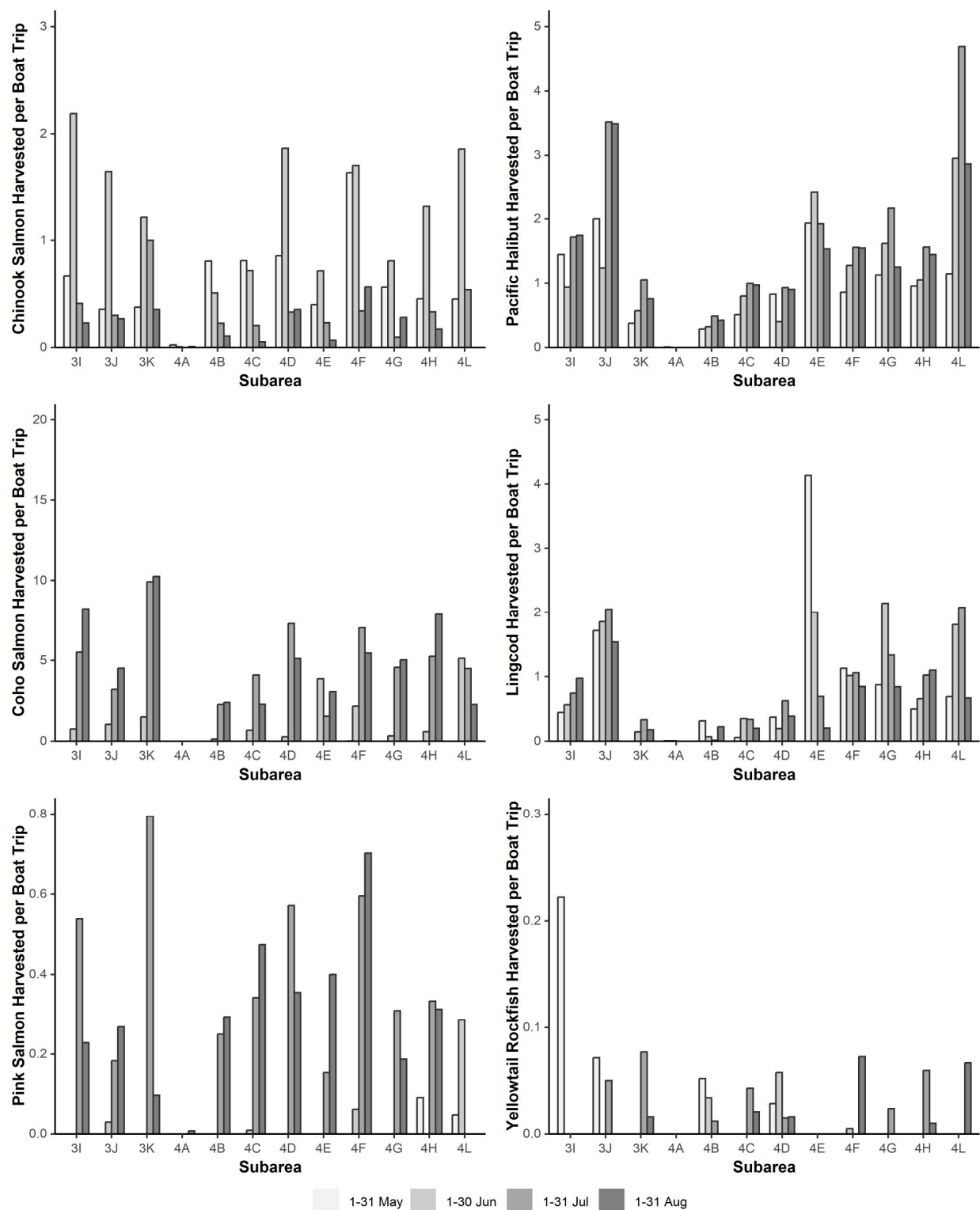
For lodge boat trips, HPE (and RPE for Chinook Salmon) values are shown in Table 11.

**Table 7. Sample sizes for CPE (harvest per effort) estimates by period, for guided and not guided non-lodge boat trips, 2023.**

|        |           | Subarea |    |    |     |    |     |     |    |     |    |    |    |
|--------|-----------|---------|----|----|-----|----|-----|-----|----|-----|----|----|----|
| Period | Trip Type | 3I      | 3J | 3K | 4A  | 4B | 4C  | 4D  | 4E | 4F  | 4G | 4H | 4L |
| May    | Guided    | 3       | 6  | 0  | 58  | 5  | 13  | 11  | 3  | 46  | 2  | 10 | 3  |
|        | Unguided  | 6       | 8  | 8  | 104 | 72 | 77  | 24  | 12 | 108 | 14 | 12 | 1  |
|        | Combined  | 9       | 14 | 8  | 162 | 77 | 90  | 35  | 15 | 154 | 16 | 22 | 4  |
| June   | Guided    | 4       | 14 | 2  | 110 | 2  | 17  | 16  | 2  | 83  | 4  | 12 | 16 |
|        | Unguided  | 12      | 20 | 12 | 67  | 57 | 89  | 36  | 5  | 113 | 17 | 29 | 5  |
|        | Combined  | 16      | 34 | 14 | 177 | 59 | 106 | 52  | 7  | 196 | 21 | 41 | 21 |
| July   | Guided    | 11      | 33 | 22 | 95  | 2  | 35  | 33  | 3  | 132 | 11 | 35 | 21 |
|        | Unguided  | 28      | 27 | 17 | 112 | 82 | 152 | 100 | 10 | 162 | 31 | 49 | 5  |
|        | Combined  | 39      | 60 | 39 | 207 | 84 | 187 | 133 | 13 | 294 | 42 | 84 | 26 |
| August | Guided    | 13      | 16 | 26 | 66  | 5  | 10  | 5   | 2  | 36  | 13 | 60 | 9  |
|        | Unguided  | 22      | 25 | 36 | 61  | 80 | 87  | 57  | 13 | 88  | 19 | 39 | 6  |
|        | Combined  | 35      | 41 | 62 | 127 | 85 | 97  | 62  | 15 | 124 | 32 | 99 | 15 |

**Table 8. Subarea data (pooled over several adjacent areas, in some cases) used to calculate CPE in each period and subarea, for guided and not guided non-lodge boat trips, 2023.**

|        |           | Subarea  |    |          |    |        |    |    |        |    |        |    |        |
|--------|-----------|----------|----|----------|----|--------|----|----|--------|----|--------|----|--------|
| Period | Trip Type | 3I       | 3J | 3K       | 4A | 4B     | 4C | 4D | 4E     | 4F | 4G     | 4H | 4L     |
| May    | Guided    | 3?,I,J,K | 3J | 3?,I,J,K | 4A | 4B     | 4C | 4D | 4D,E,F | 4F | 4G,H,L | 4H | 4G,H,L |
|        | Unguided  | 3I       | 3J | 3K       | 4A | 4B     | 4C | 4D | 4E     | 4F | 4G     | 4H | 4G,H,L |
|        | Combined  | 3I       | 3J | 3K       | 4A | 4B     | 4C | 4D | 4E     | 4F | 4G     | 4H | 4G,H,L |
| June   | Guided    | 3?,I,J,K | 3J | 3?,I,J,K | 4A | 4A,B,C | 4C | 4D | 4D,E,F | 4F | 4G,H,L | 4H | 4L     |
|        | Unguided  | 3I       | 3J | 3K       | 4A | 4B     | 4C | 4D | 4E     | 4F | 4G     | 4H | 4L     |
|        | Combined  | 3I       | 3J | 3K       | 4A | 4B     | 4C | 4D | 4E     | 4F | 4G     | 4H | 4L     |
| July   | Guided    | 3I       | 3J | 3K       | 4A | 4A,B,C | 4C | 4D | 4D,E,F | 4F | 4G     | 4H | 4L     |
|        | Unguided  | 3I       | 3J | 3K       | 4A | 4B     | 4C | 4D | 4E     | 4F | 4G     | 4H | 4L     |
|        | Combined  | 3I       | 3J | 3K       | 4A | 4B     | 4C | 4D | 4E     | 4F | 4G     | 4H | 4L     |
| August | Guided    | 3I       | 3J | 3K       | 4A | 4B     | 4C | 4D | 4D,E,F | 4F | 4G     | 4H | 4L     |
|        | Unguided  | 3I       | 3J | 3K       | 4A | 4B     | 4C | 4D | 4E     | 4F | 4G     | 4H | 4L     |
|        | Combined  | 3I       | 3J | 3K       | 4A | 4B     | 4C | 4D | 4E     | 4F | 4G     | 4H | 4L     |



**Figure 5.** Average number of fish harvested per non-lodge boat trip, for six sportfish species, by period and subarea, 2023. Data are for guided and unguided trips combined. Note that Y axis scale varies among plots.

**Table 9. Kruskal-Wallis tests for differences in HPE (fish harvested per non-lodge boat trip) among periods, by species, 2023. Subareas treated as replicates. Red letters indicate statistical differences within species (periods not connected by a common letter were statistically different).**

| Species               |                                   | Kruskal-Wallis Statistics |    |          | Median HPE   |              |              |              |
|-----------------------|-----------------------------------|---------------------------|----|----------|--------------|--------------|--------------|--------------|
|                       |                                   | $\chi^2$                  | df | P        | May          | June         | July         | August       |
| Chinook Salmon        | <i>Oncorhynchus tshawytscha</i>   | 22.5                      | 3  | < 0.0001 | <b>B</b> 0.5 | <b>A</b> 1.3 | <b>C</b> 0.3 | <b>C</b> 0.2 |
| Chum Salmon           | <i>Oncorhynchus keta</i>          | 5.0                       | 3  | 0.17     | 0            | 0            | 0            | 0            |
| Coho Salmon           | <i>Oncorhynchus kisutch</i>       | 29.0                      | 3  | < 0.0001 | <b>C</b> 0   | <b>B</b> 0.7 | <b>A</b> 4.5 | <b>A</b> 4.8 |
| Pink Salmon           | <i>Oncorhynchus gorbuscha</i>     | 24.8                      | 3  | < 0.0001 | <b>B</b> 0   | <b>B</b> 0   | <b>A</b> 0.3 | <b>A</b> 0.3 |
| Sockeye Salmon        | <i>Oncorhynchus nerka</i>         | 4.0                       | 3  | 0.26     | 0            | 0            | 0            | 0            |
| Arrow-tooth Flounder  | <i>Atheresthes stomias</i>        | 4.0                       | 3  | 0.26     | 0            | 0            | 0            | 0            |
| Big Skate             | <i>Beringraja binoculata</i>      | 2.0                       | 3  | 0.56     | 0            | 0            | 0            | 0            |
| Cabezon               | <i>Scorpaenichthys marmoratus</i> | 2.1                       | 3  | 0.54     | 0            | 0            | 0            | 0            |
| English Sole          | <i>Parophrys vetulus</i>          | 3.0                       | 3  | 0.39     | 0            | 0            | 0            | 0            |
| Greenlings            | <i>Hexagrammos</i> spp.           | 2.5                       | 3  | 0.48     | 0            | 0            | 0            | 0            |
| Lingcod               | <i>Ophiodon elongatus</i>         | 0.9                       | 3  | 0.84     | 0.5          | 0.6          | 0.7          | 0.5          |
| Longnose Skate        | <i>Caliraja rhina</i>             | 0.0                       | 3  | 1        | 0            | 0            | 0            | 0            |
| Pacific Cod           | <i>Gadus macrocephalus</i>        | 0.6                       | 3  | 0.91     | 0.09         | 0.04         | 0.07         | 0.04         |
| Pacific Halibut       | <i>Hippoglossus stenolepis</i>    | 3.5                       | 3  | 0.32     | 0.9          | 1.0          | 1.6          | 1.3          |
| Pacific Rock Sole     | <i>Lepidopsetta bilineata</i>     | 3.7                       | 3  | 0.29     | 0            | 0            | 0            | 0            |
| Pacific Spiny Dogfish | <i>Squalus suckleyi</i>           | 3.7                       | 3  | 0.29     | 0            | 0            | 0            | 0            |
| Sablefish             | <i>Anoplopoma fimbria</i>         | 2.0                       | 3  | 0.56     | 0            | 0            | 0            | 0            |
| Spotted Ratfish       | <i>Hydrolagus colliei</i>         | 0.0                       | 3  | 1        | 0            | 0            | 0            | 0            |
| Starry Flounder       | <i>Platichthys stellatus</i>      | 0.0                       | 3  | 1        | 0            | 0            | 0            | 0            |
| Black Rockfish        | <i>Sebastes melanops</i>          | 3.5                       | 3  | 0.32     | 0.05         | 0.1          | 0.1          | 0.2          |
| Boccacio Rockfish     | <i>Sebastes paucispinis</i>       | 8.3                       | 3  | 0.040    | 0            | 0            | 0            | 0            |
| Canary Rockfish       | <i>Sebastes pinniger</i>          | 2.0                       | 3  | 0.56     | 0            | 0            | 0            | 0            |
| China Rockfish        | <i>Sebastes nebulosus</i>         | 5.1                       | 3  | 0.16     | 0            | 0            | 0            | 0.01         |
| Copper Rockfish       | <i>Sebastes caurinus</i>          | 1.3                       | 3  | 0.72     | 0            | 0.02         | 0.005        | 0.005        |
| Dusky Rockfish        | <i>Sebastes ciliatus</i>          | 0.0                       | 3  | 1        | 0            | 0            | 0            | 0            |
| Quillback Rockfish    | <i>Sebastes maliger</i>           | 2.6                       | 3  | 0.46     | 0.08         | 0.1          | 0.07         | 0.1          |
| Redbanded Rockfish    | <i>Sebastes babcocki</i>          | 3.0                       | 3  | 0.39     | 0            | 0            | 0            | 0            |
| Redstripe Rockfish    | <i>Sebastes proriger</i>          | 0.0                       | 3  | 1        | 0            | 0            | 0            | 0            |
| Tiger Rockfish        | <i>Sebastes nigrocinctus</i>      | 2.0                       | 3  | 0.56     | 0            | 0            | 0            | 0            |
| Widow Rockfish        | <i>Sebastes entomelas</i>         | 2.9                       | 3  | 0.40     | 0            | 0            | 0            | 0            |
| Yelloweye Rockfish    | <i>Sebastes ruberrimus</i>        | 1.0                       | 3  | 0.79     | 0            | 0            | 0            | 0            |
| Yellowtail Rockfish   | <i>Sebastes flavidus</i>          | 2.9                       | 3  | 0.41     | 0            | 0            | 0.01         | 0.005        |

**Table 10. Paired t-tests for differences in HPE (fish harvested per non-lodge boat trip) between guided and non-guided boats, by species, 2023. Periods and subareas treated as replicates.**

| Species               | Paired t-test Statistics |    |          | Median HPE |       | Paired Mean Difference |
|-----------------------|--------------------------|----|----------|------------|-------|------------------------|
|                       | t                        | df | P        | Guided     | Not   |                        |
| Chinook Salmon        | 5.2                      | 47 | < 0.0001 | 0.5        | 0.3   | 0.5                    |
| Chum Salmon           | -1.5                     | 47 | 0.14     | 0          | 0     | -0.0262                |
| Coho Salmon           | 5.7                      | 47 | < 0.0001 | 2.5        | 1.1   | 2.1                    |
| Pink Salmon           | 0.7                      | 47 | 0.47     | 0.05       | 0.006 | 0.03                   |
| Sockeye Salmon        | -1.0                     | 47 | 0.34     | 0          | 0     | -0.0018                |
| Arrow-tooth Flounder  | 1.4                      | 47 | 0.17     | 0          | 0     | 0.005                  |
| Big Skate             | -1.4                     | 47 | 0.17     | 0          | 0     | -0.0009                |
| Cabezon               | 0.5                      | 47 | 0.60     | 0          | 0     | 0.0009                 |
| English Sole          | -1.0                     | 47 | 0.32     | 0          | 0     | -0.0007                |
| Greenlings            | -0.4                     | 47 | 0.72     | 0          | 0     | -0.0022                |
| Lingcod               | 4.4                      | 47 | < 0.0001 | 1.2        | 0.4   | 0.6                    |
| Longnose Skate        | 0.0                      | 47 | 1        | 0          | 0     | 0                      |
| Pacific Cod           | 2.1                      | 47 | 0.039    | 0.07       | 0.03  | 0.10                   |
| Pacific Halibut       | 6.5                      | 47 | < 0.0001 | 1.7        | 0.9   | 0.8                    |
| Pacific Rock Sole     | -1.5                     | 47 | 0.13     | 0          | 0     | -0.0015                |
| Pacific Spiny Dogfish | -0.9                     | 47 | 0.35     | 0          | 0     | -0.0014                |
| Sablefish             | 1.2                      | 47 | 0.25     | 0          | 0     | 0.004                  |
| Spotted Ratfish       | 0.0                      | 47 | 1        | 0          | 0     | 0                      |
| Starry Flounder       | 0.0                      | 47 | 1        | 0          | 0     | 0                      |
| Black Rockfish        | 1.6                      | 47 | 0.12     | 0.2        | 0.08  | 0.1                    |
| Boccacio Rockfish     | 0.9                      | 47 | 0.35     | 0          | 0     | 0.007                  |
| Canary Rockfish       | -1.3                     | 47 | 0.20     | 0          | 0     | -0.0007                |
| China Rockfish        | 1.6                      | 47 | 0.11     | 0          | 0     | 0.02                   |
| Copper Rockfish       | 0.1                      | 47 | 0.96     | 0          | 0     | 0.0007                 |
| Dusky Rockfish        | 0.0                      | 47 | 1        | 0          | 0     | 0                      |
| Quillback Rockfish    | 4.5                      | 47 | < 0.0001 | 0.2        | 0.06  | 0.2                    |
| Redbanded Rockfish    | 1.0                      | 47 | 0.32     | 0          | 0     | 0.0007                 |
| Redstripe Rockfish    | 0.0                      | 47 | 1        | 0          | 0     | 0                      |
| Tiger Rockfish        | 2.7                      | 47 | 0.009    | 0          | 0     | 0.01                   |
| Widow Rockfish        | 2.1                      | 47 | 0.045    | 0          | 0     | 0.02                   |
| Yelloweye Rockfish    | 1.0                      | 47 | 0.33     | 0          | 0     | 0.003                  |
| Yellowtail Rockfish   | 0.1                      | 47 | 0.90     | 0          | 0     | 0.001                  |

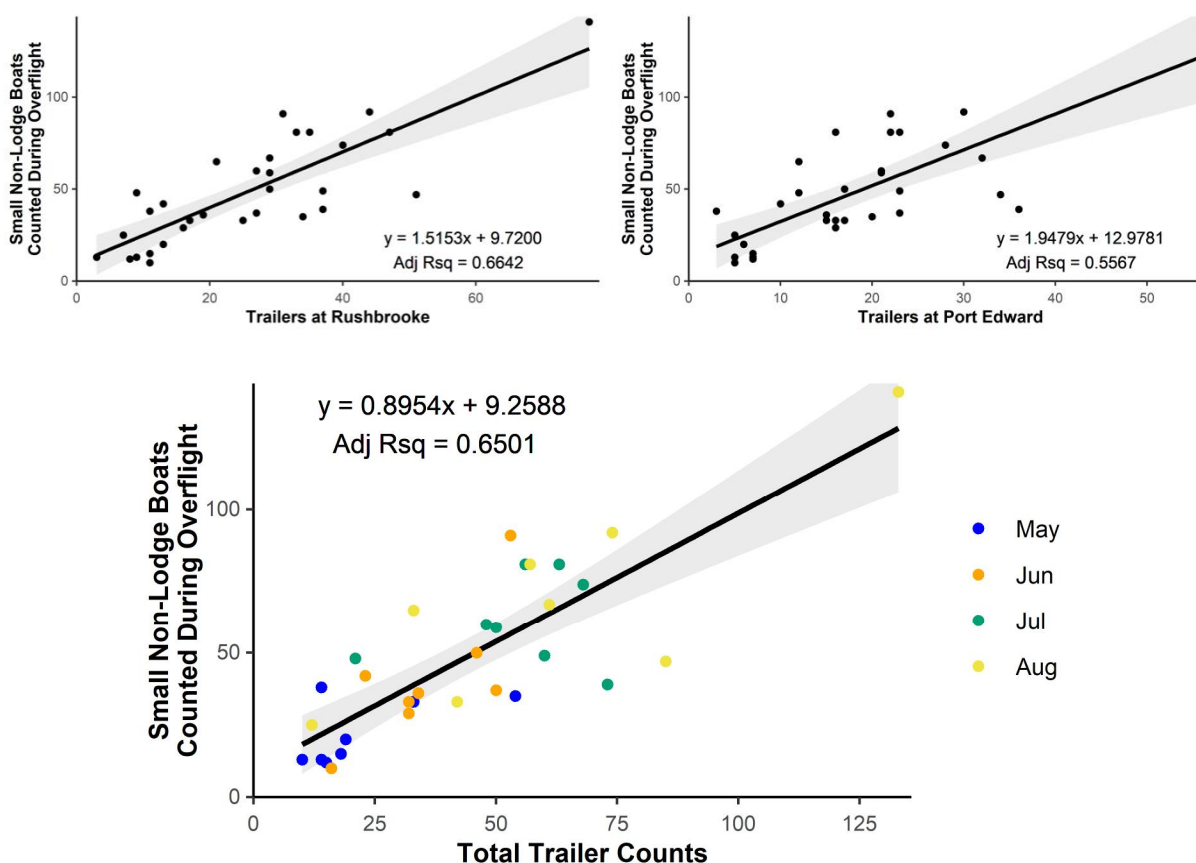
**Table 11. Average number of harvested fish (and for Chinook Salmon, released) per boat trip, by month, from the census of Area 3 lodges, 2023.**

| Species            | May | June   | July | August | Total  |
|--------------------|-----|--------|------|--------|--------|
| Chinook Salmon:    |     |        |      |        |        |
| retained           | 0.8 | 0.8    | 0.3  | 0.2    | 0.5    |
| released           | 0.7 | 1.0    | 1.2  | 0.2    | 0.8    |
| Coho Salmon        | 0   | 0.2    | 2.3  | 5.9    | 2.6    |
| Pink Salmon        | 0   | 0.0005 | 0.04 | 0.03   | 0.02   |
| Chum Salmon        | 0   | 0      | 0    | 0      | 0      |
| Sockeye Salmon     | 0   | 0      | 0    | 0      | 0      |
| Pacific Halibut    | 0.3 | 0.6    | 0.7  | 0.8    | 0.7    |
| Lingcod            | 0.7 | 0.8    | 1.0  | 0.8    | 0.8    |
| Other groundfish   | 0   | 0      | 0    | 0      | 0      |
| Yelloweye Rockfish | 0   | 0      | 0    | 0.002  | 0.0005 |
| Other rockfish     | 0.5 | 0.3    | 0.3  | 0.4    | 0.4    |

## Angler Effort Estimates

Over the four month study period, a total of 32 effort surveys were conducted, including 20 weekday surveys and 12 conducted on weekends/holidays (Table 2, Appendix 3). The numbers of lodge boats that were visible during the overflights (Appendix 3) were partitioned among the subareas in Management Area 3, and subtracted from the overflight counts of small boats. The remaining small boat counts (“the non-lodge small boat counts”) were significantly correlated to trailer counts at both survey launch sites (Rushbrooke:  $F_{1,30} = 62.3$ ,  $P < 0.0001$ ; Port Edwards:  $F_{1,30} = 39.9$ ,  $P < 0.0001$ ), and when the two launch sites were combined ( $F_{1,30} = 58.6$ ,  $P < 0.0001$ ). The fit of the Port Edward regression (Port Edward  $R^2 = 0.56$ ) was lower than that of the Rushbrooke regression ( $R^2 = 0.66$ ) or that of combined-site regression ( $R^2 = 0.65$ ; Figure 6).

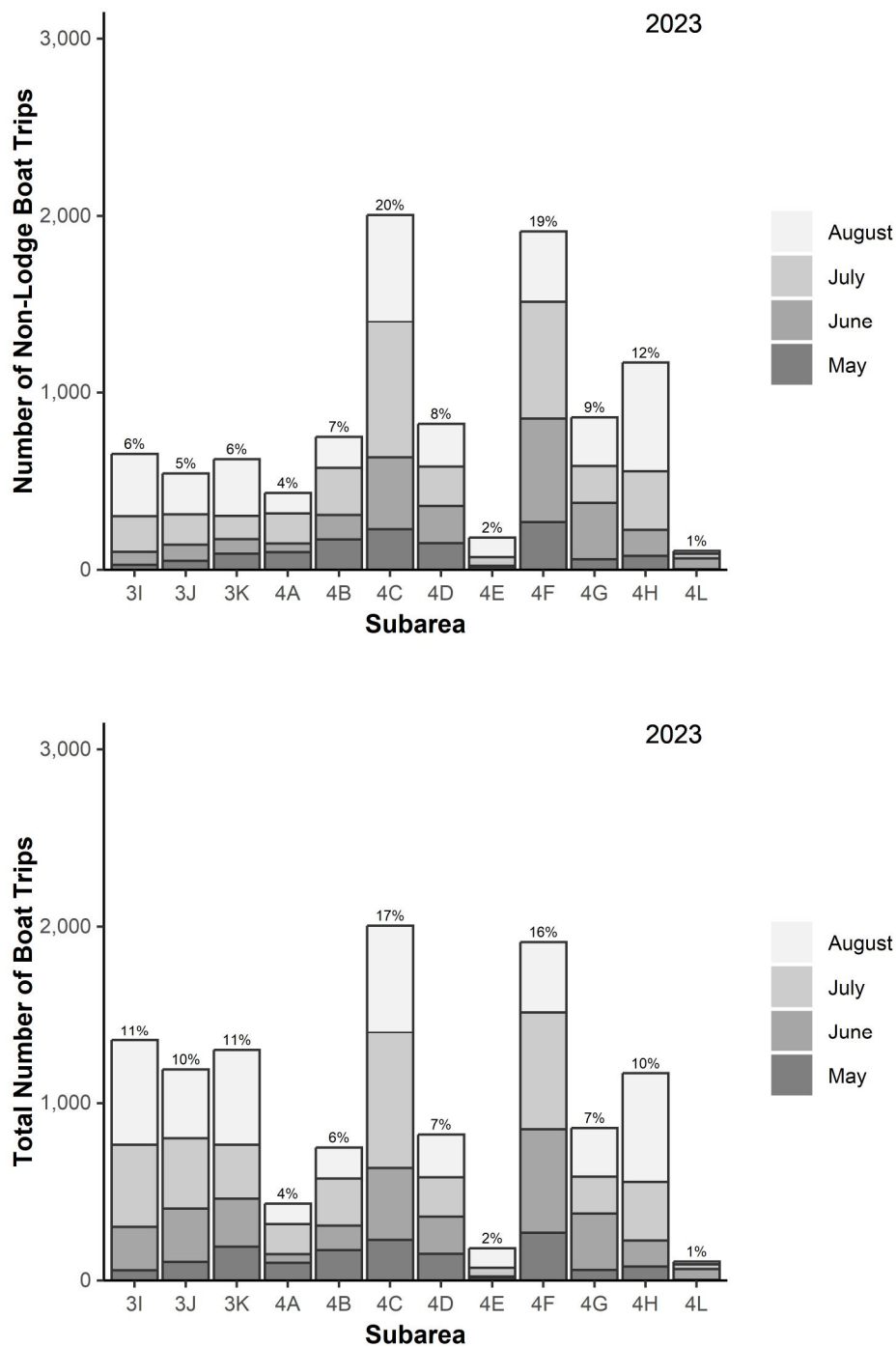
The non-lodge boat counts were adjusted for non-lodge angler effort patterns in order to estimate the total monthly non-lodge angling effort for each subarea in 2023 (Table 12, Figures 7 and 8). Median daily non-lodge effort on weekdays (4.6 trips per day per subarea) was lower than on weekends/holidays (6.3 trips per day per subarea) but the effect was not statistically significant ( $\chi^2 = 2.7$ ,  $df = 1$ ,  $P = 0.10$ ). Effort in May (2.2 trips per day per subarea) was significantly ( $\chi^2 = 15.1$ ,  $df = 3$ ,  $P = 0.002$ ) lower than during other months (range from 5.0 to 8.9).



**Figure 6.** Linear regressions showing the relationships between small non-lodge boats counted during overflights and trailers concurrently counted at landing sites. Shaded areas show the 95% confidence interval of the predicted values. Note that X axis scale varies among plots.

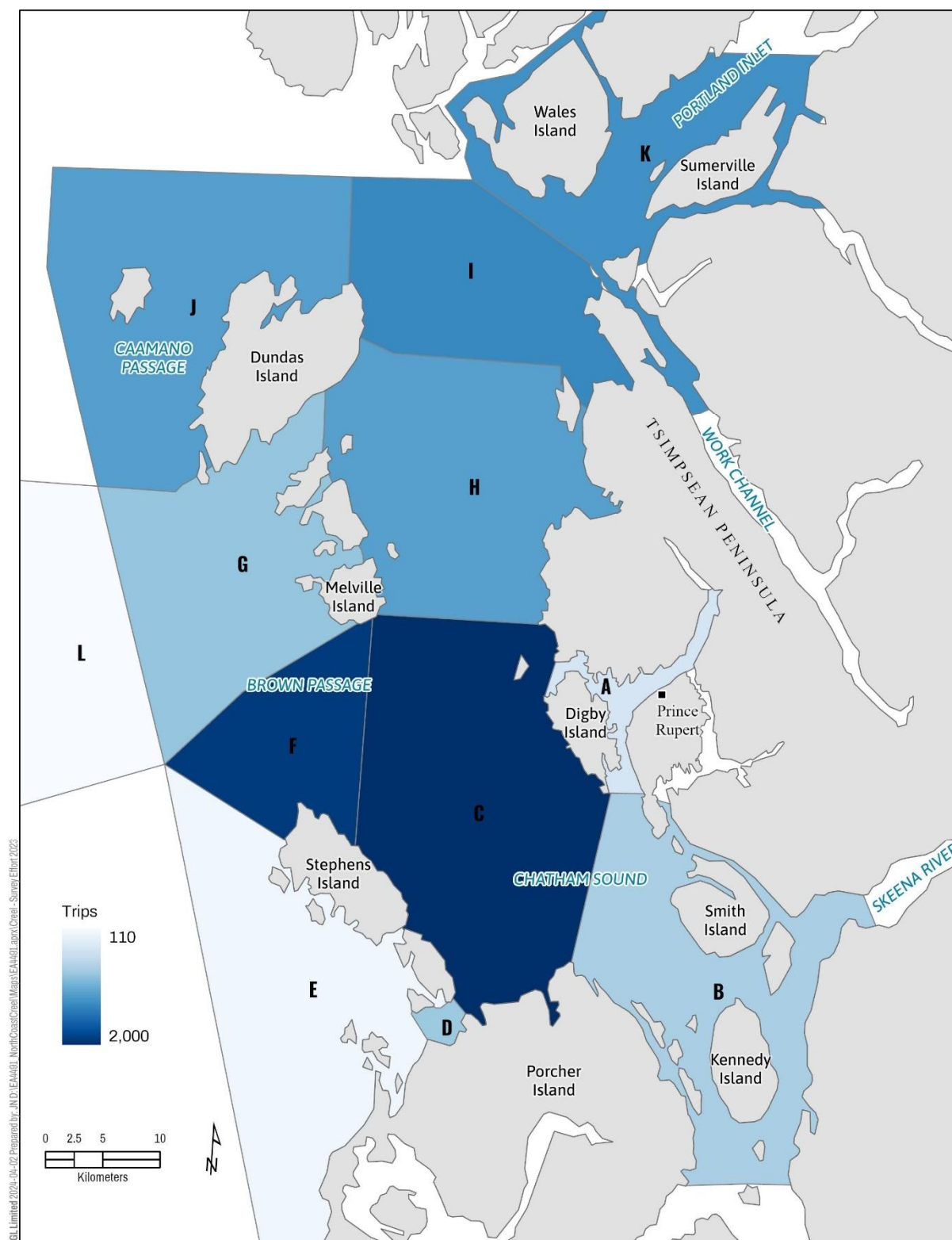
**Table 12. Lodge, non-lodge, and total effort estimates (boat trips per month), by month and subarea, 2023.**

| Month             |                | Subarea |       |       |       |       |       |       |       |       |       |       |      | TOTALS |        |         | SE of the Totals |        |         |
|-------------------|----------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--------|--------|---------|------------------|--------|---------|
|                   |                | 3I      | 3J    | 3K    | 4A    | 4B    | 4C    | 4D    | 4E    | 4F    | 4G    | 4H    | 4L   | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| May               | Creel Estimate | 27.6    | 50.0  | 91.0  | 98.9  | 171.6 | 227.9 | 150.3 | 15.0  | 268.6 | 60.1  | 79.0  | 4.0  | 169    | 1,075  | 1,244   | 3                | 8      | 8       |
|                   | Lodge Report   | 30.1    | 54.6  | 99.3  |       |       |       |       |       |       |       |       |      | 184    | 0      | 184     |                  |        |         |
|                   | TOTAL MAY      | 57.8    | 104.6 | 190.2 | 98.9  | 171.6 | 227.9 | 150.3 | 15.0  | 268.6 | 60.1  | 79.0  | 4.0  | 353    | 1,075  | 1,428   | 3                | 8      | 8       |
|                   | SE             | 0.7     | 1.3   | 2.1   | 1.8   | 2.9   | 3.0   | 3.1   | 0.0   | 4.8   | 1.9   | 2.2   | 0.0  |        |        |         |                  |        |         |
| June              | Creel Estimate | 74.4    | 91.0  | 82.4  | 48.8  | 138.1 | 407.0 | 209.2 | 7.0   | 583.8 | 316.3 | 146.6 | 61.3 | 248    | 1,918  | 2,166   | 3                | 13     | 13      |
|                   | Lodge Report   | 170.8   | 209.0 | 189.2 |       |       |       |       |       |       |       |       |      | 569    | 0      | 569     |                  |        |         |
|                   | TOTAL JUNE     | 245.2   | 300.0 | 271.7 | 48.8  | 138.1 | 407.0 | 209.2 | 7.0   | 583.8 | 316.3 | 146.6 | 61.3 | 817    | 1,918  | 2,735   | 3                | 13     | 13      |
|                   | SE             | 1.7     | 1.8   | 1.7   | 0.8   | 2.3   | 3.6   | 3.2   | 0.0   | 9.5   | 5.4   | 2.4   | 1.7  |        |        |         |                  |        |         |
| July              | Creel Estimate | 200.0   | 171.8 | 131.3 | 170.4 | 265.8 | 765.6 | 221.7 | 50.2  | 664.1 | 209.6 | 330.3 | 26.0 | 503    | 2,704  | 3,207   | 3                | 11     | 11      |
|                   | Lodge Report   | 262.8   | 225.7 | 172.5 |       |       |       |       |       |       |       |       |      | 661    | 0      | 661     |                  |        |         |
|                   | TOTAL JULY     | 462.8   | 397.5 | 303.8 | 170.4 | 265.8 | 765.6 | 221.7 | 50.2  | 664.1 | 209.6 | 330.3 | 26.0 | 1,164  | 2,704  | 3,868   | 3                | 11     | 11      |
|                   | SE             | 2.3     | 2.0   | 1.6   | 1.5   | 2.8   | 5.6   | 2.5   | 0.9   | 6.6   | 2.1   | 4.0   | 0.0  |        |        |         |                  |        |         |
| August            | Creel Estimate | 351.4   | 231.2 | 318.6 | 115.6 | 174.1 | 605.6 | 241.3 | 109.1 | 396.6 | 274.3 | 612.3 | 15.0 | 901    | 2,544  | 3,445   | 7                | 14     | 15      |
|                   | Lodge Report   | 237.8   | 156.5 | 215.7 |       |       |       |       |       |       |       |       |      | 610    | 0      | 610     |                  |        |         |
|                   | TOTAL AUGUST   | 589.2   | 387.7 | 534.3 | 115.6 | 174.1 | 605.6 | 241.3 | 109.1 | 396.6 | 274.3 | 612.3 | 15.0 | 1,511  | 2,544  | 4,055   | 7                | 14     | 15      |
|                   | SE             | 4.5     | 3.9   | 4.0   | 1.6   | 2.5   | 5.8   | 3.2   | 1.4   | 6.7   | 4.0   | 8.4   | 0.0  |        |        |         |                  |        |         |
| FULL SEASON TOTAL |                | 1,355   | 1,190 | 1,300 | 434   | 750   | 2,006 | 823   | 181   | 1,913 | 860   | 1,168 | 106  | 3,845  | 8,241  | 12,086  | 9                | 23     | 24      |
| FULL SEASON SE    |                | 5       | 5     | 5     | 3     | 5     | 9     | 6     | 2     | 14    | 7     | 10    | 2    |        |        |         |                  |        |         |



**Figure 7.** Total estimated numbers of boat trips, by month and subarea, 2023. Upper panel: estimated monthly non-lodge boat trips. Lower panel: total effort, calculated as the sum of lodge and non-lodge boat trips.





**Figure 8.** Heat map showing the estimated total effort (numbers of boat trips, lodge + non-lodge) in each subarea in 2023. See Table 12 for details.

Total lodge effort in 2023 was 2,024 boat trips (184 in May, 569 in June, 661 in July, and 610 in August; Table 12). To calculate total (lodge + non-lodge) fishing effort by subarea, lodge boat trips were distributed among the three subareas in Management Area 3 according to the relative non-lodge effort in each (Table 12; Figure 7). In all, there were 1,428 (SE = 8) boat trips in May, 2,735 (SE = 13) boat trips in June, 3,868 (SE = 11) boat trips in July, and 4,055 (SE = 15) boat trips in August. The full-season (May-August) Area 3-4 effort estimate for 2023 was 12,086 boat trips (SE = 24). For comparison with our previous study years, which did not include surveys in May, the June-August effort total was 10,658 boat trips (SE = 23).

### **Catch Estimates**

Monthly and full-season estimates were made of the numbers of fish released and retained, by species (Tables 13-22; Appendix 4). Non-lodge estimates were generated from the product of effort and CPE estimates. Lodge catches were extracted from census documents that were submitted to DFO by the lodges, and which were then presented by DFO to NCSFNSS with identifying information removed for Privacy Act reasons. The results presented in this section are total (lodge + non-lodge) numbers of fish retained and released.

An estimated 6,443 (SE = 182) Chinook Salmon were retained by anglers in Areas 3 or 4 during the 2023 study period (Table 14; Figure 9). The largest proportional harvest came from subareas 4F (29%), 3K, and 4D (11% each). Harvest was highest in June (49% of total harvest), but the among-month differences were not statistically significant ( $\chi^2 = 6.0$ ,  $df = 3$ ,  $P = 0.11$ ; Table 23). During the study, the number of released Chinook Salmon was estimated at 8,812 (SE = 295), or 42% of the total catch. Percent retention (calculated for non-lodge anglers) was 55% in May, 46% in June, 25% in July, and 59% in August (Appendix 4).

The salmonid that was harvested in the greatest numbers during the 2023 study period was, by far, Coho Salmon (42,525 fish retained, SE = 738; Table 15; Figure 10). The largest proportional harvest came from subarea 4F (19%), followed by subareas 3K (17%), 4H (16%), 3I (14%), and 4C (11%). Harvest differed significantly among months: harvest in May was significantly lower than all other months, and that in June was significantly lower than July or August ( $\chi^2 = 28.2$ ,  $df = 3$ ,  $P < 0.0001$ ; Table 23). During the study period, an estimated 9,529 (SE = 531) Coho Salmon were released by anglers (18% of the total catch).

Pink salmon was the least likely salmonid to be retained in 2023 (Table 16). We estimate that 22,639 (SE = 1,112) fish were caught, of which 11% were retained (2,548 fish, SE = 152), and 89% were released (20,091 fish, SE = 1,012). Harvest of Pink Salmon varied significantly among months ( $\chi^2 = 25.7$ ,  $df = 3$ ,  $P < 0.0001$ ; Table 23): harvest in May and June was significantly lower than in July or August.

Very few Chum (*Oncorhynchus keta*, 259 fish, SE = 88) or Sockeye (*O. nerka*, 31 fish, SE = 10) salmon were harvested in Area 3 and 4 during the study period (Tables 17 and 18). In 2023, a total of 83 harvested fish (SE = 57) were recorded as unknown salmon species.

**Table 13. Total (lodge + non-lodge) estimated harvest, by subarea, for popular sportfish taxa, 2023.**

| Species            | Subarea |       |       |    |       |       |       |     |       |       |       |     | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|-------|-------|----|-------|-------|-------|-----|-------|-------|-------|-----|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J    | 3K    | 4A | 4B    | 4C    | 4D    | 4E  | 4F    | 4G    | 4H    | 4L  | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| Chinook Salmon     | 664     | 553   | 719   | 4  | 287   | 664   | 678   | 30  | 1,884 | 387   | 444   | 130 | 1,937  | 4,507  | 6,443   | 81               | 163    | 182     |
| Coho Salmon        | 6,090   | 2,541 | 7,060 | 0  | 1,035 | 4,790 | 2,913 | 439 | 8,103 | 2,444 | 6,644 | 466 | 15,691 | 26,834 | 42,525  | 413              | 611    | 738     |
| Pink Salmon        | 206     | 106   | 148   | 1  | 118   | 553   | 212   | 51  | 709   | 116   | 309   | 18  | 460    | 2,088  | 2,548   | 76               | 132    | 152     |
| Chum Salmon        | 10      | 0     | 62    | 0  | 0     | 57    | 3     | 73  | 20    | 15    | 19    | 0   | 72     | 187    | 259     | 45               | 75     | 88      |
| Sockeye Salmon     | 0       | 0     | 15    | 0  | 6     | 4     | 0     | 0   | 5     | 0     | 0     | 0   | 15     | 16     | 31      | 8                | 6      | 10      |
| Unknown Salmon     | 0       | 0     | 0     | 0  | 0     | 0     | 0     | 0   | 2     | 0     | 80    | 0   | 0      | 83     | 83      | 0                | 57     | 57      |
| Pacific Halibut    | 1,505   | 2,317 | 671   | 1  | 297   | 1,791 | 634   | 310 | 2,623 | 1,377 | 1,629 | 350 | 4,494  | 9,012  | 13,505  | 183              | 256    | 315     |
| Lingcod            | 1,029   | 2,034 | 222   | 1  | 105   | 531   | 328   | 133 | 1,935 | 1,241 | 1,148 | 178 | 3,284  | 5,599  | 8,884   | 153              | 301    | 337     |
| Other Groundfish * | 98      | 56    | 39    | 0  | 48    | 116   | 53    | 2   | 206   | 320   | 381   | 25  | 193    | 1,151  | 1,345   | 35               | 126    | 131     |
| Rockfish †         | 785     | 987   | 166   | 0  | 252   | 645   | 391   | 127 | 1,235 | 465   | 726   | 144 | 1,938  | 3,984  | 5,922   | 140              | 194    | 239     |

\* includes Big and Longnose skates, Pacific Spiny Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, greenlings, Spotted Ratfish, Arrow-tooth and Starry flounders, Pacific Sanddab, Pacific Rock and English soles, and unknown groundfish

† includes Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail and unknown rockfish

**Table 14. Total (lodge + non-lodge) estimated Chinook Salmon catch, by subarea, shown separately for released and retained fish by month, 2023.**

| Month / Catch Type | Subarea |       |       |     |       |       |       |      |        |       |       |       | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|-------|-------|-----|-------|-------|-------|------|--------|-------|-------|-------|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J    | 3K    | 4A  | 4B    | 4C    | 4D    | 4E   | 4F     | 4G    | 4H    | 4L    | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| May                |         |       |       |     |       |       |       |      |        |       |       |       |        |        |         |                  |        |         |
| Retained           | 57.4    | 55.7  | 106.3 | 2.4 | 138.2 | 184.9 | 128.8 | 6.0  | 439.5  | 33.8  | 35.9  | 1.8   | 219    | 971    | 1,191   | 34               | 56     | 65      |
| Released           | 71.0    | 82.6  | 0     | 3.1 | 80.2  | 152.0 | 141.7 | 1.0  | 362.8  | 15.0  | 68.2  | 2.2   | 154    | 826    | 980     | 15               | 67     | 69      |
| June               |         |       |       |     |       |       |       |      |        |       |       |       |        |        |         |                  |        |         |
| Retained           | 341.0   | 314.1 | 209.7 | 0.3 | 70.2  | 291.8 | 390.3 | 5.0  | 994.9  | 256.0 | 193.0 | 113.8 | 865    | 2,315  | 3,180   | 53               | 137    | 147     |
| Released           | 158.7   | 316.7 | 522.3 | 0.3 | 39.8  | 368.6 | 261.5 | 24.0 | 1358.3 | 286.2 | 214.5 | 239.2 | 998    | 2,792  | 3,790   | 84               | 216    | 232     |
| July               |         |       |       |     |       |       |       |      |        |       |       |       |        |        |         |                  |        |         |
| Retained           | 150.2   | 94.3  | 240.4 | 0   | 60.1  | 155.6 | 73.3  | 11.6 | 225.9  | 20.0  | 110.1 | 14.0  | 485    | 671    | 1,155   | 30               | 48     | 57      |
| Released           | 321.8   | 428.4 | 617.5 | 0   | 123.4 | 589.5 | 151.7 | 3.9  | 616.6  | 219.6 | 330.3 | 12.0  | 1,368  | 2,047  | 3,415   | 72               | 135    | 153     |
| August             |         |       |       |     |       |       |       |      |        |       |       |       |        |        |         |                  |        |         |
| Retained           | 115.5   | 89.2  | 162.6 | 0.9 | 18.4  | 31.2  | 85.6  | 7.3  | 223.9  | 77.1  | 105.1 | 0     | 367    | 550    | 917     | 40               | 49     | 63      |
| Released           | 0       | 146.0 | 76.0  | 3.6 | 18.4  | 81.2  | 50.6  | 87.3 | 64.0   | 25.7  | 74.2  | 0     | 222    | 405    | 627     | 32               | 65     | 72      |
| TOTAL              |         |       |       |     |       |       |       |      |        |       |       |       |        |        |         |                  |        |         |
| Retained           | 664     | 553   | 719   | 4   | 287   | 664   | 678   | 30   | 1,884  | 387   | 444   | 130   | 1,937  | 4,507  | 6,443   | 81               | 163    | 182     |
| Released           | 551     | 974   | 1,216 | 7   | 262   | 1,191 | 606   | 116  | 2,402  | 547   | 687   | 253   | 2,741  | 6,071  | 8,812   | 116              | 271    | 295     |
| Catch              | 1,216   | 1,527 | 1,935 | 11  | 549   | 1,855 | 1,284 | 146  | 4,286  | 934   | 1,131 | 383   | 4,678  | 10,578 | 15,255  | 142              | 316    | 346     |

**Table 15. Total (lodge + non-lodge) estimated Coho Salmon catch, by subarea, shown separately for released and retained fish by month, 2023.**

| Month / Catch Type | Subarea |        |        |     |       |        |        |       |        |        |        |       | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|--------|--------|-----|-------|--------|--------|-------|--------|--------|--------|-------|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J     | 3K     | 4A  | 4B    | 4C     | 4D     | 4E    | 4F     | 4G     | 4H     | 4L    | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| <b>May</b>         |         |        |        |     |       |        |        |       |        |        |        |       |        |        |         |                  |        |         |
| Retained           | 0       | 0      | 0      | 0   | 0     | 0      | 0      | 0     | 1.7    | 0      | 0      | 0     | 0      | 2      | 2       | 0                | 1      | 1       |
| Released           | 3.1     | 0      | 0      | 0   | 0     | 0      | 0      | 0     | 0      | 0      | 0      | 0     | 3      | 0      | 3       | 3                | 0      | 3       |
| <b>June</b>        |         |        |        |     |       |        |        |       |        |        |        |       |        |        |         |                  |        |         |
| Retained           | 83.9    | 140.9  | 186.0  | 0   | 16.4  | 272.6  | 56.3   | 27.0  | 1265.9 | 105.4  | 85.8   | 315.1 | 411    | 2,145  | 2,555   | 76               | 158    | 175     |
| Released           | 0       | 5.4    | 0      | 0   | 9.4   | 15.4   | 0      | 0     | 199.6  | 0      | 14.3   | 11.7  | 5      | 250    | 256     | 4                | 43     | 44      |
| <b>July</b>        |         |        |        |     |       |        |        |       |        |        |        |       |        |        |         |                  |        |         |
| Retained           | 1658.0  | 831.0  | 1959.2 | 0   | 601.2 | 3131.9 | 1618.4 | 77.2  | 4673.4 | 958.4  | 1734.3 | 117.0 | 4,448  | 12,912 | 17,360  | 195              | 361    | 410     |
| Released           | 677.0   | 134.6  | 589.2  | 1.6 | 180.4 | 393.0  | 561.7  | 0     | 614.4  | 294.5  | 350.0  | 4.0   | 1,401  | 2,400  | 3,800   | 205              | 191    | 280     |
| <b>August</b>      |         |        |        |     |       |        |        |       |        |        |        |       |        |        |         |                  |        |         |
| Retained           | 4347.9  | 1568.7 | 4915.2 | 0   | 417.8 | 1385.9 | 1237.9 | 334.7 | 2161.9 | 1380.0 | 4824.1 | 34.0  | 10,832 | 11,776 | 22,608  | 357              | 468    | 588     |
| Released           | 722.9   | 507.5  | 827.5  | 0.9 | 32.8  | 212.3  | 210.2  | 36.4  | 390.2  | 865.7  | 1663.7 | 0     | 2,058  | 3,412  | 5,470   | 262              | 435    | 508     |
| <b>TOTAL</b>       |         |        |        |     |       |        |        |       |        |        |        |       |        |        |         |                  |        |         |
| Retained           | 6,090   | 2,541  | 7,060  | 0   | 1,035 | 4,790  | 2,913  | 439   | 8,103  | 2,444  | 6,644  | 466   | 15,691 | 26,834 | 42,525  | 413              | 611    | 738     |
| Released           | 1,403   | 647    | 1,417  | 3   | 222   | 621    | 772    | 36    | 1,204  | 1,160  | 2,028  | 16    | 3,467  | 6,062  | 9,529   | 333              | 477    | 581     |
| Catch              | 7,493   | 3,188  | 8,477  | 3   | 1,258 | 5,411  | 3,684  | 475   | 9,307  | 3,604  | 8,672  | 482   | 19,158 | 32,896 | 52,054  | 531              | 775    | 939     |

**Table 16. Total (lodge + non-lodge) estimated Pink Salmon catch, by subarea, shown separately for released and retained fish by month, 2023.**

| Month / Catch Type | Subarea |       |        |     |       |        |        |       |        |        |        |      | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|-------|--------|-----|-------|--------|--------|-------|--------|--------|--------|------|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J    | 3K     | 4A  | 4B    | 4C     | 4D     | 4E    | 4F     | 4G     | 4H     | 4L   | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| <b>May</b>         |         |       |        |     |       |        |        |       |        |        |        |      |        |        |         |                  |        |         |
| Retained           | 0       | 0     | 0      | 0   | 0     | 0      | 0      | 0     | 0      | 0      | 7.2    | 0.2  | 0      | 7      | 7       | 0                | 6      | 6       |
| Released           | 0       | 0     | 0      | 0   | 0     | 0      | 0      | 0     | 0      | 0      | 0      | 0    | 0      | 0      | 0       | 0                | 0      | 0       |
| <b>June</b>        |         |       |        |     |       |        |        |       |        |        |        |      |        |        |         |                  |        |         |
| Retained           | 0       | 2.9   | 0      | 0   | 0     | 3.8    | 0      | 0     | 35.7   | 0      | 0      | 17.5 | 3      | 57     | 60      | 2                | 14     | 14      |
| Released           | 4.7     | 5.4   | 0      | 0   | 0     | 15.4   | 0      | 0     | 0      | 0      | 0      | 17.5 | 10     | 33     | 43      | 5                | 17     | 18      |
| <b>July</b>        |         |       |        |     |       |        |        |       |        |        |        |      |        |        |         |                  |        |         |
| Retained           | 118.2   | 34.6  | 114.6  | 0   | 66.4  | 262.0  | 126.7  | 7.7   | 395.3  | 64.9   | 110.1  | 0    | 267    | 1,033  | 1,301   | 64               | 76     | 99      |
| Released           | 702.6   | 489.6 | 700.3  | 0   | 496.8 | 917.0  | 775.0  | 38.6  | 1664.7 | 344.4  | 424.7  | 4.0  | 1,893  | 4,665  | 6,558   | 235              | 256    | 348     |
| <b>August</b>      |         |       |        |     |       |        |        |       |        |        |        |      |        |        |         |                  |        |         |
| Retained           | 88.2    | 68.1  | 33.9   | 0.9 | 51.2  | 287.2  | 85.6   | 43.7  | 278.2  | 51.4   | 191.7  | 0    | 190    | 990    | 1,180   | 41               | 107    | 114     |
| Released           | 1506.0  | 507.5 | 1937.6 | 2.7 | 561.2 | 1123.7 | 1790.6 | 269.2 | 2900.6 | 1045.7 | 1843.0 | 2.0  | 3,951  | 9,539  | 13,490  | 435              | 950    | 1,045   |
| <b>TOTAL</b>       |         |       |        |     |       |        |        |       |        |        |        |      |        |        |         |                  |        |         |
| Retained           | 206     | 106   | 148    | 1   | 118   | 553    | 212    | 51    | 709    | 116    | 309    | 18   | 460    | 2,088  | 2,548   | 76               | 132    | 152     |
| Released           | 2,213   | 1,003 | 2,638  | 3   | 1,058 | 2,056  | 2,566  | 308   | 4,565  | 1,390  | 2,268  | 24   | 5,854  | 14,237 | 20,091  | 495              | 984    | 1,102   |
| Catch              | 2,420   | 1,108 | 2,786  | 4   | 1,176 | 2,609  | 2,778  | 359   | 5,275  | 1,506  | 2,577  | 41   | 6,314  | 16,325 | 22,639  | 501              | 993    | 1,112   |

**Table 17. Total (lodge + non-lodge) estimated Chum Salmon catch, by subarea, shown separately for released and retained fish by month, 2023.**

| Month / Catch Type | Subarea |    |      |    |    |      |     |      |      |      |      |    | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|----|------|----|----|------|-----|------|------|------|------|----|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J | 3K   | 4A | 4B | 4C   | 4D  | 4E   | 4F   | 4G   | 4H   | 4L | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| May                |         |    |      |    |    |      |     |      |      |      |      |    |        |        |         |                  |        |         |
| Retained           | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0    | 0    | 0    | 0    | 0  | 0      | 0      | 0       | 0                | 0      | 0       |
| Released           | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0    | 0    | 0    | 0    | 0  | 0      | 0      | 0       | 0                | 0      | 0       |
| June               |         |    |      |    |    |      |     |      |      |      |      |    |        |        |         |                  |        |         |
| Retained           | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0    | 8.9  | 15.1 | 0    | 0  | 0      | 24     | 24      | 0                | 14     | 14      |
| Released           | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0    | 8.9  | 0    | 0    | 0  | 0      | 9      | 9       | 0                | 5      | 5       |
| July               |         |    |      |    |    |      |     |      |      |      |      |    |        |        |         |                  |        |         |
| Retained           | 0       | 0  | 0    | 0  | 0  | 57.3 | 3.3 | 0    | 11.3 | 0    | 0    | 0  | 0      | 72     | 72      | 0                | 41     | 41      |
| Released           | 0       | 0  | 30.3 | 0  | 0  | 28.7 | 3.3 | 0    | 29.4 | 0    | 7.9  | 0  | 30     | 69     | 100     | 15               | 24     | 28      |
| August             |         |    |      |    |    |      |     |      |      |      |      |    |        |        |         |                  |        |         |
| Retained           | 10.0    | 0  | 61.7 | 0  | 0  | 0    | 0   | 72.8 | 0    | 0    | 18.6 | 0  | 72     | 91     | 163     | 45               | 62     | 77      |
| Released           | 0       | 0  | 15.4 | 0  | 0  | 49.9 | 0   | 0    | 0    | 0    | 18.6 | 0  | 15     | 68     | 84      | 8                | 37     | 38      |
| TOTAL              |         |    |      |    |    |      |     |      |      |      |      |    |        |        |         |                  |        |         |
| Retained           | 10      | 0  | 62   | 0  | 0  | 57   | 3   | 73   | 20   | 15   | 19   | 0  | 72     | 187    | 259     | 45               | 75     | 88      |
| Released           | 0       | 0  | 46   | 0  | 0  | 79   | 3   | 0    | 38   | 0    | 26   | 0  | 46     | 147    | 192     | 17               | 44     | 47      |
| Catch              | 10      | 0  | 107  | 0  | 0  | 136  | 7   | 73   | 59   | 15   | 45   | 0  | 117    | 334    | 451     | 48               | 87     | 100     |

**Table 18. Total (lodge + non-lodge) estimated Sockeye Salmon catch, by subarea, shown separately for released and retained fish by month, 2023.**

| Month / Catch Type | Subarea |    |      |    |     |     |    |    |     |    |     |    | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|----|------|----|-----|-----|----|----|-----|----|-----|----|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J | 3K   | 4A | 4B  | 4C  | 4D | 4E | 4F  | 4G | 4H  | 4L | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| May                |         |    |      |    |     |     |    |    |     |    |     |    |        |        |         |                  |        |         |
| Retained           | 0       | 0  | 0    | 0  | 0   | 0   | 0  | 0  | 0   | 0  | 0   | 0  | 0      | 0      | 0       | 0                | 0      | 0       |
| Released           | 0       | 0  | 0    | 0  | 0   | 0   | 0  | 0  | 0   | 0  | 0   | 0  | 0      | 0      | 0       | 0                | 0      | 0       |
| June               |         |    |      |    |     |     |    |    |     |    |     |    |        |        |         |                  |        |         |
| Retained           | 0       | 0  | 0    | 0  | 0   | 0   | 0  | 0  | 3.0 | 0  | 0   | 0  | 0      | 3      | 3       | 0                | 2      | 2       |
| Released           | 0       | 0  | 0    | 0  | 0   | 0   | 0  | 0  | 0   | 0  | 0   | 0  | 0      | 0      | 0       | 0                | 0      | 0       |
| July               |         |    |      |    |     |     |    |    |     |    |     |    |        |        |         |                  |        |         |
| Retained           | 0       | 0  | 0    | 0  | 6.3 | 4.1 | 0  | 0  | 2.3 | 0  | 0   | 0  | 0      | 13     | 13      | 0                | 6      | 6       |
| Released           | 0       | 0  | 3.4  | 0  | 0   | 4.1 | 0  | 0  | 0   | 0  | 7.9 | 0  | 3      | 12     | 15      | 2                | 6      | 7       |
| August             |         |    |      |    |     |     |    |    |     |    |     |    |        |        |         |                  |        |         |
| Retained           | 0       | 0  | 15.4 | 0  | 0   | 0   | 0  | 0  | 0   | 0  | 0   | 0  | 15     | 0      | 15      | 8                | 0      | 8       |
| Released           | 0       | 0  | 0    | 0  | 0   | 0   | 0  | 0  | 0   | 0  | 0   | 0  | 0      | 0      | 0       | 0                | 0      | 0       |
| TOTAL              |         |    |      |    |     |     |    |    |     |    |     |    |        |        |         |                  |        |         |
| Retained           | 0       | 0  | 15   | 0  | 6   | 4   | 0  | 0  | 5   | 0  | 0   | 0  | 15     | 16     | 31      | 8                | 6      | 10      |
| Released           | 0       | 0  | 3    | 0  | 0   | 4   | 0  | 0  | 0   | 0  | 8   | 0  | 3      | 12     | 15      | 2                | 6      | 7       |
| Catch              | 0       | 0  | 19   | 0  | 6   | 8   | 0  | 0  | 5   | 0  | 8   | 0  | 19     | 28     | 46      | 9                | 9      | 12      |

**Table 19. Total (lodge + non-lodge) estimated Pacific Halibut catch, by subarea, shown separately for released and retained fish by month, 2023.**

| Month / Catch Type | Subarea |        |       |     |       |       |       |       |        |       |       |       | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|--------|-------|-----|-------|-------|-------|-------|--------|-------|-------|-------|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J     | 3K    | 4A  | 4B    | 4C    | 4D    | 4E    | 4F     | 4G    | 4H    | 4L    | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| <b>May</b>         |         |        |       |     |       |       |       |       |        |       |       |       |        |        |         |                  |        |         |
| Retained           | 54.1    | 135.7  | 46.3  | 0.6 | 49.0  | 116.5 | 124.6 | 29.0  | 230.2  | 67.6  | 75.4  | 4.6   | 236    | 698    | 934     | 36               | 54     | 65      |
| Released           | 15.3    | 96.5   | 0     | 0   | 8.9   | 30.4  | 34.4  | 4.0   | 144.8  | 71.4  | 18.0  | 2.7   | 112    | 314    | 426     | 41               | 42     | 58      |
| <b>June</b>        |         |        |       |     |       |       |       |       |        |       |       |       |        |        |         |                  |        |         |
| Retained           | 174.1   | 280.7  | 117.6 | 0   | 44.5  | 326.4 | 84.5  | 17.0  | 744.7  | 512.1 | 153.7 | 180.9 | 572    | 2,064  | 2,636   | 36               | 115    | 121     |
| Released           | 23.3    | 160.7  | 82.4  | 0   | 11.7  | 92.2  | 60.3  | 100.0 | 399.1  | 225.9 | 128.7 | 332.6 | 266    | 1,351  | 1,617   | 76               | 187    | 202     |
| <b>July</b>        |         |        |       |     |       |       |       |       |        |       |       |       |        |        |         |                  |        |         |
| Retained           | 495.8   | 871.8  | 199.2 | 0   | 129.7 | 761.5 | 206.7 | 96.5  | 1034.5 | 454.2 | 515.2 | 122.0 | 1,567  | 3,320  | 4,887   | 92               | 141    | 168     |
| Released           | 138.5   | 463.9  | 30.3  | 0   | 60.1  | 110.5 | 58.3  | 73.3  | 370.4  | 144.8 | 357.9 | 118.0 | 633    | 1,293  | 1,926   | 98               | 111    | 148     |
| <b>August</b>      |         |        |       |     |       |       |       |       |        |       |       |       |        |        |         |                  |        |         |
| Retained           | 781.4   | 1028.8 | 308.2 | 0   | 73.7  | 586.8 | 218.0 | 167.4 | 614.0  | 342.9 | 884.4 | 43.0  | 2,118  | 2,930  | 5,049   | 150              | 172    | 228     |
| Released           | 20.1    | 394.7  | 159.3 | 0   | 16.4  | 174.8 | 120.7 | 50.9  | 342.2  | 111.4 | 358.7 | 14.0  | 574    | 1,189  | 1,763   | 81               | 137    | 159     |
| <b>TOTAL</b>       |         |        |       |     |       |       |       |       |        |       |       |       |        |        |         |                  |        |         |
| Retained           | 1,505   | 2,317  | 671   | 1   | 297   | 1,791 | 634   | 310   | 2,623  | 1,377 | 1,629 | 350   | 4,494  | 9,012  | 13,505  | 183              | 256    | 315     |
| Released           | 197     | 1,116  | 272   | 0   | 97    | 408   | 274   | 228   | 1,257  | 553   | 863   | 467   | 1,585  | 4,147  | 5,732   | 154              | 261    | 302     |
| Catch              | 1,703   | 3,433  | 943   | 1   | 394   | 2,199 | 907   | 538   | 3,880  | 1,930 | 2,492 | 818   | 6,079  | 13,159 | 19,238  | 239              | 365    | 436     |

**Table 20. Total (lodge + non-lodge) estimated Lingcod catch, by subarea, shown separately for released and retained fish by month, 2023.**

| Month / Catch Type | Subarea |       |      |     |      |       |       |      |       |       |       |       | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|-------|------|-----|------|-------|-------|------|-------|-------|-------|-------|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J    | 3K   | 4A  | 4B   | 4C    | 4D    | 4E   | 4F    | 4G    | 4H    | 4L    | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| <b>May</b>         |         |       |      |     |      |       |       |      |       |       |       |       |        |        |         |                  |        |         |
| Retained           | 27.4    | 191.6 | 0    | 0.6 | 53.5 | 12.7  | 55.8  | 62.0 | 303.5 | 52.6  | 39.5  | 2.8   | 219    | 583    | 802     | 32               | 57     | 66      |
| Released           | 9.2     | 71.5  | 0    | 0   | 4.5  | 7.6   | 8.6   | 2.0  | 40.1  | 18.8  | 3.6   | 0.6   | 81     | 86     | 166     | 40               | 18     | 44      |
| <b>June</b>        |         |       |      |     |      |       |       |      |       |       |       |       |        |        |         |                  |        |         |
| Retained           | 124.9   | 503.3 | 35.1 | 0.3 | 9.4  | 142.1 | 40.2  | 14.0 | 592.8 | 677.8 | 96.5  | 110.9 | 663    | 1,684  | 2,347   | 47               | 209    | 214     |
| Released           | 0       | 26.8  | 0    | 0   | 2.3  | 11.5  | 0     | 0    | 35.7  | 30.1  | 0     | 0     | 27     | 80     | 107     | 17               | 21     | 27      |
| <b>July</b>        |         |       |      |     |      |       |       |      |       |       |       |       |        |        |         |                  |        |         |
| Retained           | 321.8   | 762.1 | 94.7 | 0   | 3.2  | 257.9 | 138.3 | 34.7 | 702.5 | 279.5 | 338.2 | 54.0  | 1,179  | 1,808  | 2,987   | 73               | 137    | 155     |
| Released           | 0       | 100.2 | 3.4  | 0   | 0    | 4.1   | 26.7  | 0    | 79.1  | 5.0   | 74.7  | 3.0   | 104    | 193    | 296     | 38               | 40     | 55      |
| <b>August</b>      |         |       |      |     |      |       |       |      |       |       |       |       |        |        |         |                  |        |         |
| Retained           | 554.4   | 577.0 | 91.8 | 0   | 38.9 | 118.6 | 93.4  | 21.8 | 335.8 | 231.4 | 674.1 | 10.0  | 1,223  | 1,524  | 2,747   | 122              | 158    | 199     |
| Released           | 0       | 73.3  | 0    | 0   | 2.0  | 0     | 0     | 0    | 41.6  | 0     | 30.9  | 0     | 73     | 75     | 148     | 33               | 24     | 40      |
| <b>TOTAL</b>       |         |       |      |     |      |       |       |      |       |       |       |       |        |        |         |                  |        |         |
| Retained           | 1,029   | 2,034 | 222  | 1   | 105  | 531   | 328   | 133  | 1,935 | 1,241 | 1,148 | 178   | 3,284  | 5,599  | 8,884   | 153              | 301    | 337     |
| Released           | 9       | 272   | 3    | 0   | 9    | 23    | 35    | 2    | 196   | 54    | 109   | 4     | 284    | 433    | 717     | 67               | 54     | 86      |
| Catch              | 1,038   | 2,306 | 225  | 1   | 114  | 554   | 363   | 135  | 2,131 | 1,295 | 1,258 | 181   | 3,569  | 6,032  | 9,600   | 167              | 306    | 348     |

**Table 21. Total (lodge + non-lodge) estimated catch of other groundfish by subarea, shown separately for released and retained fish by month, 2023. Included are Big and Longnose skates, Pacific Spiny Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, greenlings, Spotted Ratfish, Arrow-tooth and Starry flounders, Pacific Sanddab, Pacific Rock and English soles, and other miscellaneous groundfishes.**

| Month / Catch Type | Subarea |      |      |     |      |       |      |      |      |       |       |      | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|------|------|-----|------|-------|------|------|------|-------|-------|------|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J   | 3K   | 4A  | 4B   | 4C    | 4D   | 4E   | 4F   | 4G    | 4H    | 4L   | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| <i>May</i>         |         |      |      |     |      |       |      |      |      |       |       |      |        |        |         |                  |        |         |
| Retained           | 27.6    | 10.7 | 0    | 0   | 11.1 | 12.7  | 0    | 2.0  | 19.2 | 15.0  | 10.8  | 0.7  | 38     | 71     | 110     | 21               | 16     | 26      |
| Released           | 6.1     | 7.1  | 0    | 3.1 | 11.1 | 5.1   | 12.9 | 0    | 29.7 | 3.8   | 10.8  | 0.7  | 13     | 77     | 90      | 8                | 14     | 16      |
| <i>June</i>        |         |      |      |     |      |       |      |      |      |       |       |      |        |        |         |                  |        |         |
| Retained           | 0       | 8.0  | 11.8 | 0   | 2.3  | 23.0  | 32.2 | 0    | 29.8 | 105.4 | 71.5  | 17.5 | 20     | 282    | 302     | 7                | 85     | 85      |
| Released           | 0       | 0    | 11.8 | 0   | 9.4  | 11.5  | 16.1 | 0    | 26.8 | 0     | 7.1   | 0    | 12     | 71     | 83      | 8                | 14     | 17      |
| <i>July</i>        |         |      |      |     |      |       |      |      |      |       |       |      |        |        |         |                  |        |         |
| Retained           | 20.5    | 37.2 | 6.7  | 0   | 22.1 | 61.4  | 16.7 | 0    | 83.6 | 79.9  | 169.1 | 7.0  | 64     | 440    | 504     | 15               | 64     | 66      |
| Released           | 20.5    | 5.7  | 16.8 | 0   | 25.3 | 69.6  | 31.7 | 19.3 | 54.2 | 89.8  | 19.7  | 2.0  | 43     | 312    | 355     | 12               | 58     | 59      |
| <i>August</i>      |         |      |      |     |      |       |      |      |      |       |       |      |        |        |         |                  |        |         |
| Retained           | 50.2    | 0    | 20.6 | 0   | 12.3 | 18.7  | 3.9  | 0    | 73.6 | 120.0 | 129.9 | 0    | 71     | 358    | 429     | 23               | 66     | 70      |
| Released           | 90.4    | 0    | 15.4 | 0   | 24.6 | 199.8 | 3.9  | 0    | 44.8 | 120.0 | 259.8 | 4.0  | 106    | 657    | 763     | 38               | 133    | 139     |
| <i>TOTAL</i>       |         |      |      |     |      |       |      |      |      |       |       |      |        |        |         |                  |        |         |
| Retained           | 98      | 56   | 39   | 0   | 48   | 116   | 53   | 2    | 206  | 320   | 381   | 25   | 193    | 1,151  | 1,345   | 35               | 126    | 131     |
| Released           | 117     | 13   | 44   | 3   | 70   | 286   | 65   | 19   | 155  | 214   | 297   | 7    | 174    | 1,116  | 1,290   | 41               | 147    | 152     |
| Catch              | 215     | 69   | 83   | 3   | 118  | 402   | 117  | 21   | 362  | 534   | 679   | 32   | 367    | 2,268  | 2,635   | 54               | 193    | 201     |

**Table 22. Total (lodge + non-lodge) estimated rockfish catch, by subarea, shown separately for released and retained fish by month, 2023. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, and Yellowtail rockfish, and other miscellaneous rockfishes.**

| Month / Catch Type | Subarea |       |      |     |       |       |       |      |       |       |       |      | TOTALS |        |         | SE of the Totals |        |         |
|--------------------|---------|-------|------|-----|-------|-------|-------|------|-------|-------|-------|------|--------|--------|---------|------------------|--------|---------|
|                    | 3I      | 3J    | 3K   | 4A  | 4B    | 4C    | 4D    | 4E   | 4F    | 4G    | 4H    | 4L   | Area 3 | Area 4 | Overall | Area 3           | Area 4 | Overall |
| <i>May</i>         |         |       |      |     |       |       |       |      |       |       |       |      |        |        |         |                  |        |         |
| Retained           | 54.2    | 127.4 | 0    | 0   | 71.3  | 38.0  | 73.0  | 21.0 | 129.1 | 18.8  | 46.7  | 1.9  | 182    | 400    | 581     | 38               | 41     | 56      |
| Released           | 24.6    | 157.2 | 0    | 4.9 | 122.6 | 134.2 | 193.3 | 31.0 | 254.7 | 67.6  | 111.3 | 4.9  | 182    | 924    | 1,106   | 76               | 84     | 113     |
| <i>June</i>        |         |       |      |     |       |       |       |      |       |       |       |      |        |        |         |                  |        |         |
| Retained           | 4.7     | 267.0 | 11.8 | 0   | 18.7  | 119.0 | 76.4  | 13.0 | 440.8 | 210.9 | 78.6  | 67.1 | 283    | 1,025  | 1,308   | 23               | 97     | 100     |
| Released           | 0       | 69.6  | 5.9  | 0   | 30.4  | 122.9 | 160.9 | 0    | 375.3 | 150.6 | 39.3  | 14.6 | 76     | 894    | 970     | 28               | 108    | 111     |
| <i>July</i>        |         |       |      |     |       |       |       |      |       |       |       |      |        |        |         |                  |        |         |
| Retained           | 191.7   | 270.2 | 97.6 | 0   | 82.3  | 212.9 | 105.0 | 27.0 | 415.6 | 89.8  | 106.2 | 51.0 | 560    | 1,090  | 1,649   | 69               | 88     | 112     |
| Released           | 123.1   | 220.5 | 10.1 | 0   | 88.6  | 237.5 | 120.0 | 23.2 | 280.1 | 104.8 | 212.4 | 13.0 | 354    | 1,079  | 1,433   | 52               | 100    | 113     |
| <i>August</i>      |         |       |      |     |       |       |       |      |       |       |       |      |        |        |         |                  |        |         |
| Retained           | 534.8   | 321.9 | 56.5 | 0   | 79.9  | 274.7 | 136.2 | 65.5 | 249.4 | 145.7 | 494.8 | 24.0 | 913    | 1,470  | 2,383   | 114              | 136    | 177     |
| Released           | 471.9   | 535.7 | 56.5 | 0   | 73.7  | 262.2 | 159.6 | 7.3  | 297.4 | 102.9 | 383.5 | 3.0  | 1,064  | 1,290  | 2,354   | 201              | 166    | 261     |
| <i>TOTAL</i>       |         |       |      |     |       |       |       |      |       |       |       |      |        |        |         |                  |        |         |
| Retained           | 785     | 987   | 166  | 0   | 252   | 645   | 391   | 127  | 1,235 | 465   | 726   | 144  | 1,938  | 3,984  | 5,922   | 140              | 194    | 239     |
| Released           | 620     | 983   | 73   | 5   | 315   | 757   | 634   | 61   | 1,207 | 426   | 746   | 35   | 1,675  | 4,188  | 5,863   | 223              | 237    | 325     |
| Catch              | 1,405   | 1,970 | 238  | 5   | 568   | 1,401 | 1,025 | 188  | 2,442 | 891   | 1,473 | 179  | 3,613  | 8,172  | 11,785  | 263              | 306    | 404     |



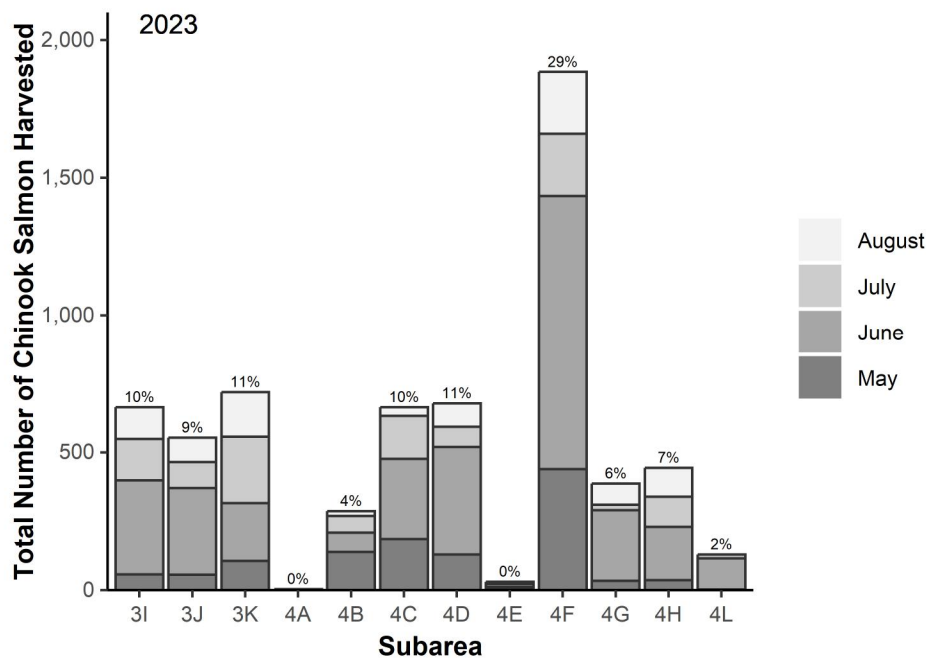


Figure 9. Total (lodge + non-lodge) estimated Chinook Salmon harvest, by month and subarea, 2023.

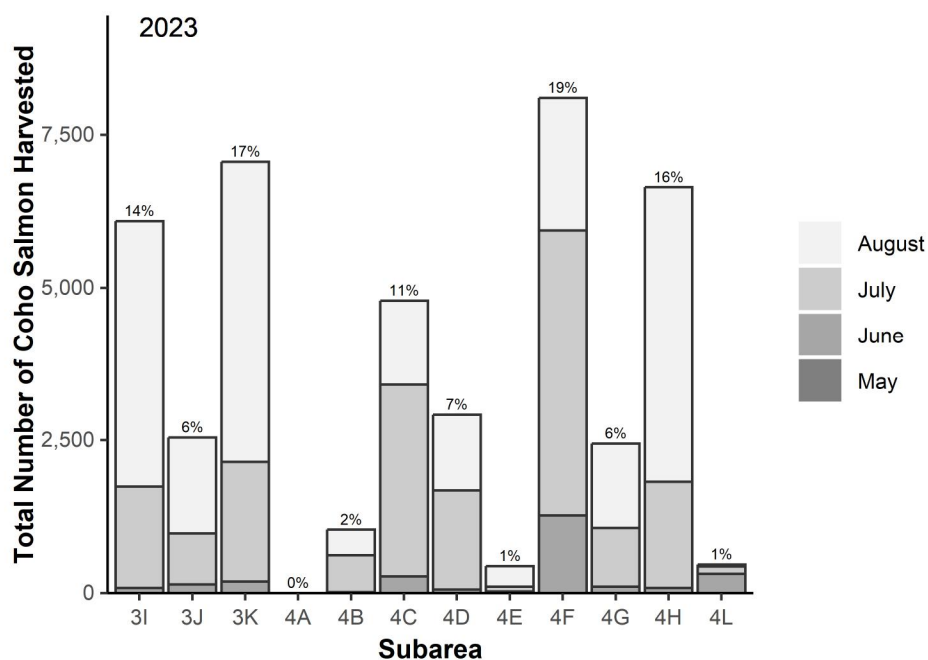


Figure 10. Total (lodge + non-lodge) estimated Coho Salmon harvest, by month and subarea, 2023.

**Table 23. Kruskal Wallis tests for differences in harvest (fish caught per boat trip per subarea) among months, by species, 2023. Subareas treated as replicates. Red letters indicate statistical differences within species (months not connected by a common letter were statistically different).**

| Species               | Kruskal-Wallis Statistics |    |          | Median Monthly Harvest |                |                 |                 |
|-----------------------|---------------------------|----|----------|------------------------|----------------|-----------------|-----------------|
|                       | $\chi^2$                  | df | P        | May                    | June           | July            | August          |
| Chinook Salmon        | 6.0                       | 3  | 0.11     | 34.0                   | 156.4          | 66.7            | 69.6            |
| Chum Salmon           | 5.1                       | 3  | 0.17     | 0                      | 0              | 0               | 0               |
| Coho Salmon           | 28.2                      | 3  | < 0.0001 | 0 <b>C</b>             | 89.8 <b>B</b>  | 1030.5 <b>A</b> | 1308.9 <b>A</b> |
| Pink Salmon           | 25.7                      | 3  | < 0.0001 | 0 <b>B</b>             | 0 <b>B</b>     | 85.4 <b>A</b>   | 56.7 <b>A</b>   |
| Sockeye Salmon        | 4.0                       | 3  | 0.26     | 0                      | 0              | 0               | 0               |
| Arrow-tooth Flounder  | 3.8                       | 3  | 0.28     | 0                      | 0              | 0               | 0               |
| Big Skate             | 2.0                       | 3  | 0.56     | 0                      | 0              | 0               | 0               |
| Cabezon               | 2.1                       | 3  | 0.54     | 0                      | 0              | 0               | 0               |
| English Sole          | 3.0                       | 3  | 0.39     | 0                      | 0              | 0               | 0               |
| Greenlings            | 2.2                       | 3  | 0.53     | 0                      | 0              | 0               | 0               |
| Lingcod               | 4.4                       | 3  | 0.22     | 46.0                   | 69.2           | 143.5           | 106.0           |
| Longnose Skate        | 0.0                       | 3  | 1        | 0                      | 0              | 0               | 0               |
| Pacific Cod           | 2.9                       | 3  | 0.40     | 4.8                    | 7.4            | 17.2            | 14.5            |
| Pacific Halibut       | 11.8                      | 3  | 0.008    | 58.3 <b>B</b>          | 98.5 <b>AB</b> | 275.1 <b>A</b>  | 292.2 <b>A</b>  |
| Pacific Rock Sole     | 3.7                       | 3  | 0.29     | 0                      | 0              | 0               | 0               |
| Pacific Spiny Dogfish | 3.7                       | 3  | 0.29     | 0                      | 0              | 0               | 0               |
| Sablefish             | 2.0                       | 3  | 0.56     | 0                      | 0              | 0               | 0               |
| Spotted Ratfish       | 0.0                       | 3  | 1        | 0                      | 0              | 0               | 0               |
| Starry Flounder       | 0.0                       | 3  | 1        | 0                      | 0              | 0               | 0               |
| Black Rockfish        | 7.7                       | 3  | 0.053    | 8.0                    | 13.2           | 39.2            | 55.3            |
| Boccacio Rockfish     | 8.2                       | 3  | 0.043    | 0                      | 0              | 0               | 0               |
| Canary Rockfish       | 2.0                       | 3  | 0.56     | 0                      | 0              | 0               | 0               |
| China Rockfish        | 7.6                       | 3  | 0.055    | 0                      | 0              | 0               | 3.9             |
| Copper Rockfish       | 1.4                       | 3  | 0.70     | 0                      | 2.0            | 1.4             | 3.1             |
| Dusky Rockfish        | 0.0                       | 3  | 1        | 0                      | 0              | 0               | 0               |
| Quillback Rockfish    | 5.6                       | 3  | 0.13     | 3.7                    | 28.0           | 8.7             | 31.6            |
| Redbanded Rockfish    | 3.0                       | 3  | 0.39     | 0                      | 0              | 0               | 0               |
| Redstripe Rockfish    | 0.0                       | 3  | 1        | 0                      | 0              | 0               | 0               |
| Tiger Rockfish        | 2.4                       | 3  | 0.49     | 0                      | 0              | 0               | 0               |
| Widow Rockfish        | 3.5                       | 3  | 0.32     | 0                      | 0              | 0               | 0               |
| Yelloweye Rockfish    | 1.0                       | 3  | 0.79     | 0                      | 0              | 0               | 0               |
| Yellowtail Rockfish   | 4.0                       | 3  | 0.26     | 0                      | 0              | 3.2             | 0.5             |

An estimated 13,505 (SE = 315) Pacific Halibut were harvested in Areas 3 or 4 during the 2023 study period (Table 19; Figure 11). Most of the harvest could be attributed to effort in 4F (19%) and 3J (17%), and in subareas 3I, 4C, and 4H (together another 36%). Harvest varied significantly among months (lowest in May, highest in July and August;  $\chi^2 = 11.8$ , df = 3,  $P = 0.008$ ; Table 23). In all, 5,732 (SE = 302) Pacific Halibut were released, representing 30% of the total catch.

From May to August 2023, an estimated 8,884 (SE = 337) Lingcod were harvested, occurring mainly from subareas 3J (23%) and 4F (22%), followed by subareas 4G (14%), 4H (13%), and 3I (12%). There was relatively low harvest in May (9% of total harvest) as compared to other months (26-34%), though the differences were not statistically significant. ( $\chi^2 = 4.4$ , df = 3,  $P = 0.22$ ; Table 20; Figure 12). Of the captured Lingcod, 7% were released (717 fish, SE = 86).

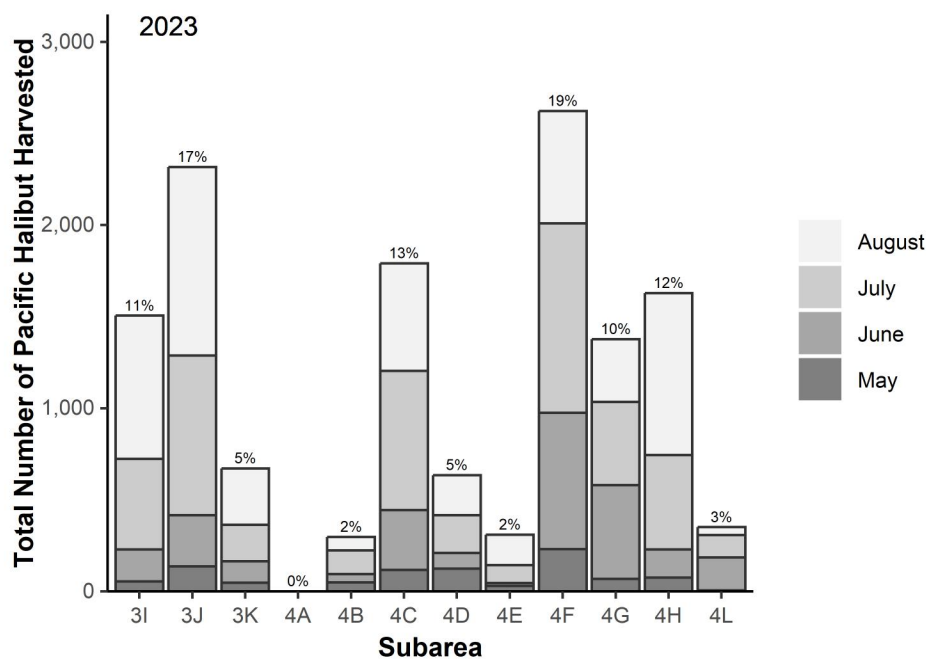


Figure 11. Total (lodge + non-lodge) estimated Pacific Halibut harvest, by month and subarea, 2023.

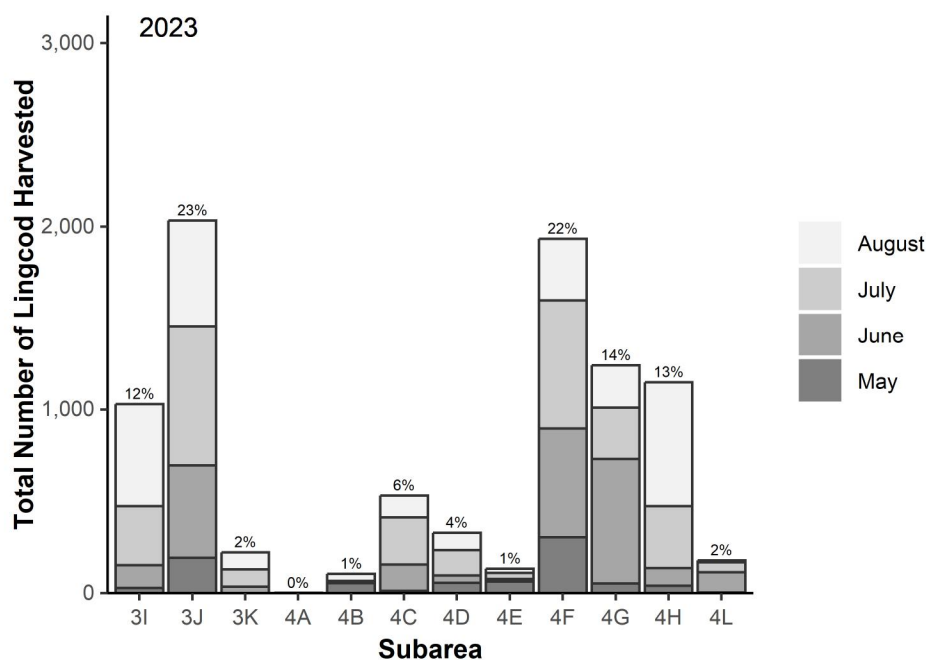


Figure 12. Total (lodge + non-lodge) estimated Lingcod harvest, by month and subarea, 2023.

An estimated 1,345 (SE = 131) fish from other groundfish species were harvested from May to August 2023, of which 81% (i.e., 1,091 fish, SE = 126) were Pacific Cod. The remaining harvest included greenlings (*Hexagrammos* spp.; 5%), Cabezon (*Scorpaenichthys marmoratus*, 3%), Sablefish (*Anoplopoma fimbria*; 3%), and a variety of other species (<35 individuals retained per species). Over the study period, it is estimated that 181 (SE = 53) Pacific Cod were released (14% of the total catch of Pacific Cod).

Rockfish harvest was estimated at 5,922 fish (SE = 239; Table 22), composed mostly of Black (35%), Quillback (23%), Copper (*S. caurinus*, 4%), Yellowtail (*Sebastes flavidus*, 3%), Tiger (3%), and China (*S. nebulosus*, 2%), unknown (28%), and five other rockfish species. In all, 5,863 (SE = 325) were released. Of the 856 Yelloweye Rockfish caught, all but 41 (95%) were released.

**Table 24. Biosampling results, 2023.**

|                      | Chinook Salmon |         | Coho Salmon            |         | Pacific Halibut |         |
|----------------------|----------------|---------|------------------------|---------|-----------------|---------|
|                      | Number         | Percent | Number                 | Percent | Number          | Percent |
| Number Examined      | 483            |         | 1653                   |         | 724             |         |
| <b>Subarea</b>       |                |         |                        |         |                 |         |
| 3I                   | 13             | 3%      | 17                     | 1%      | 7               | 1%      |
| 3J                   | 23             | 5%      | 38                     | 2%      | 72              | 10%     |
| 3K                   | 22             | 5%      | 136                    | 8%      | 16              | 2%      |
| 4A                   | 4              | 1%      |                        |         |                 |         |
| 4B                   | 63             | 13%     | 132                    | 8%      | 43              | 6%      |
| 4C                   | 52             | 11%     | 122                    | 7%      | 131             | 18%     |
| 4D                   | 74             | 15%     | 306                    | 19%     | 48              | 7%      |
| 4E                   | 2              | 0%      | 16                     | 1%      | 22              | 3%      |
| 4F                   | 192            | 40%     | 573                    | 35%     | 234             | 32%     |
| 4G                   | 12             | 2%      | 47                     | 3%      | 48              | 7%      |
| 4H                   | 12             | 2%      | 211                    | 13%     | 57              | 8%      |
| 4L                   | 12             | 2%      | 55                     | 3%      | 43              | 6%      |
| <b>Sex</b>           |                |         |                        |         |                 |         |
| Male                 | 2              | 25%     | 6                      | 46%     |                 |         |
| Female               | 6              | 75%     | 7                      | 54%     |                 |         |
| Undeterminable       | 73             |         |                        |         | 31              |         |
| Unknown              | 402            |         | 1640                   |         | 693             |         |
| <b>Trip Type</b>     |                |         |                        |         |                 |         |
| Guided               | 192            | 40%     | 755                    | 46%     | 255             | 35%     |
| Unguided             | 291            | 60%     | 898                    | 54%     | 469             | 65%     |
| <b>Adipose Clips</b> |                |         |                        |         |                 |         |
| Clipped              | 41             | 9%      | 6                      | 0%      |                 |         |
| Unmarked             | 421            | 91%     | 1638                   | 100%    |                 |         |
| Unknown              |                |         |                        |         |                 |         |
| <b>Head Tagged</b>   |                |         |                        |         |                 |         |
| Yes                  | 34             | 7%      | 6                      | 0%      |                 |         |
| No                   | 449            | 93%     | 1647                   | 100%    |                 |         |
| <b>Scale Samples</b> |                |         | <b>Whatman Samples</b> |         |                 |         |
| Yes                  | 470            | 97%     | 1653                   | 100%    |                 |         |
| No                   | 13             | 3%      |                        |         |                 |         |
| <b>Flesh Colour</b>  |                |         |                        |         |                 |         |
| Red                  | 8              | 47%     |                        |         |                 |         |
| White                | 9              | 53%     |                        |         |                 |         |
| Marble               |                |         |                        |         |                 |         |
| Unknown              |                |         |                        |         |                 |         |

## **Biosampling**

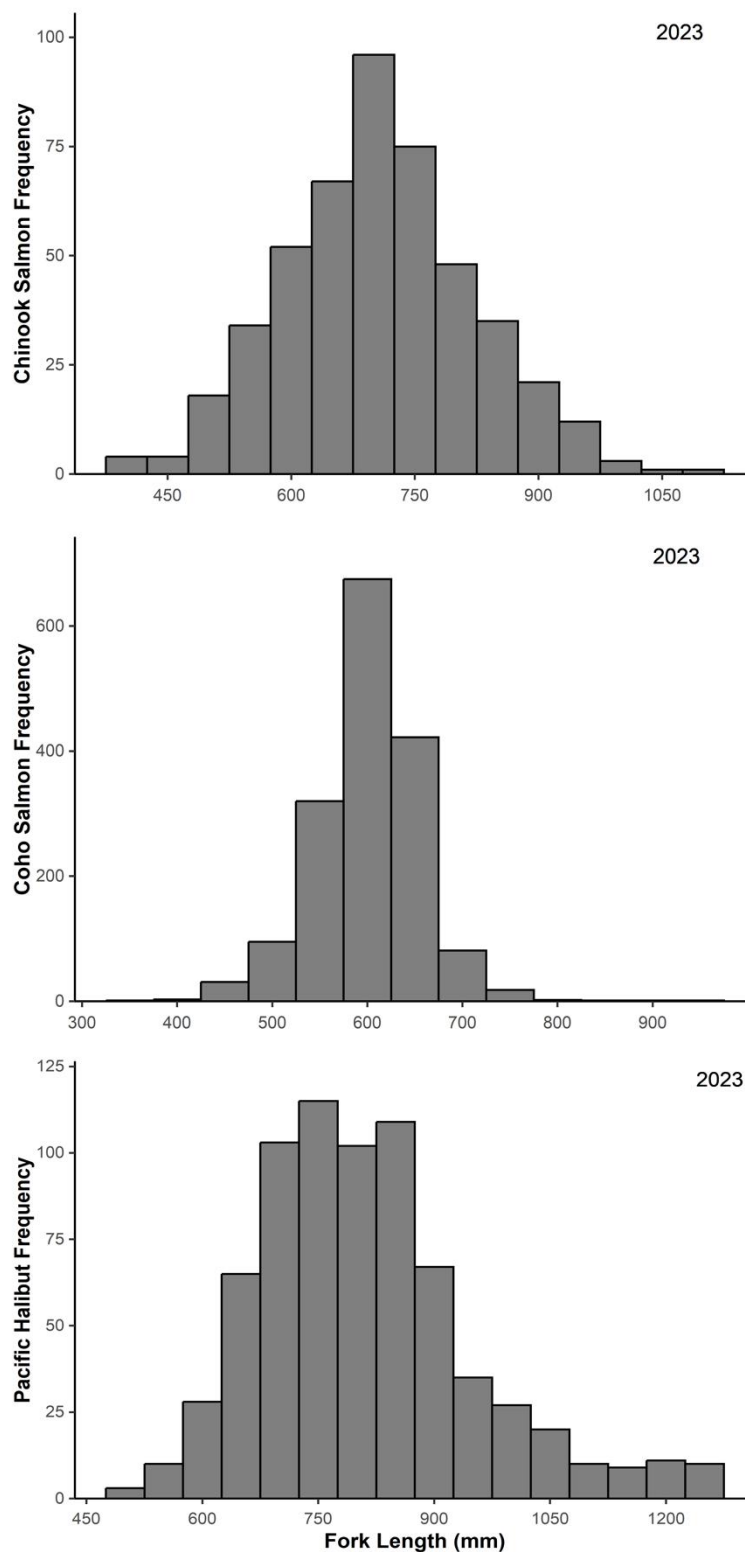
During interviews, observers were told about 1,756 retained Chinook Salmon, and were permitted to handle 483 fish. Too few fish were sexed ( $n=8$ ) to provide a robust estimate of sex ratio (Table 24). The median fork length of biosampled Chinook Salmon ( $n = 471$ ) was 71 cm (95% of observations were between 49 and 94 cm; Figure 13). In all, 41 adipose-clipped individuals were encountered (Table 24), including 22 from standardized sampling methods (complete samples collected from anglers who were asked if their catch could be biosampled before any other interactions took place; Appendix 5) and 19 from non-standard interactions (“non-standard interactions” were interviews for which anything other than complete access was given to all harvested fish). Of the 41, 83% ( $n=34$ ) had their heads collected and labelled for submission to the Salmon Head Recovery Program (Table 24). Of the standardized samples ( $n=271$ ), 8.1% were adipose-clipped (Appendix 5). Scale samples were obtained from 97% ( $n=470$ ) of the biosampled Chinook Salmon (Table 24).

For Coho Salmon, observers were told about 9,999 retained fish, and were permitted to handle 1,653 fish. Too few fish were sexed ( $n=13$ ) to provide a robust estimate of sex ratio (Table 24). The median Coho Salmon fork length ( $n = 1,651$ ) was 61 cm (95% of observations were between 48 and 70 cm; Figure 13). In all, 6 adipose-clipped individuals were encountered (Table 24), including three from standardized sampling interviews. All of the six heads were collected for submission to the Salmon Head Recovery Program (Table 24). Of the standardized samples ( $n=763$ ), 0.4% were adipose-clipped (Appendix 5). Tissue samples were obtained from all ( $n=1,653$ ) of the biosampled Coho Salmon (Table 24).

Of the 3,410 Pacific Halibut that anglers reported as retained, 724 were handled (Table 24). All were measured, but none were sexed. The median size was 79 cm (95% of observations were between 59 and 1119 cm; Figure 13). Average net and round weights, calculated from average lengths using regression formulas, were 6.0 kg (13.2 lb) and 7.9 kg (17.5 lb) in Area 3, and 5.3 kg (11.6 lb) and 7.0 kg (15.4 lb) in Area 4, respectively (Table 25).

**Table 25. Average weight, and harvest (numbers and biomass) of Pacific Halibut in Area 3 and 4, 2023.**

|                                                     | Stat Area 3 |        |              |        | Stat Area 4 |         |              |         |
|-----------------------------------------------------|-------------|--------|--------------|--------|-------------|---------|--------------|---------|
|                                                     | Net Weight  |        | Round Weight |        | Net Weight  |         | Round Weight |         |
|                                                     | kg          | lb     | kg           | lb     | kg          | lb      | kg           | lb      |
| <i>Creel Interview (Non-Lodge) Average Weights</i>  |             |        |              |        |             |         |              |         |
| <i>Guided</i>                                       | 6.5         | 14.4   | 8.7          | 19.1   | 5.8         | 12.9    | 7.7          | 17.0    |
| <i>Unguided</i>                                     | 4.8         | 10.5   | 6.3          | 13.9   | 5.0         | 11.1    | 6.7          | 14.7    |
| <i>Overall</i>                                      | 6.0         | 13.2   | 7.9          | 17.5   | 5.3         | 11.6    | 7.0          | 15.4    |
| <i>Estimated Non-Lodge Harvest (number of fish)</i> |             |        |              |        |             |         |              |         |
| <i>Overall</i>                                      |             | 3,150  |              |        |             | 9,012   |              |         |
| <i>Estimated Lodge Harvest (number of fish)</i>     |             |        |              |        |             |         |              |         |
| <i>Overall</i>                                      |             | 1,344  |              |        |             | 0       |              |         |
| <i>Biomass Harvested</i>                            |             |        |              |        |             |         |              |         |
| <i>Non-Lodge</i>                                    | 18,883      | 41,630 | 25,008       | 55,134 | 47,606      | 104,954 | 63,048       | 138,998 |
| <i>Lodge</i>                                        | 8,058       | 17,764 | 10,671       | 23,527 | 0           | 0       | 0            | 0       |
| <i>Overall</i>                                      | 26,941      | 59,394 | 35,680       | 78,660 | 47,606      | 104,954 | 63,048       | 138,998 |



**Figure 13. Length-frequency distributions of biosampled Chinook ( $n = 471$ ) and Coho ( $n = 1,651$ ) salmon, and Pacific Halibut ( $n = 724$ ), 2023. Note that X and Y axis scales vary among plots.**

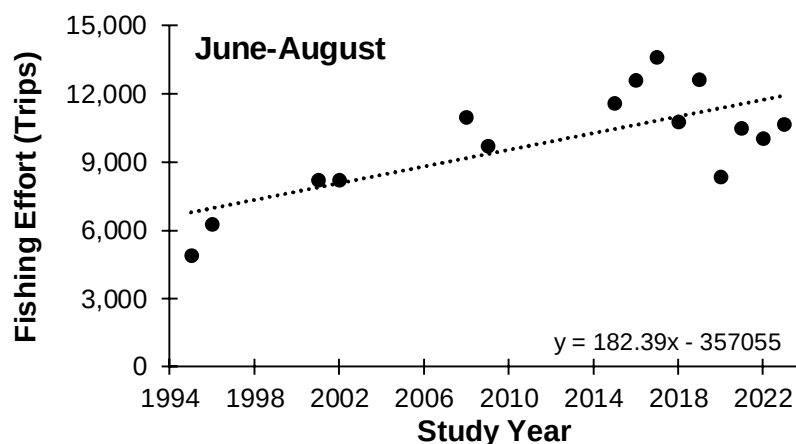
## **Harvested Halibut Biomass**

Based on the average weights of biosampled Pacific Halibut, and the estimated harvest, it was estimated that total harvested Pacific Halibut net biomass was 74.5 tonnes (164,348 lb), including 26.9 t (59,394 lb) in Area 3 and 47.6 t (104,954 lb) in Area 4. In terms of round weight, Pacific Halibut harvest was 98.7 t (217,658 lb) including 35.7 t (78,660 lb) in Area 3 and 63.0 t (138,998 lb) in Area 4 (Table 25).

## **DISCUSSION**

### **Creel Estimates**

In general, recreational fishing effort and salmon catch in Areas 3 & 4 have been increasing over the last 29 years (Tables 26 and 27). Using published creel data collected since 1995, we found 15 years (1995-96, 2001-02, 2008-09<sup>1</sup>, and 2015-2023) for which effort data were collected in June, July, and August (Van Tongeren and Winther 2010, Robichaud et al. 2016, 2017, Robichaud and Addison 2018, 2019, 2020, 2021, 2022, 2023), and we have therefore limited this meta analyses to those three months. Meta-analysis revealed (Figure 14) that fishing effort has increased significantly (by 182 trips per year on average;  $F_{1,13} = 15.0$ ,  $P = 0.002$ ) since 1995. Effort since 2020 has been lower than what would have been predicted from the long-term trend, and has yet to recover to 2019 levels.



**Figure 14. Fishing effort from 1995 to 2023. Annual estimates only include data from June, July and August. Only years with data from all three months are included. Long-term linear regression equation is shown in bottom right of the panel.**

<sup>1</sup> Creel data from 2010-2014 were collected and analyzed by DFO, and preliminary results were presented as part of their post-season review process. We based our meta-analysis on published accounts (written accounts showing methods, results, and discussing the caveats of the results), and hence have not included data from these years.

Over the same period, Coho Salmon harvests have increased significantly by 925 fish per year, on average ( $F_{1,13} = 12.0$ ,  $P = 0.004$ , Table 27, Figure 15). The 2023 Coho Salmon harvest was close to that expected given the long-term trend, which is concerning given the lower-than-expected effort observed in 2023. The number of Chinook Salmon being released by anglers has been increasing strongly since 2009 (the first year with data available; ln scale,  $F_{1,8} = 21.9$ ,  $P = 0.002$ ; Figure 16). Chinook Salmon harvest had been on a strongly increasing trend until 2019, but now appears to be declining over time. The long-term trend (72 additional fish per year on average) is not statistically significant ( $F_{1,13} = 0.5$ ,  $P = 0.49$ ), whereas the trend since 2015 (979 fewer fish per year on average) is strongly and significantly negative ( $F_{1,7} = 10.0$ ,  $P = 0.02$ ). The dome-shaped long term trend may be the result of management measures to control harvest (as evidenced by the increasing trend in Chinook salmon release numbers), or the result of declining stocks.

Pacific Halibut harvest did not show a strong among-year trend (Figure 15). While fishing effort increased over the 29-year period, the total harvest of Pacific Halibut in recent years (2009, 2015-2023) has been very similar to that from 2001-2002; Table 27). This is very likely the result of effective management measures that were enacted between 2008 and 2009 (including the reduction in the daily bag limit from two to one), that appeared to have reduced the catch of halibut by nearly 50% (Figure 15).

For the third consecutive year, Lingcod harvest (June to August) in 2023 was the highest observed for the available time series (Figure 15). In general, the harvest from 2018 to 2023 had been higher than in other recent years (e.g., 1,803 to 3,052 in 2015-2017). The long-term trend is statically significant (ln scale,  $F_{1,12} = 4.9$ ,  $P = 0.047$ ; Figure 15). This, despite a coast-wide reduction in the Lingcod bag limit from three to two fish starting in May 2018. The elevated and increasing harvest values from 2021 and 2023 could potentially cause concern.

The meta-analysis results make three important points: 1) the effort and CPUE of certain species has increased tremendously over the past 29 years (even despite the COVID-19 pandemic), potentially imperilling stocks if left unchecked; 2) management decisions for Pacific Halibut and possibly Chinook Salmon appear to have been effective, indicating that tools are available to keep effort and CPUE increases from going unchecked; and 3) despite that, Lingcod harvest is increasing to a concerning degree.

### **May Effort**

In our report for the 2018 study year (Robichaud and Addison 2019), we made the case that substantial angling activity was being missed by starting the creel survey in June each year. Surveys in 1995 and 1996 (Van Tongeren and Winther 2010) included May, which accounted for 14% and 5% of the May-August effort, and for 8% and 1% of the May-August Chinook harvest. May surveys occurred again in 2001, 2002, and 2008, representing 5%, 10%, and 8% of the May-August effort, respectively (and 5%, 7%, and 4% of the May-August Chinook harvest). Beginning in 2019, we did start the survey in May, and found that May effort (1,553 boat trips in 2019, 1,320 in 2020, 1,785 in 2021, 1,587 in 2022, and 1,428 in 2023) accounted for 11-15% of the May-August total. May effort from 2019 to 2021 and in 2023 accounted for 15-21% of the May-August total Chinook Salmon harvest, whereas effort in May 2022 accounted for 32% of the Chinook Salmon harvest.



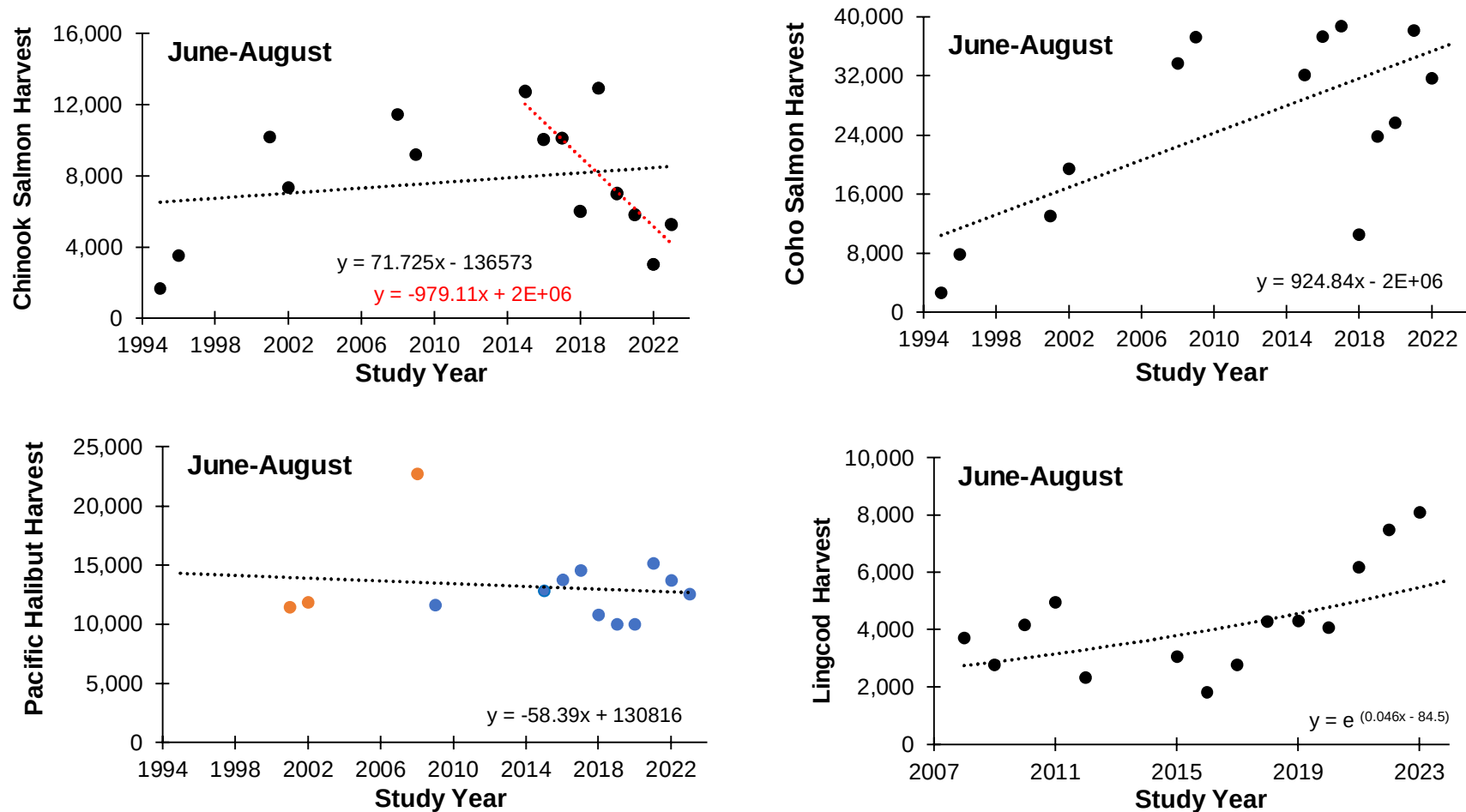
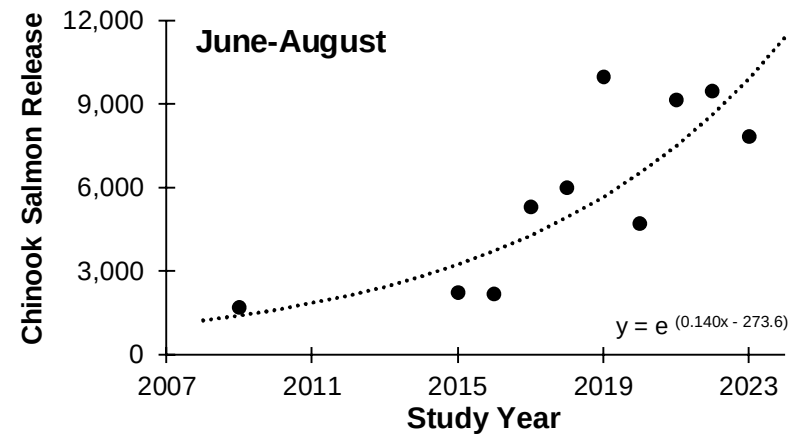


Figure 15. Harvest of Chinook and Coho salmon, Pacific Halibut, and Lingcod from 1995 to 2023. Annual estimates only include data from June, July and August. Only years with data from all three months are included. Long-term linear regression equations are shown in black within each panel, and the Chinook Salmon harvest trend since 2015 is in red. For Pacific Halibut, bag limits changed after 2008 from two fish per day (orange dots) to one fish per day (blue dots).



**Figure 16.** Number of Chinook Salmon released from June to August by anglers in Areas 3 and 4 from 1995 to 2023. Only years with data from all three months are included. Long-term linear regression equation (calculated on ln scale) is shown in the bottom right.

**Table 26.** Monthly recreational fishing effort (boat trips) in Area 3 and 4 from 1995 to 2023, as estimated using creel survey methods.

|              | Year         |              |              |              |              |              |              |               |               |               |               |               |               |               |              |               |               |               |
|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|
| Month        | 1995         | 1996         | 1998         | 2000         | 2001         | 2002         | 2006         | 2008          | 2009          | 2015          | 2016          | 2017          | 2018          | 2019          | 2020         | 2021          | 2022          | 2023          |
| May          | 965          | 391          | --           | --           | 449          | 960          | --           | 983           | --            | --            | --            | --            | --            | 1,553         | 1,320        | 1,785         | 1,587         | 1,428         |
| June         | 1,590        | 2,490        | --           | --           | 3,801        | 3,463        | --           | 4,696         | 3,043         | 3,424         | 3,961         | 4,574         | 2,873         | 3,790         | 1,970        | 2,507         | 2,712         | 2,735         |
| July         | 1,394        | 2,385        | 1,307        | 1,842        | 2,588        | 2,892        | --           | 3,750         | 3,804         | 4,279         | 5,295         | 6,044         | 4,593         | 4,644         | 3,745        | 4,318         | 4,040         | 3,868         |
| Aug          | 1,927        | 1,400        | 1,351        | 1,327        | 1,808        | 1,843        | 2,593        | 2,529         | 2,865         | 3,881         | 3,325         | 2,980         | 3,304         | 4,175         | 2,631        | 3,642         | 3,285         | 4,055         |
| Sept         | 864          | 488          | 504          | 954          | 540          | 676          | 758          | 394           | 817           | --            | --            | --            | --            | --            | --           | --            | --            | --            |
| <b>TOTAL</b> | <b>6,740</b> | <b>7,154</b> | <b>3,162</b> | <b>4,123</b> | <b>9,186</b> | <b>9,834</b> | <b>3,351</b> | <b>12,352</b> | <b>10,529</b> | <b>11,585</b> | <b>12,581</b> | <b>13,598</b> | <b>10,770</b> | <b>14,163</b> | <b>9,665</b> | <b>12,252</b> | <b>11,624</b> | <b>12,086</b> |

**Table 27. Monthly Chinook Salmon, Coho Salmon, Pacific Halibut, and Lingcod harvest, and Chinook Salmon releases, in the Area 3 and 4 recreational fishery from 1995 to 2023, as estimated using creel survey methods.**

|                         | Year  |       |       |       |        |        |        |        |        |       |       |       |        |        |        |        |        |        |        |        |        |
|-------------------------|-------|-------|-------|-------|--------|--------|--------|--------|--------|-------|-------|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Month                   | 1995  | 1996  | 1998  | 2000  | 2001   | 2002   | 2006   | 2008   | 2009   | 2010  | 2011  | 2012  | 2015   | 2016   | 2017   | 2018   | 2019   | 2020   | 2021   | 2022   | 2023   |
| Chinook Salmon Harvest  |       |       |       |       |        |        |        |        |        |       |       |       |        |        |        |        |        |        |        |        |        |
| May                     | 141   | 45    | --    | --    | 509    | 522    | --     | 474    | --     | --    | --    | --    | --     | --     | --     | --     | 2,234  | 1,256  | 1,580  | 1,428  | 1,191  |
| June                    | 907   | 2,165 | --    | --    | 6,630  | 4,028  | --     | 7,708  | 4,797  | --    | --    | --    | 7,071  | 5,678  | 5,229  | 1,273  | 5,675  | 2,574  | 1,834  | 1,564  | 3,180  |
| July                    | 505   | 1,097 | 254   | 1,611 | 3,260  | 2,822  | --     | 3,080  | 3,873  | --    | --    | --    | 4,016  | 3,474  | 3,800  | 2,964  | 4,806  | 2,856  | 2,797  | 9      | 1,155  |
| Aug                     | 245   | 257   | 35    | 108   | 299    | 474    | 1,207  | 688    | 503    | --    | --    | --    | 1,669  | 891    | 1,078  | 1,593  | 2,437  | 1,560  | 1,172  | 1,431  | 917    |
| Sept                    | 15    | 62    | 6     | 93    | 61     | 59     | 203    | 20     | 4      | --    | --    | --    | --     | --     | --     | --     | --     | --     | --     | --     | --     |
| TOTAL                   | 1,813 | 3,626 | 295   | 1,812 | 10,759 | 7,905  | 1,410  | 11,970 | 9,177  |       |       |       | 12,756 | 10,043 | 10,107 | 5,830  | 15,152 | 8,247  | 7,383  | 4,432  | 6,443  |
| Chinook Salmon Release  |       |       |       |       |        |        |        |        |        |       |       |       |        |        |        |        |        |        |        |        |        |
| May                     | --    | --    | --    | --    | --     | --     | --     | --     | --     | --    | --    | --    | --     | --     | --     | --     | 1,145  | 933    | 497    | 536    | 980    |
| June                    | --    | --    | --    | --    | --     | --     | --     | --     | 808    | --    | --    | --    | 1,534  | 975    | 2,800  | 2,121  | 4,735  | 2,088  | 3,652  | 2,297  | 3,790  |
| July                    | --    | --    | --    | --    | --     | --     | --     | --     | 705    | --    | --    | --    | 594    | 710    | 1,811  | 2,612  | 2,382  | 1,560  | 3,694  | 6,213  | 3,415  |
| Aug                     | --    | --    | --    | --    | --     | --     | --     | --     | 175    | --    | --    | --    | 103    | 506    | 697    | 1,264  | 2,866  | 1,061  | 1,799  | 966    | 627    |
| Sept                    | --    | --    | --    | --    | --     | --     | --     | --     | 15     | --    | --    | --    | --     | --     | --     | --     | --     | --     | --     | --     | --     |
| TOTAL                   |       |       |       |       |        |        |        |        | 1,703  |       |       |       | 2,231  | 2,190  | 5,308  | 5,997  | 11,129 | 5,642  | 9,643  | 10,012 | 8,812  |
| Coho Salmon Harvest     |       |       |       |       |        |        |        |        |        |       |       |       |        |        |        |        |        |        |        |        |        |
| May                     | 0     | 0     | --    | --    | 0      | 0      | --     | 0      | --     | --    | --    | --    | --     | --     | --     | --     | 29     | 0      | 0      | 3      | 2      |
| June                    | 335   | 474   | --    | --    | 1,771  | 1,332  | --     | 3,913  | 3,686  | --    | --    | --    | 4,436  | 4,389  | 1,839  | 126    | 1,475  | 1,006  | 3,038  | 1,356  | 2,555  |
| July                    | 402   | 3,565 | 1,937 | 568   | 4,736  | 9,798  | --     | 15,982 | 17,043 | --    | --    | --    | 15,101 | 21,761 | 22,871 | 5,758  | 10,341 | 16,304 | 17,685 | 13,929 | 17,360 |
| Aug                     | 1,907 | 3,771 | 1,500 | 2,125 | 6,488  | 8,292  | 15,647 | 13,791 | 16,459 | --    | --    | --    | 12,612 | 11,200 | 14,004 | 4,580  | 11,974 | 8,288  | 17,403 | 16,331 | 22,608 |
| Sept                    | 1,115 | 1,417 | 79    | 790   | 1,006  | 1,010  | 3,215  | 733    | 3,136  | --    | --    | --    | --     | --     | --     | --     | --     | --     | --     | --     | --     |
| TOTAL                   | 3,759 | 9,227 | 3,516 | 3,483 | 14,001 | 20,432 | 18,862 | 34,419 | 40,324 |       |       |       | 32,148 | 37,350 | 38,714 | 10,464 | 23,820 | 25,599 | 38,126 | 31,620 | 42,525 |
| Pacific Halibut Harvest |       |       |       |       |        |        |        |        |        |       |       |       |        |        |        |        |        |        |        |        |        |
| May                     | --    | --    | --    | --    | 281    | 467    | --     | 1,435  | --     | --    | --    | --    | --     | --     | --     | --     | 941    | 867    | 1,499  | 1,385  | 934    |
| June                    | --    | --    | --    | --    | 4,872  | 3,912  | --     | 8,367  | 3,029  | --    | --    | --    | 3,551  | 4,032  | 4,632  | 3,407  | 2,698  | 2,344  | 3,725  | 4,229  | 2,636  |
| July                    | --    | --    | --    | 3,963 | 3,535  | 4,716  | --     | 8,507  | 4,466  | --    | --    | --    | 4,096  | 6,069  | 6,563  | 4,263  | 3,894  | 4,150  | 6,518  | 5,558  | 4,887  |
| Aug                     | --    | --    | 1,801 | 2,530 | 3,085  | 3,247  | 8,178  | 5,865  | 4,142  | --    | --    | --    | 5,201  | 3,670  | 3,410  | 3,183  | 3,435  | 3,504  | 4,929  | 3,953  | 5,049  |
| Sept                    | --    | --    | 328   | 1,224 | 222    | 615    | 1,812  | 774    | 1,205  | --    | --    | --    | --     | --     | --     | --     | --     | --     | --     | --     | --     |
| TOTAL                   |       |       | 2,129 | 7,717 | 11,995 | 12,957 | 9,990  | 24,948 | 12,842 |       |       |       | 12,848 | 13,771 | 14,605 | 10,853 | 10,967 | 10,865 | 16,671 | 15,125 | 13,505 |
| Lingcod Harvest         |       |       |       |       |        |        |        |        |        |       |       |       |        |        |        |        |        |        |        |        |        |
| May                     | --    | --    | --    | --    | --     | --     | --     | 81     | --     | --    | --    | --    | --     | --     | --     | --     | 219    | 635    | 1,178  | 1,661  | 802    |
| June                    | --    | --    | --    | --    | --     | --     | --     | 1,462  | 760    | 1,093 | 2,241 | 692   | 1,042  | 732    | 1,124  | 1,754  | 1,201  | 964    | 2,120  | 2,722  | 2,347  |
| July                    | --    | --    | --    | --    | --     | --     | --     | 1,348  | 1,019  | 1,536 | 1,180 | 1,342 | 951    | 701    | 1,173  | 1,740  | 1,389  | 2,157  | 2,312  | 2,972  | 2,987  |
| Aug                     | --    | --    | --    | --    | --     | --     | --     | 884    | 999    | 1,521 | 1,528 | 274   | 1,059  | 370    | 475    | 776    | 1,704  | 943    | 1,733  | 1,792  | 2,747  |
| Sept                    | --    | --    | --    | --    | --     | --     | --     | 167    | 208    | --    | --    | --    | --     | --     | --     | --     | --     | --     | --     | --     | --     |
| TOTAL                   |       |       |       |       |        |        |        | 3,942  | 2,986  | 4,150 | 4,949 | 2,308 | 3,052  | 1,803  | 2,772  | 4,270  | 4,514  | 4,699  | 7,343  | 9,147  | 8,884  |

Much of the early season angling targets Chinook Salmon, and occurs during a period in which the Area 3 and 4 stock composition is unknown or poorly resolved. As such, the Department of Fisheries and Oceans Canada (DFO) expressed interest in a May creel survey because genetic samples collected from Chinook Salmon would help them resolve stock composition and help determine exploitation rates on early-run-timing Chinook Salmon stocks. In 2019, during the month of May, 269 Chinook Salmon were biosampled, including 230 scale samples, and 22 tagged heads. Since then, 144-194 Chinook Salmon scale samples have been collected each May, representing 33-52% of the annual total. Since 2020, May accounted for 11-16 of the heads submitted to the Salmon Head Recovery Program, 31-56 % of the season-wide totals.

### **Precision of the Results**

Overall, our level of survey effort, both for CPE and Effort estimation, were adequate, as evidenced by the relatively good precision of the harvest estimates for most of the taxa surveyed. In 2023 our precision goal was to estimate total harvests within 15% of the true value 19 times out of 20, for the main taxa (Chinook Salmon, Coho Salmon and Pacific Halibut), for which harvest estimates were precise to within 3-6%. Pink Salmon, Lingcod, and rockfish estimates were precise to within 12%, 7%, and 8%, respectively. Chum and Sockeye salmon estimates, and those for other groundfish, were not precise to within 15%, though the survey was not designed to meet any level of precision for these last taxa.

The precision of our harvest estimates depended largely on the variability in the CPE and effort data. Differing levels of angler expertise, and day-to-day changes in abundance of the target species, fishing effort, weather, or fishing conditions mean that fishing trips have a wide range of outcomes. Similarly, day-to-day differences in weather or other factors (such as leisure time, day of week, the price of gasoline, etc.) affect the number of anglers that decide to go out on the water on any given day, resulting in variability within the boat-count dataset. Sampling error is another main source of estimation imprecision; and by meeting our precision goals we have demonstrated that we are sampling an adequate proportion of the population.

### **Accuracy of the Results**

The accuracy of the results depends on the quality of data provided by the anglers to the interviewers. This is partly why we have a target to sample 20% of estimated effort. Even so, each time an angler is relied upon for unconfirmed data, there is opportunity for results to become inaccurate. For example, in this study, 1,358 of the 1,756 Chinook Salmon (77%) and 7,278 of the 9,999 Coho Salmon (73%) reported as kept during angler interviews were recorded as observed by the interviewer, and 483 and 1,653 (36% and 23%) were handled for biosampling. While not every fish can be observed<sup>2</sup>, each one that goes unconfirmed is nevertheless a possible source of inaccuracy in the reported harvest totals. Angler compliance with requests to examine their catch is voluntary.

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<sup>2</sup> Fish may not be observed when anglers refuse to provide access to their fish storage containers for various reasons. Reluctance to provide access may be due to fish already being processed and packed in ice, boat launch traffic congestion causing tension, or COVID-19 concerns, among other reasons.

In this study, the angler effort estimates depend on the accuracy of the aerial survey boat counts and the activity patterns for each month. For example, we have no estimate of the proportion of angling boats that are missed during overflights, where missed boats would contribute to the underestimation of recreational fishing effort. Also, we do not know if all fishing areas are adequately covered by the current flightpath – perhaps there are fishing locations that are hidden behind islands (e.g., Dunira Island or the Moffat Islands), too far from the flightpath for the surveyors to see active fishing boats. On the other hand, our estimate of fishing effort may be overestimated due to our inability to survey and count boats on foggy or bad weather days. We know that during bad weather, fewer anglers go out on the water; yet these same conditions can cause flights to be cancelled, and thus the lowest effort values are sometimes not sampled.

Another potential source of inaccuracy is the removal of lodge boats from the overflight counts. There is uncertainty around whether all of the active lodge boats were observed by the overflights (e.g., vessels outside of the survey area), as evidenced by surveys for which more lodge boats were operating than there were total boats observed in Area 3 during the flight. There is also uncertainty around the proportion of lodge vessels that were active during the overflights. These factors may influence the accuracy of the estimates of non-lodge fishing effort.

## RECOMMENDATIONS

- The catch and effort estimates for the Area 3 and 4 recreational creel surveys for 2010-2014 should be compiled and included in summary tables similar to Table 26 and 27 in this report. If sufficient funding can be obtained, the results from 2010-14 surveys should be compiled into a manuscript report. This additional information will help provide context for the current results, and help clarify how rates of effort and harvest have changed in recent years.
- We recommend that future surveys continue to run for four months (May to August) rather than three (June to August). May surveys will help document and quantify angling effort and harvest during that month. The Department of Fisheries and Oceans Canada (DFO) has expressed interest in a May creel survey because genetic samples collected from Chinook Salmon will help resolve the currently unknown early-season Area 3 and 4 stock composition, which will help determine exploitation rates on early-run-timing Chinook Salmon stocks.
- Trailer counts should be conducted in April 2024 for comparison with April trailer counts in 2019-2020 and continued throughout the creel survey season as in recent years.
- Factors that are resulting increased Lingcod harvest should be investigated to inform managers should additional measures be enacted.

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## LITERATURE CITED

- Benjamini, Y., and Y. Hochberg. 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society. Series B* 57: 289-300.
- Clark, W. G. 1992. Validation of the IPHC length-weight relationship for halibut. *International Pacific Halibut Commission Report of Assessment and Research Activities* 1991: 113-116.
- English, K. K., G. F. Searing, and D. A. Nagtegaal. 2002. Review of Georgia Strait recreational creel survey, 1983-1999. *Canadian Technical Report of Fisheries and Aquatic Sciences* 2414: 81 p.
- Goodman, L. 1960. On the exact variance of products. *Journal of the American Statistical Association* 55: 708-713.
- Robichaud, D., and A. Addison. 2018. North Coast (Areas 3 & 4) recreational angling creel survey 2017. Report for North Coast - Skeena First Nations Stewardship Society, Prince Rupert, BC.
- Robichaud, D., and A. Addison. 2019. North Coast (Areas 3 & 4) recreational angling creel survey 2018. Report for North Coast - Skeena First Nations Stewardship Society, Prince Rupert, BC.
- Robichaud, D., and A. Addison. 2020. North Coast (Areas 3 & 4) recreational angling creel survey 2019. Report for North Coast - Skeena First Nations Stewardship Society, Prince Rupert, BC.
- Robichaud, D., and A. Addison. 2021. North Coast (Areas 3 & 4) recreational angling creel survey 2020. Report for North Coast - Skeena First Nations Stewardship Society, Prince Rupert, BC.
- Robichaud, D., and A. Addison. 2022. North Coast (Areas 3 & 4) recreational angling creel survey 2021. Report for North Coast - Skeena First Nations Stewardship Society, Prince Rupert, BC.
- Robichaud, D., and A. Addison. 2023. North Coast (Areas 3 & 4) recreational angling creel survey 2022. Report for North Coast - Skeena First Nations Stewardship Society, Prince Rupert, BC.
- Robichaud, D., A. Addison, and M. Imrie. 2016. North Coast (Areas 3 & 4) recreational angling creel survey 2015. Report for North Coast - Skeena First Nations Stewardship Society, Prince Rupert, BC.

- Robichaud, D., A. Addison, and M. Imrie. 2017. North Coast (Areas 3 & 4) recreational angling creel survey 2016. Report for North Coast - Skeena First Nations Stewardship Society, Prince Rupert, BC.
- Sokal, R.R. and F.J. Rohlf. 1995. Biometry. W. H. Freeman and Company, New York.
- Van Tongeren, V., and I. Winther. 2010. 2009 North Coast (Areas 3 & 4) creel survey statistics for salmon and groundfish. *Canadian Manuscript Report of Fisheries and Aquatic Sciences* 2907.
- Zar, J. H. 1984. Biostatistical Analysis. Prentice-Hall Inc., Englewood Cliffs, NJ.

## APPENDICES

## APPENDIX 1: DATA FORMS

2023 AREA 3/4 CREEL SURVEY  
AERIAL SURVEY VESSEL ENUMERATION FORM

DATE: \_\_\_\_\_ FLIGHT#: \_\_\_\_\_

SURVEYOR: \_\_\_\_\_

*Note: only survey shaded subareas when conditions permit*

| Sub Area | RIGHT FRONT COUNT  |                   |                    |                   | TOTAL COUNT |          |          |          | FLIGHT TIME |        | SEA STATE (1-4) |
|----------|--------------------|-------------------|--------------------|-------------------|-------------|----------|----------|----------|-------------|--------|-----------------|
|          | SPORT<br>(Fishing) | SPORT<br>(Moving) | COMM.<br>(Fishing) | COMM.<br>(Moving) | S<br>(F)    | S<br>(M) | C<br>(F) | C<br>(M) | START       | FINISH |                 |
| A        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| C        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| B        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| C        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| D        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| E        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| C        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| F        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| E        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| F        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| G        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| L        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| G        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| J        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| I        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| K        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| I        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| H        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| C        |                    |                   |                    |                   |             |          |          |          |             |        |                 |
| A        |                    |                   |                    |                   |             |          |          |          |             |        |                 |

WEATHER / SEA STATE / COMMENTS:

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

SIGNATURE: \_\_\_\_\_



### SHIFT SUMMARY SHEET

**Landing Site:** \_\_\_\_\_ **Interviewer Name:** \_\_\_\_\_

**Interviewer Code:** \_\_\_\_\_ **Date:** \_\_\_\_\_

**Shift Code:** \_\_\_\_\_ **Shift Times:** \_\_\_\_\_

| Time Blocks   |    |
|---------------|----|
| From - To     | TB |
| Before 7:00   | 1  |
| 7:00 - 7:59   | 2  |
| 8:00 - 8:59   | 3  |
| 9:00 - 9:59   | 4  |
| 10:00 - 10:59 | 5  |
| 11:00 - 11:59 | 6  |
| 12:00 - 12:59 | 7  |
| 13:00 - 13:59 | 8  |
| 14:00 - 14:59 | 9  |
| 15:00 - 15:59 | 10 |
| 16:00 - 16:59 | 11 |
| 17:00 - 17:59 | 12 |
| 18:00 - 18:59 | 13 |
| 19:00 - 19:59 | 14 |
| 20:00 - 20:59 | 15 |
| 21:00 - 21:59 | 16 |

\* Fill in using numbers only.

| Time Block | Fishing Conditions | Boats Missed | Not Fishing | Fishing |
|------------|--------------------|--------------|-------------|---------|
|            |                    |              |             |         |
|            |                    |              |             |         |
|            |                    |              |             |         |
|            |                    |              |             |         |
|            |                    |              |             |         |
|            |                    |              |             |         |
|            |                    |              |             |         |
|            |                    |              |             |         |
|            |                    |              |             |         |

**Totals:**

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

**Comments:** \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_

**Trailer Count:** \_\_\_\_\_ **Time:** \_\_\_\_\_

|                           |                   |
|---------------------------|-------------------|
| <b>Database Tally # :</b> | (office use only) |
|---------------------------|-------------------|

2023 Area 3 and 4 Creel Survey

Site# \_\_\_\_\_ Obs# \_\_\_\_\_ Date      /      /      Interview# \_\_\_\_\_  
DD/MM/YY

# of Anglers: \_\_\_\_\_ Guided?: Y N Depart. Time: \_\_\_\_\_ Land Time: \_\_\_\_\_

**Pick ONE**

☐ Refusal      ☐ Partial

☐ Catch data given by angler (Catch not verified)

☐ Fish observed (Catch & Species verified)

☐ Adipose checked by interviewer

☐ Fish biosampled (e.g., length, sex, scales, etc.)

**Pick ONE Target**

|                          |                       |
|--------------------------|-----------------------|
| <input type="checkbox"/> | Shellfish Only        |
| <input type="checkbox"/> | Finfish Only          |
| <input type="checkbox"/> | Finfish and Shellfish |

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## APPENDIX 2: HPE ESTIMATES (AVERAGE NUMBER OF FISH HARVESTED PER NON-LODGE BOAT TRIP) BY PERIOD, TRIP TYPE, SUBAREA AND SPECIES.

**Average number of salmon harvested per boat trip in May 2023, by trip type and subarea. Zeros hidden.**

|                     | Subarea |      |      |      |      |      |      |      |      |      |      |      |
|---------------------|---------|------|------|------|------|------|------|------|------|------|------|------|
| Species / Trip Type | 3I      | 3J   | 3K   | 4A   | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| Chinook Salmon      |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              | 0.44    | 0.33 | 0.44 |      | 2.20 | 2.15 | 1.45 | 2.10 | 2.39 | 0.67 | 0.60 | 0.67 |
| Unguided            | 0.67    | 0.38 | 0.38 | 0.04 | 0.71 | 0.58 | 0.58 | 0.50 | 1.31 | 0.36 | 0.33 | 0.33 |
| Combined            | 0.67    | 0.36 | 0.38 | 0.02 | 0.81 | 0.81 | 0.86 | 0.40 | 1.64 | 0.56 | 0.45 | 0.45 |
| Chum Salmon         |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |      |      |      |      |      |      |      |      |      |
| Coho Salmon         |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      | 0.02 | 0.02 |      |      |      |
| Unguided            |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |      |      |      |      |      | 0.01 |      |      |      |
| Pink Salmon         |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |      |      |      |      |      |      |      | 0.17 | 0.07 |
| Combined            |         |      |      |      |      |      |      |      |      |      | 0.09 | 0.05 |
| Sockeye Salmon      |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |      |      |      |      |      |      |      |      |      |
| Unknown Salmon      |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |      |      |      |      |      |      |      |      |      |

**Average number of salmon harvested per boat trip in June 2023, by trip type and subarea. Zeros hidden.**

|                     | Subarea |      |      |      |      |      |      |      |      |      |      |      |
|---------------------|---------|------|------|------|------|------|------|------|------|------|------|------|
| Species / Trip Type | 3I      | 3J   | 3K   | 4A   | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| Chinook Salmon      |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              | 2.67    | 2.50 | 2.67 |      | 0.33 | 2.29 | 3.56 | 2.56 | 2.43 | 1.97 | 1.58 | 2.25 |
| Unguided            | 1.33    | 1.05 | 1.25 | 0.01 | 0.46 | 0.42 | 1.11 | 1.00 | 1.17 | 0.53 | 1.21 | 0.60 |
| Combined            | 2.19    | 1.65 | 1.21 | 0.01 | 0.51 | 0.72 | 1.87 | 0.71 | 1.70 | 0.81 | 1.32 | 1.86 |
| Chum Salmon         |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      | 0.03 |      |      |
| Unguided            |         |      |      |      |      |      |      |      | 0.03 |      |      |      |
| Combined            |         |      |      |      |      |      |      |      | 0.02 | 0.05 |      |      |
| Coho Salmon         |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              | 1.05    | 0.71 | 1.05 |      | 0.29 | 2.18 | 0.31 | 2.37 | 2.63 | 3.53 | 0.50 | 6.31 |
| Unguided            | 1.00    | 1.25 | 0.75 |      | 0.12 | 0.38 | 0.25 | 2.20 | 1.83 | 0.06 | 0.62 | 1.40 |
| Combined            | 0.75    | 1.03 | 1.50 |      | 0.12 | 0.67 | 0.27 | 3.86 | 2.17 | 0.33 | 0.59 | 5.14 |
| Pink Salmon         |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              | 0.05    | 0.07 | 0.05 |      |      |      |      | 0.10 | 0.12 | 0.19 |      | 0.38 |
| Unguided            |         |      |      |      |      | 0.01 |      |      | 0.02 |      |      |      |
| Combined            |         | 0.03 |      |      |      | 0.01 |      |      | 0.06 |      |      | 0.29 |
| Sockeye Salmon      |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |      |      |      |      |      | 0.01 |      |      |      |
| Combined            |         |      |      |      |      |      |      |      | 0.01 |      |      |      |
| Unknown Salmon      |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |      |      |      |      |      |      |      |      |      |

**Average number of salmon harvested per boat trip in July 2023, by trip type and subarea. Zeros hidden.**

| Species / Trip Type | Subarea |      |       |    |      |      |       |      |      |      |      |      |
|---------------------|---------|------|-------|----|------|------|-------|------|------|------|------|------|
|                     | 3I      | 3J   | 3K    | 4A | 4B   | 4C   | 4D    | 4E   | 4F   | 4G   | 4H   | 4L   |
| Chinook Salmon      |         |      |       |    |      |      |       |      |      |      |      |      |
| Guided              | 0.55    | 0.42 | 1.18  |    | 0.06 | 0.09 | 0.48  | 0.44 | 0.43 | 0.09 | 0.43 | 0.43 |
| Unguided            | 0.36    | 0.15 | 0.76  |    | 0.17 | 0.23 | 0.28  | 0.20 | 0.27 | 0.10 | 0.27 | 1.00 |
| Combined            | 0.41    | 0.30 | 1.00  |    | 0.23 | 0.20 | 0.33  | 0.23 | 0.34 | 0.10 | 0.33 | 0.54 |
| Chum Salmon         |         |      |       |    |      |      |       |      |      |      |      |      |
| Guided              |         |      |       |    |      |      |       | 0.01 | 0.01 |      |      |      |
| Unguided            |         |      |       |    |      | 0.09 | 0.02  |      | 0.02 |      |      |      |
| Combined            |         |      |       |    |      | 0.07 | 0.02  |      | 0.02 |      |      |      |
| Coho Salmon         |         |      |       |    |      |      |       |      |      |      |      |      |
| Guided              | 9.00    | 3.36 | 11.95 |    | 2.39 | 8.40 | 12.12 | 9.82 | 9.35 | 8.27 | 7.51 | 4.67 |
| Unguided            | 4.14    | 3.04 | 7.29  |    | 2.05 | 3.10 | 5.71  | 0.50 | 5.15 | 3.26 | 3.63 | 3.80 |
| Combined            | 5.51    | 3.22 | 9.92  |    | 2.26 | 4.09 | 7.30  | 1.54 | 7.04 | 4.57 | 5.25 | 4.50 |
| Pink Salmon         |         |      |       |    |      |      |       |      |      |      |      |      |
| Guided              | 1.09    | 0.18 | 1.41  |    | 0.18 | 0.63 | 0.33  | 0.40 | 0.43 | 0.36 | 0.40 |      |
| Unguided            | 0.32    | 0.19 |       |    | 0.23 | 0.28 | 0.65  | 0.20 | 0.73 | 0.29 | 0.29 |      |
| Combined            | 0.54    | 0.18 | 0.79  |    | 0.25 | 0.34 | 0.57  | 0.15 | 0.60 | 0.31 | 0.33 |      |
| Sockeye Salmon      |         |      |       |    |      |      |       |      |      |      |      |      |
| Guided              |         |      |       |    | 0.01 | 0.03 |       |      |      |      |      |      |
| Unguided            |         |      |       |    | 0.02 |      |       |      | 0.01 |      |      |      |
| Combined            |         |      |       |    | 0.02 | 0.01 |       |      | 0.00 |      |      |      |
| Unknown Salmon      |         |      |       |    |      |      |       |      |      |      |      |      |
| Guided              |         |      |       |    |      |      |       |      |      |      |      |      |
| Unguided            |         |      |       |    |      |      |       |      | 0.01 |      |      |      |
| Combined            |         |      |       |    |      |      |       |      | 0.00 |      |      |      |

**Average number of salmon harvested per boat trip in August 2023, by trip type and subarea. Zeros hidden.**

| Species / Trip Type | Subarea |      |       |      |      |      |      |      |      |      |       |      |
|---------------------|---------|------|-------|------|------|------|------|------|------|------|-------|------|
|                     | 3I      | 3J   | 3K    | 4A   | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H    | 4L   |
| Chinook Salmon      |         |      |       |      |      |      |      |      |      |      |       |      |
| Guided              | 0.15    | 0.31 | 0.50  |      | 0.20 | 0.10 | 0.60 | 1.00 | 1.11 | 0.38 | 0.17  |      |
| Unguided            | 0.27    | 0.24 | 0.25  | 0.02 | 0.10 | 0.05 | 0.33 | 0.08 | 0.34 | 0.21 | 0.18  |      |
| Combined            | 0.23    | 0.27 | 0.35  | 0.01 | 0.11 | 0.05 | 0.35 | 0.07 | 0.56 | 0.28 | 0.17  |      |
| Chum Salmon         |         |      |       |      |      |      |      |      |      |      |       |      |
| Guided              |         |      |       |      |      |      |      |      |      |      | 0.03  |      |
| Unguided            | 0.05    |      | 0.33  |      |      |      |      | 0.77 |      |      | 0.03  |      |
| Combined            | 0.03    |      | 0.19  |      |      |      |      | 0.67 |      |      | 0.03  |      |
| Coho Salmon         |         |      |       |      |      |      |      |      |      |      |       |      |
| Guided              | 10.85   | 4.94 | 12.81 |      | 5.80 | 4.60 | 7.80 | 7.84 | 7.61 | 8.08 | 11.07 | 3.78 |
| Unguided            | 6.68    | 4.24 | 8.42  |      | 2.19 | 2.02 | 4.89 | 1.69 | 4.57 | 2.95 | 2.97  |      |
| Combined            | 8.23    | 4.51 | 10.26 |      | 2.40 | 2.29 | 5.13 | 3.07 | 5.45 | 5.03 | 7.88  | 2.27 |
| Pink Salmon         |         |      |       |      |      |      |      |      |      |      |       |      |
| Guided              | 0.23    | 0.25 | 0.08  |      | 0.40 | 0.20 |      | 0.35 | 0.42 |      | 0.30  |      |
| Unguided            | 0.23    | 0.28 | 0.11  | 0.02 | 0.29 | 0.51 | 0.39 | 0.46 | 0.82 | 0.32 | 0.33  |      |
| Combined            | 0.23    | 0.27 | 0.10  | 0.01 | 0.29 | 0.47 | 0.35 | 0.40 | 0.70 | 0.19 | 0.31  |      |
| Sockeye Salmon      |         |      |       |      |      |      |      |      |      |      |       |      |
| Guided              |         |      |       |      |      |      |      |      |      |      |       |      |
| Unguided            |         |      | 0.08  |      |      |      |      |      |      |      |       |      |
| Combined            |         |      | 0.05  |      |      |      |      |      |      |      |       |      |
| Unknown Salmon      |         |      |       |      |      |      |      |      |      |      |       |      |
| Guided              |         |      |       |      |      |      |      |      |      |      |       |      |
| Unguided            |         |      |       |      |      |      |      |      |      |      | 0.33  |      |
| Combined            |         |      |       |      |      |      |      |      |      |      | 0.13  |      |

**Average number of groundfish harvested per boat trip in May 2023, by trip type and subarea. Zeros hidden.**

| Species / Trip Type   | Subarea |      |      |      |      |      |      |      |      |      |      |      |
|-----------------------|---------|------|------|------|------|------|------|------|------|------|------|------|
|                       | 3I      | 3J   | 3K   | 4A   | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| Arrow-tooth Flounder  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined              |         |      |      |      |      |      |      |      |      |      |      |      |
| Big Skate             |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |         |      |      |      |      | 0.03 |      |      |      |      |      |      |
| Combined              |         |      |      |      |      | 0.02 |      |      |      |      |      |      |
| Cabezon               |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined              |         |      |      |      |      |      |      |      |      |      |      |      |
| English Sole          |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined              |         |      |      |      |      |      |      |      |      |      |      |      |
| Greenlings            |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |         |      |      |      |      |      |      |      | 0.03 |      |      |      |
| Combined              |         |      |      |      |      |      |      |      | 0.02 |      |      |      |
| Lingcod               |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                | 2.56    | 3.17 | 2.56 |      |      | 0.23 | 0.18 | 1.48 | 1.37 | 0.87 | 0.50 | 0.87 |
| Unguided              |         | 0.63 |      | 0.01 | 0.33 | 0.03 | 0.46 | 3.17 | 1.03 | 0.64 | 0.50 | 0.59 |
| Combined              | 0.44    | 1.71 |      | 0.01 | 0.31 | 0.06 | 0.37 | 4.13 | 1.13 | 0.88 | 0.50 | 0.69 |
| Longnose Skate        |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined              |         |      |      |      |      |      |      |      |      |      |      |      |
| Pacific Cod           |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                | 0.67    |      | 0.67 |      |      | 0.15 |      | 0.03 | 0.04 | 0.20 | 0.30 | 0.20 |
| Unguided              | 0.50    | 0.38 |      |      |      | 0.01 |      | 0.17 | 0.06 | 0.29 |      | 0.15 |
| Combined              | 1.00    | 0.21 |      |      |      | 0.03 |      | 0.13 | 0.05 | 0.25 | 0.14 | 0.17 |
| Pacific Halibut       |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                | 2.78    | 3.17 | 2.78 |      |      | 0.46 | 0.91 | 1.50 | 1.61 | 2.13 | 2.00 | 2.13 |
| Unguided              | 1.17    | 1.13 | 0.38 | 0.01 | 0.31 | 0.52 | 0.79 | 1.92 | 0.54 | 0.93 | 0.08 | 0.59 |
| Combined              | 1.44    | 2.00 | 0.38 | 0.01 | 0.29 | 0.51 | 0.83 | 1.93 | 0.86 | 1.13 | 0.95 | 1.14 |
| Pacific Rock Sole     |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |         |      |      |      | 0.04 |      |      |      |      |      |      |      |
| Combined              |         |      |      |      | 0.04 |      |      |      |      |      |      |      |
| Pacific Spiny Dogfish |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined              |         |      |      |      |      |      |      |      |      |      |      |      |
| Sablefish             |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined              |         |      |      |      |      |      |      |      |      |      |      |      |
| Spotted Ratfish       |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined              |         |      |      |      |      |      |      |      |      |      |      |      |
| Starry Flounder       |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined              |         |      |      |      |      |      |      |      |      |      |      |      |
| Unknown Groundfish    |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |         |      |      |      | 0.03 |      |      |      |      |      |      |      |
| Combined              |         |      |      |      | 0.03 |      |      |      |      |      |      |      |

**Average number of groundfish harvested per boat trip in June 2023, by trip type and subarea. Zeros hidden.**

|                       |  | Subarea |      |      |      |      |      |      |      |      |      |      |      |
|-----------------------|--|---------|------|------|------|------|------|------|------|------|------|------|------|
| Species / Trip Type   |  | 3I      | 3J   | 3K   | 4A   | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| Arrow-tooth Flounder  |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |  |         |      |      |      |      |      |      |      | 0.01 |      |      |      |
| Combined              |  |         |      |      |      |      |      |      |      | 0.01 |      |      |      |
| Big Skate             |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined              |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Cabezon               |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |  |         |      |      |      |      |      |      |      |      | 0.03 |      | 0.06 |
| Unguided              |  |         |      |      |      |      |      |      |      | 0.02 |      |      |      |
| Combined              |  |         |      |      |      |      |      |      |      | 0.01 |      |      | 0.05 |
| English Sole          |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |  |         |      |      |      |      |      |      |      | 0.04 |      |      |      |
| Combined              |  |         |      |      |      |      |      |      |      | 0.02 |      |      |      |
| Greenlings            |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |  |         |      |      |      |      |      | 0.06 | 0.01 |      | 0.03 |      | 0.06 |
| Unguided              |  |         | 0.10 |      |      |      |      |      |      | 0.01 |      |      | 0.20 |
| Combined              |  |         | 0.06 |      |      |      |      | 0.02 |      | 0.01 |      |      | 0.10 |
| Lingcod               |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |  | 2.48    | 3.29 | 2.48 |      | 0.09 | 0.65 | 0.44 | 1.27 | 1.45 | 2.03 | 1.00 | 2.25 |
| Unguided              |  | 0.75    | 0.85 | 0.17 | 0.01 | 0.07 | 0.29 | 0.08 | 2.60 | 0.70 | 1.65 | 0.52 | 0.40 |
| Combined              |  | 0.56    | 1.85 | 0.14 | 0.01 | 0.07 | 0.35 | 0.19 | 2.00 | 1.02 | 2.14 | 0.66 | 1.81 |
| Longnose Skate        |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined              |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Pacific Cod           |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |  | 0.05    |      | 0.05 |      |      |      | 0.25 | 0.04 |      | 0.91 | 1.67 | 0.19 |
| Unguided              |  |         | 0.05 | 0.08 |      | 0.02 | 0.07 | 0.03 |      | 0.02 |      |      |      |
| Combined              |  |         | 0.03 | 0.14 |      | 0.02 | 0.06 | 0.10 |      | 0.01 | 0.29 | 0.49 | 0.14 |
| Pacific Halibut       |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |  | 1.43    | 1.86 | 1.43 |      | 0.16 | 1.12 | 0.94 | 1.72 | 1.82 | 2.56 | 2.17 | 3.06 |
| Unguided              |  | 1.25    | 0.80 | 0.33 |      | 0.32 | 0.74 | 0.17 | 1.80 | 0.88 | 1.59 | 0.59 | 2.60 |
| Combined              |  | 0.94    | 1.24 | 0.57 |      | 0.32 | 0.80 | 0.40 | 2.43 | 1.28 | 1.62 | 1.05 | 2.95 |
| Pacific Rock Sole     |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined              |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Pacific Spiny Dogfish |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |  |         |      |      |      |      |      |      |      |      | 0.06 |      |      |
| Combined              |  |         |      |      |      |      |      |      |      |      | 0.05 |      |      |
| Sablefish             |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined              |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Spotted Ratfish       |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined              |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Starry Flounder       |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Unguided              |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined              |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Unknown Groundfish    |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Guided                |  |         |      |      |      |      |      | 0.13 | 0.02 |      |      |      |      |
| Unguided              |  |         |      |      |      |      |      |      |      |      |      |      |      |
| Combined              |  |         |      |      |      |      |      | 0.04 |      |      |      |      |      |

**Average number of groundfish harvested per boat trip in July 2023, by trip type and subarea. Zeros hidden.**

|                       |      | Subarea |      |    |      |      |      |      |      |      |      |      |
|-----------------------|------|---------|------|----|------|------|------|------|------|------|------|------|
| Species / Trip Type   | 3I   | 3J      | 3K   | 4A | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| Arrow-tooth Flounder  |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      | 0.15    |      |    |      |      |      |      |      |      | 0.06 |      |
| Unguided              |      |         |      |    |      |      | 0.01 |      |      |      |      |      |
| Combined              |      | 0.08    |      |    |      |      | 0.01 |      |      |      | 0.02 |      |
| Big Skate             |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      |      |      |
| Unguided              |      |         |      |    |      |      |      |      |      |      |      |      |
| Combined              |      |         |      |    |      |      |      |      |      |      |      |      |
| Cabezon               |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      |      |      |
| Unguided              |      |         |      |    |      |      |      |      | 0.01 |      |      |      |
| Combined              |      |         |      |    |      |      |      |      | 0.00 |      |      |      |
| English Sole          |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      |      |      |
| Unguided              |      |         |      |    |      |      |      |      |      |      |      |      |
| Combined              |      |         |      |    |      |      |      |      |      |      |      |      |
| Greenlings            |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      | 0.01 | 0.01 |      |      | 0.19 |
| Unguided              |      |         |      |    | 0.01 |      |      |      | 0.01 |      | 0.02 |      |
| Combined              |      |         |      |    | 0.01 |      |      |      | 0.01 |      | 0.01 | 0.15 |
| Lingcod               |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                | 1.45 | 3.15    | 0.36 |    | 0.11 | 0.43 | 1.58 | 1.55 | 1.57 | 2.82 | 1.69 | 1.90 |
| Unguided              | 0.46 | 0.70    | 0.29 |    | 0.01 | 0.32 | 0.31 | 0.70 | 0.64 | 0.81 | 0.55 | 2.80 |
| Combined              | 0.74 | 2.05    | 0.33 |    | 0.01 | 0.34 | 0.62 | 0.69 | 1.06 | 1.33 | 1.02 | 2.08 |
| Longnose Skate        |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      |      |      |
| Unguided              |      |         |      |    |      |      |      |      |      |      |      |      |
| Combined              |      |         |      |    |      |      |      |      |      |      |      |      |
| Pacific Cod           |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                | 0.09 | 0.09    | 0.09 |    | 0.02 | 0.09 | 0.15 | 0.13 | 0.13 | 0.73 | 0.60 |      |
| Unguided              | 0.07 | 0.19    |      |    | 0.07 | 0.08 | 0.03 |      | 0.09 | 0.26 | 0.22 |      |
| Combined              | 0.08 | 0.13    | 0.05 |    | 0.07 | 0.08 | 0.06 |      | 0.11 | 0.38 | 0.38 |      |
| Pacific Halibut       |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                | 2.27 | 3.88    | 1.59 |    | 0.42 | 1.34 | 1.48 | 1.68 | 1.73 | 4.09 | 1.89 | 5.05 |
| Unguided              | 1.50 | 3.07    | 0.35 |    | 0.40 | 0.91 | 0.75 | 1.90 | 1.42 | 1.48 | 1.33 | 3.20 |
| Combined              | 1.72 | 3.52    | 1.05 |    | 0.49 | 0.99 | 0.93 | 1.92 | 1.56 | 2.17 | 1.56 | 4.69 |
| Pacific Rock Sole     |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      |      |      |
| Unguided              |      |         |      |    |      |      | 0.01 |      |      |      | 0.02 |      |
| Combined              |      |         |      |    |      |      | 0.01 |      |      |      | 0.01 |      |
| Pacific Spiny Dogfish |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      | 0.01 | 0.02 |      |      |      |
| Unguided              | 0.04 |         |      |    |      |      |      |      |      |      |      |      |
| Combined              | 0.03 |         |      |    |      |      |      |      | 0.01 |      |      |      |
| Sablefish             |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      | 0.17 |      |
| Unguided              |      |         |      |    |      |      |      |      |      |      |      |      |
| Combined              |      |         |      |    |      |      |      |      |      |      | 0.07 |      |
| Spotted Ratfish       |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      |      |      |
| Unguided              |      |         |      |    |      |      |      |      |      |      |      |      |
| Combined              |      |         |      |    |      |      |      |      |      |      |      |      |
| Starry Flounder       |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      |      |      |
| Unguided              |      |         |      |    |      |      |      |      |      |      |      |      |
| Combined              |      |         |      |    |      |      |      |      |      |      |      |      |
| Unknown Groundfish    |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      |      | 0.14 |
| Unguided              |      |         |      |    |      |      |      |      |      |      | 0.02 |      |
| Combined              |      |         |      |    |      |      |      |      |      |      | 0.01 | 0.12 |

**Average number of groundfish harvested per boat trip in August 2023, by trip type and subarea. Zeros hidden.**

|                       |      | Subarea |      |    |      |      |      |      |      |      |      |      |
|-----------------------|------|---------|------|----|------|------|------|------|------|------|------|------|
| Species / Trip Type   | 3I   | 3J      | 3K   | 4A | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| Arrow-tooth Flounder  |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      | 0.02 | 0.03 |      |      |      |
| Unguided              |      |         |      |    | 0.01 |      |      |      |      |      |      |      |
| Combined              |      |         |      |    | 0.01 |      |      |      | 0.01 |      |      |      |
| Big Skate             |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      |      |      |
| Unguided              |      |         |      |    |      |      | 0.02 |      |      |      |      |      |
| Combined              |      |         |      |    |      |      | 0.02 |      |      |      |      |      |
| Cabezon               |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      | 0.08 |      |      |
| Unguided              |      |         |      |    |      |      |      |      |      | 0.11 |      |      |
| Combined              |      |         |      |    |      |      |      |      |      | 0.09 |      |      |
| English Sole          |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      |      |      |
| Unguided              |      |         |      |    |      |      |      |      |      |      |      |      |
| Combined              |      |         |      |    |      |      |      |      |      |      |      |      |
| Greenlings            |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      | 0.02 |      |
| Unguided              |      |         |      |    |      |      |      |      | 0.01 | 0.11 |      |      |
| Combined              |      |         |      |    |      |      |      |      | 0.01 | 0.06 | 0.01 |      |
| Lingcod               |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                | 0.69 | 1.19    | 0.31 |    |      | 0.80 | 1.80 | 1.42 | 1.44 | 1.08 | 1.65 | 1.11 |
| Unguided              | 1.14 | 1.76    | 0.08 |    | 0.24 | 0.13 | 0.26 | 0.23 | 0.60 | 0.68 | 0.26 |      |
| Combined              | 0.97 | 1.54    | 0.18 |    | 0.22 | 0.20 | 0.39 | 0.20 | 0.85 | 0.84 | 1.10 | 0.67 |
| Longnose Skate        |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      |      |      |
| Unguided              |      |         |      |    |      |      |      |      |      |      |      |      |
| Combined              |      |         |      |    |      |      |      |      |      |      |      |      |
| Pacific Cod           |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                | 0.15 |         | 0.12 |    | 0.40 |      |      |      |      | 0.15 | 0.15 |      |
| Unguided              | 0.14 |         | 0.03 |    | 0.04 | 0.03 |      |      | 0.22 | 0.37 | 0.23 |      |
| Combined              | 0.14 |         | 0.06 |    | 0.06 | 0.03 |      |      | 0.15 | 0.28 | 0.18 |      |
| Pacific Halibut       |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                | 2.46 | 4.88    | 0.92 |    | 2.60 | 1.10 | 1.40 | 2.53 | 2.78 | 0.92 | 1.78 | 2.89 |
| Unguided              | 1.32 | 2.60    | 0.64 |    | 0.29 | 0.95 | 0.86 | 1.62 | 1.05 | 1.47 | 0.92 | 2.83 |
| Combined              | 1.74 | 3.49    | 0.76 |    | 0.42 | 0.97 | 0.90 | 1.53 | 1.55 | 1.25 | 1.44 | 2.87 |
| Pacific Rock Sole     |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      |      |      |
| Unguided              |      |         |      |    |      |      |      |      |      |      |      |      |
| Combined              |      |         |      |    |      |      |      |      |      |      |      |      |
| Pacific Spiny Dogfish |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      |      |      |
| Unguided              |      |         |      |    |      |      |      |      |      |      |      |      |
| Combined              |      |         |      |    |      |      |      |      |      |      |      |      |
| Sablefish             |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      | 0.03 |      |
| Unguided              |      |         |      |    |      |      |      |      |      |      |      |      |
| Combined              |      |         |      |    |      |      |      |      |      |      | 0.02 |      |
| Spotted Ratfish       |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      |      |      |
| Unguided              |      |         |      |    |      |      |      |      |      |      |      |      |
| Combined              |      |         |      |    |      |      |      |      |      |      |      |      |
| Starry Flounder       |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      |      |      |
| Unguided              |      |         |      |    |      |      |      |      |      |      |      |      |
| Combined              |      |         |      |    |      |      |      |      |      |      |      |      |
| Unknown Groundfish    |      |         |      |    |      |      |      |      |      |      |      |      |
| Guided                |      |         |      |    |      |      |      |      |      |      |      |      |
| Unguided              |      |         |      |    |      |      |      |      | 0.02 |      |      |      |
| Combined              |      |         |      |    |      |      |      |      | 0.02 |      |      |      |



**Average number of rockfish harvested per boat trip in May 2023 by trip type and subarea. Zeros hidden.**

|                     | Subarea |      |      |    |      |      |      |      |      |      |      |      |
|---------------------|---------|------|------|----|------|------|------|------|------|------|------|------|
| Species / Trip Type | 3I      | 3J   | 3K   | 4A | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| Black Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              | 1.33    | 2.00 | 1.33 |    |      |      | 0.18 | 0.15 | 0.15 |      |      |      |
| Unguided            |         | 0.75 |      |    | 0.24 | 0.13 | 0.21 | 1.33 | 0.06 |      |      |      |
| Combined            |         | 1.29 |      |    | 0.22 | 0.11 | 0.20 | 1.07 | 0.09 |      |      |      |
| Boccacio Rockfish   |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Canary Rockfish     |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      | 0.01 |      |      |      |
| Combined            |         |      |      |    |      |      |      |      | 0.01 |      |      |      |
| China Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      | 0.02 | 0.02 |      |      |      |
| Unguided            |         |      |      |    | 0.01 |      |      | 0.08 | 0.06 |      |      |      |
| Combined            |         |      |      |    | 0.01 |      |      | 0.07 | 0.05 |      |      |      |
| Copper Rockfish     |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      | 0.02 | 0.02 |      |      |      |
| Unguided            |         | 0.13 |      |    | 0.01 |      |      |      | 0.02 | 0.29 |      | 0.15 |
| Combined            |         | 0.07 |      |    | 0.01 |      |      |      | 0.02 | 0.25 |      | 0.10 |
| Dusky Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Quillback Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              | 0.22    | 0.17 | 0.22 |    |      | 0.08 | 0.09 | 0.20 | 0.24 | 0.67 | 0.80 | 0.67 |
| Unguided            |         |      |      |    | 0.10 | 0.05 | 0.25 | 0.08 | 0.09 | 0.07 |      | 0.04 |
| Combined            | 0.11    | 0.07 |      |    | 0.09 | 0.06 | 0.20 | 0.07 | 0.14 | 0.06 | 0.36 | 0.26 |
| Redband Rockfish    |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Redstripe Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Tiger Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      | 0.07 | 0.10 | 0.07 |
| Unguided            |         |      |      |    |      |      |      |      | 0.02 |      |      |      |
| Combined            |         |      |      |    |      |      |      |      | 0.01 |      | 0.05 | 0.02 |
| Widow Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      | 0.20 | 0.30 | 0.20 |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      | 0.14 | 0.07 |
| Yelloweye Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Yellowtail Rockfish |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              | 0.22    |      | 0.22 |    |      |      |      |      |      |      |      |      |
| Unguided            |         | 0.13 |      |    | 0.06 |      | 0.04 |      |      |      |      |      |
| Combined            | 0.22    | 0.07 |      |    | 0.05 |      | 0.03 |      |      |      |      |      |
| Unknown rockfish    |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      | 0.20 | 0.26 |      |      |      |
| Unguided            | 0.33    | 0.25 |      |    | 0.03 |      | 0.08 | 0.25 | 0.12 |      | 0.08 | 0.04 |
| Combined            | 0.22    | 0.14 |      |    | 0.03 |      | 0.06 | 0.20 | 0.16 |      | 0.05 | 0.02 |

**Average number of rockfish harvested per boat trip in June 2023, by trip type and subarea. Zeros hidden.**

|                     | Subarea |      |      |    |      |      |      |      |      |      |      |      |
|---------------------|---------|------|------|----|------|------|------|------|------|------|------|------|
| Species / Trip Type | 3I      | 3J   | 3K   | 4A | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| Black Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              | 0.76    | 1.07 | 0.76 |    | 0.04 | 0.29 |      | 0.27 | 0.33 | 0.22 |      | 0.38 |
| Unguided            |         |      |      |    | 0.05 | 0.07 | 0.08 | 2.40 | 0.35 | 0.18 | 0.14 |      |
| Combined            | 0.06    | 0.44 |      |    | 0.05 | 0.10 | 0.06 | 1.71 | 0.34 | 0.19 | 0.10 | 0.29 |
| Boccacio Rockfish   |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Canary Rockfish     |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| China Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              | 0.05    | 0.07 | 0.05 |    |      |      |      |      |      | 0.03 |      | 0.06 |
| Unguided            |         |      |      |    |      |      | 0.03 |      | 0.01 |      |      |      |
| Combined            |         | 0.03 |      |    |      |      | 0.02 |      | 0.01 |      |      | 0.05 |
| Copper Rockfish     |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              | 0.14    | 0.21 | 0.14 |    |      |      | 0.06 | 0.11 | 0.11 |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      | 0.02 | 0.12 | 0.03 | 0.20 |
| Combined            |         | 0.09 |      |    |      |      | 0.02 | 0.14 | 0.06 | 0.10 | 0.02 | 0.05 |
| Dusky Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Quillback Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              | 0.48    | 0.57 | 0.48 |    | 0.05 | 0.35 | 0.13 | 0.21 | 0.23 | 0.69 | 1.25 | 0.44 |
| Unguided            |         | 0.10 |      |    | 0.04 | 0.06 | 0.17 |      | 0.07 | 0.12 |      | 0.60 |
| Combined            |         | 0.29 | 0.14 |    | 0.03 | 0.10 | 0.15 |      | 0.14 | 0.10 | 0.37 | 0.48 |
| Redband Rockfish    |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      | 0.03 |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      | 0.05 |      |      |
| Redstripe Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Tiger Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              | 0.05    | 0.07 | 0.05 |    |      |      |      | 0.03 | 0.04 | 0.06 |      | 0.13 |
| Unguided            |         |      |      |    |      | 0.01 |      |      | 0.01 | 0.12 |      |      |
| Combined            |         | 0.03 |      |    |      | 0.01 |      |      | 0.02 | 0.10 |      | 0.10 |
| Widow Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      | 0.06 |      | 0.13 |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      | 0.10 |
| Yelloweye Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      | 0.06 |      |      |
| Combined            |         |      |      |    |      |      |      |      |      | 0.05 |      |      |
| Yellowtail Rockfish |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      | 0.13 | 0.03 | 0.01 |      |      |      |
| Unguided            |         |      |      |    | 0.04 |      | 0.03 |      |      |      |      |      |
| Combined            |         |      |      |    | 0.03 |      | 0.06 |      | 0.01 |      |      |      |
| Unknown rockfish    |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      | 0.13 | 0.09 | 0.08 |      |      |      |
| Unguided            |         | 0.05 |      |    | 0.02 | 0.09 | 0.03 |      | 0.27 | 0.12 | 0.07 | 0.20 |
| Combined            |         | 0.03 |      |    | 0.02 | 0.08 | 0.06 |      | 0.19 | 0.10 | 0.05 | 0.05 |

**Average number of rockfish harvested per boat trip in July 2023, by trip type and subarea. Zeros hidden.**

|                     | Subarea |      |      |    |      |      |      |      |      |      |      |      |
|---------------------|---------|------|------|----|------|------|------|------|------|------|------|------|
| Species / Trip Type | 3I      | 3J   | 3K   | 4A | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| Black Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         | 0.91 |      |    | 0.05 | 0.03 | 0.03 | 0.24 | 0.30 |      | 0.09 | 1.29 |
| Unguided            | 0.64    | 0.07 |      |    | 0.11 | 0.10 | 0.13 |      | 0.21 |      | 0.14 | 2.40 |
| Combined            | 0.46    | 0.53 |      |    | 0.17 | 0.09 | 0.11 |      | 0.25 |      | 0.12 | 1.50 |
| Boccacio Rockfish   |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      | 0.18 |    |      | 0.01 |      |      |      |      |      |      |
| Combined            |         |      | 0.08 |    |      | 0.01 |      |      |      |      |      |      |
| Canary Rockfish     |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| China Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         | 0.12 |      |    |      |      |      | 0.01 | 0.02 |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      | 0.01 | 0.03 |      |      |
| Combined            |         | 0.07 |      |    |      |      |      |      | 0.01 | 0.02 |      |      |
| Copper Rockfish     |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              | 0.09    | 0.03 |      |    |      |      |      | 0.01 | 0.02 |      | 0.20 |      |
| Unguided            |         |      |      |    | 0.05 | 0.01 |      |      | 0.02 |      |      |      |
| Combined            | 0.03    | 0.02 |      |    | 0.05 | 0.01 |      |      | 0.02 |      | 0.08 |      |
| Dusky Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Quillback Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         | 0.30 |      |    | 0.01 | 0.03 | 0.09 | 0.12 | 0.13 | 0.27 | 0.03 | 0.29 |
| Unguided            |         |      |      |    | 0.04 | 0.11 | 0.25 |      | 0.11 | 0.29 | 0.02 |      |
| Combined            |         | 0.17 |      |    | 0.04 | 0.10 | 0.21 |      | 0.12 | 0.29 | 0.02 | 0.23 |
| Redband Rockfish    |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Redstripe Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Tiger Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         | 0.09 |      |    |      |      |      | 0.02 | 0.02 |      |      |      |
| Unguided            |         |      |      |    |      | 0.01 |      |      | 0.01 |      |      |      |
| Combined            |         | 0.05 |      |    |      | 0.01 |      |      | 0.02 |      |      |      |
| Widow Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    | 0.01 |      |      |      |      |      |      |      |
| Combined            |         |      |      |    | 0.01 |      |      |      |      |      |      |      |
| Yelloweye Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      | 0.04 | 0.05 |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      | 0.02 |      |      |      |
| Yellowtail Rockfish |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         | 0.06 |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         | 0.04 | 0.18 |    | 0.01 | 0.05 | 0.02 |      |      | 0.03 | 0.10 |      |
| Combined            |         | 0.05 | 0.08 |    | 0.01 | 0.04 | 0.02 |      |      | 0.02 | 0.06 |      |
| Unknown rockfish    |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         | 0.21 | 0.23 |    | 0.02 | 0.06 | 0.15 | 0.23 | 0.20 | 0.18 |      | 0.29 |
| Unguided            | 0.14    | 0.07 |      |    | 0.04 | 0.03 | 0.14 |      | 0.19 | 0.06 | 0.06 |      |
| Combined            | 0.10    | 0.15 | 0.13 |    | 0.04 | 0.03 | 0.14 | 0.54 | 0.19 | 0.10 | 0.04 | 0.23 |

**Average number of rockfish harvested per boat trip in August 2023, by trip type and subarea. Zeros hidden.**

| Species / Trip Type | Subarea |      |      |    |      |      |      |      |      |      |      |      |
|---------------------|---------|------|------|----|------|------|------|------|------|------|------|------|
|                     | 3I      | 3J   | 3K   | 4A | 4B   | 4C   | 4D   | 4E   | 4F   | 4G   | 4H   | 4L   |
| Black Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              | 0.62    | 1.31 | 0.08 |    |      |      | 1.40 | 0.86 | 0.83 | 0.54 | 0.35 | 0.67 |
| Unguided            | 0.23    | 0.80 |      |    | 0.26 | 0.11 | 0.11 | 0.38 | 0.08 |      | 0.08 |      |
| Combined            | 0.37    | 1.00 | 0.03 |    | 0.25 | 0.10 | 0.21 | 0.33 | 0.30 | 0.22 | 0.24 | 0.40 |
| Boccacio Rockfish   |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         | 0.25 |      |    |      |      | 0.20 | 0.02 |      | 0.08 |      |      |
| Unguided            |         |      |      |    |      | 0.01 |      |      |      |      |      |      |
| Combined            |         | 0.10 |      |    |      | 0.01 | 0.02 |      |      | 0.03 |      |      |
| Canary Rockfish     |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      | 0.03 |      |
| Combined            |         |      |      |    |      |      |      |      |      |      | 0.01 |      |
| China Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      | 0.20 | 0.02 |      | 0.15 |      | 0.44 |
| Unguided            | 0.05    |      |      |    | 0.04 | 0.02 |      |      | 0.01 |      | 0.03 |      |
| Combined            | 0.03    |      |      |    | 0.04 | 0.02 | 0.02 |      | 0.01 | 0.06 | 0.01 | 0.27 |
| Copper Rockfish     |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      | 0.10 |      |      |      | 0.08 | 0.02 |      |
| Unguided            |         | 0.08 |      |    | 0.05 |      |      | 0.08 |      |      | 0.08 |      |
| Combined            |         | 0.05 |      |    | 0.05 | 0.01 |      | 0.07 |      | 0.03 | 0.04 |      |
| Dusky Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Quillback Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              | 0.15    | 0.13 | 0.19 |    | 0.40 | 0.50 | 0.40 | 0.26 | 0.25 | 0.08 | 0.42 | 0.67 |
| Unguided            | 0.23    | 0.04 | 0.06 |    | 0.08 | 0.10 | 0.09 | 0.15 | 0.10 | 0.26 | 0.18 | 0.17 |
| Combined            | 0.20    | 0.07 | 0.11 |    | 0.09 | 0.14 | 0.11 | 0.13 | 0.15 | 0.19 | 0.32 | 0.47 |
| Redband Rockfish    |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Redstripe Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         |      |      |    |      |      |      |      |      |      |      |      |
| Tiger Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         | 0.06 |      |    |      |      |      | 0.05 | 0.03 |      | 0.02 |      |
| Unguided            | 0.05    |      |      |    |      |      |      |      | 0.06 |      |      |      |
| Combined            | 0.03    | 0.02 |      |    |      |      |      | 0.07 | 0.05 |      | 0.01 |      |
| Widow Rockfish      |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      | 0.04 |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    | 0.01 | 0.01 | 0.02 |      |      |      |      |      |
| Combined            |         |      | 0.02 |    | 0.01 | 0.01 | 0.02 |      |      |      |      |      |
| Yelloweye Rockfish  |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         | 0.13 |      |    |      |      |      |      |      |      |      |      |
| Unguided            |         |      |      |    |      |      |      |      |      |      |      |      |
| Combined            |         | 0.05 |      |    |      |      |      |      |      |      |      |      |
| Yellowtail Rockfish |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              |         |      |      |    |      |      |      | 0.07 | 0.08 |      |      | 0.11 |
| Unguided            |         |      | 0.03 |    |      | 0.02 | 0.02 |      | 0.07 |      | 0.03 |      |
| Combined            |         |      | 0.02 |    |      | 0.02 | 0.02 |      | 0.07 |      | 0.01 | 0.07 |
| Unknown rockfish    |         |      |      |    |      |      |      |      |      |      |      |      |
| Guided              | 0.46    | 0.06 |      |    |      |      |      | 0.05 | 0.06 |      | 0.27 | 0.44 |
| Unguided            | 0.09    |      |      |    | 0.03 | 0.15 | 0.19 |      | 0.06 |      |      | 0.33 |
| Combined            | 0.23    | 0.02 |      |    | 0.02 | 0.13 | 0.18 |      | 0.06 |      | 0.16 | 0.40 |

### APPENDIX 3: COUNTS OF ACTIVELY FISHING SPORT VESSELS IN EACH SUBAREA, BY DATE

Counts of actively fishing sport vessels in each subarea, by date, 2023, along with numbers that were estimated to be lodge boats. Weekend and holiday flights are shaded.

| Month  | Date        | Subarea |    |    |    |    |    |    |    |    |    |    |    | Area 3      |
|--------|-------------|---------|----|----|----|----|----|----|----|----|----|----|----|-------------|
|        |             | 3I      | 3J | 3K | 4A | 4B | 4C | 4D | 4E | 4F | 4G | 4H | 4L | Lodge Boats |
| May    | 4 May 2023  | 2       | 1  | 2  | 1  | 3  | 1  | 2  | 0  | 1  | 0  | 0  | 0  | 0           |
|        | 11 May 2023 | 0       | 0  | 3  | 1  | 3  | 2  | 0  | 0  | 2  | 0  | 2  | 0  | 0           |
|        | 13 May 2023 | 0       | 1  | 0  | 1  | 3  | 3  | 1  | 0  | 10 | 0  | 1  | 0  | 0           |
|        | 17 May 2023 | 1       | 7  | 1  | 0  | 3  | 11 | 2  | 0  | 7  | 5  | 0  | 0  | 4           |
|        | 21 May 2023 | 2       | 1  | 5  | 1  | 7  | 11 | 1  | 0  | 2  | 4  | 8  | 0  | 7           |
|        | 26 May 2023 | 1       | 1  | 3  | 2  | 0  | 3  | 2  | 0  | 4  | 0  | 1  | 0  | 7           |
|        | 28 May 2023 | 0       | 5  | na | 1  | 3  | 0  | 6  | 0  | 5  | 0  | 1  | 0  | 7           |
|        | 30 May 2023 | 0       | 1  | 6  | 7  | 4  | 3  | 8  | 0  | 9  | 0  | 0  | 0  | 8           |
| June   | 4 Jun 2023  | 4       | 5  | 5  | 1  | 4  | 6  | 5  | 0  | 7  | 9  | 2  | 0  | 11          |
|        | 7 Jun 2023  | 1       | 8  | 9  | 0  | 3  | 11 | 6  | 0  | 8  | 15 | 1  | 0  | 12          |
|        | 11 Jun 2023 | 7       | 12 | 10 | 2  | 7  | 11 | 10 | 0  | 33 | 4  | 7  | 0  | 10          |
|        | 13 Jun 2023 | 1       | 1  | 7  | 1  | 0  | 3  | 3  | 0  | 0  | 1  | 2  | 0  | 11          |
|        | 17 Jun 2023 | 1       | 3  | 12 | 2  | 2  | 5  | 3  | 0  | 19 | 1  | 0  | 0  | 11          |
|        | 26 Jun 2023 | 5       | 2  | 1  | 0  | 2  | 6  | 2  | 0  | 6  | 8  | 1  | 0  | 0           |
|        | 27 Jun 2023 | 2       | 4  | 1  | 1  | 2  | 9  | 1  | 0  | 13 | 0  | 3  | 0  | 6           |
|        | 29 Jun 2023 | 0       | 0  | 4  | 1  | 2  | 8  | 3  | 0  | 10 | 6  | 6  | 6  | 6           |
| July   | 3 Jul 2023  | 9       | 5  | 2  | 2  | 9  | 15 | 3  | 0  | 6  | 2  | 10 | 0  | 4           |
|        | 11 Jul 2023 | 2       | 9  | 1  | 2  | 6  | 10 | 4  | 1  | 17 | 2  | 6  | 0  | 11          |
|        | 13 Jul 2023 | 6       | 14 | 5  | 5  | 2  | 10 | 1  | 0  | 17 | 7  | 4  | 0  | 11          |
|        | 16 Jul 2023 | 3       | 9  | 7  | 2  | 3  | 24 | 4  | 0  | 25 | 3  | 9  | 0  | 11          |
|        | 19 Jul 2023 | 4       | 0  | 9  | 3  | 5  | 14 | 8  | 1  | 10 | 6  | 0  | na | 11          |
|        | 27 Jul 2023 | 10      | 6  | 1  | 1  | 3  | 7  | 9  | 0  | 5  | 3  | 5  | na | 11          |
|        | 30 Jul 2023 | 17      | 7  | 9  | 4  | 4  | 20 | 1  | 5  | 10 | 5  | 10 | 0  | 11          |
|        | 31 Jul 2023 | 4       | 6  | 7  | 7  | 10 | 20 | 4  | 1  | 15 | 5  | 6  | 0  | 4           |
| August | 2 Aug 2023  | 1       | 15 | 10 | 1  | 0  | 9  | 3  | 2  | 4  | 4  | 10 | na | 11          |
|        | 5 Aug 2023  | 12      | 18 | 10 | 3  | 9  | 31 | 12 | 0  | 30 | 11 | 19 | 0  | 13          |
|        | 10 Aug 2023 | 15      | 2  | 11 | 1  | 4  | 14 | 7  | 0  | 11 | 4  | 23 | 0  | 11          |
|        | 19 Aug 2023 | 15      | 6  | 20 | 5  | 6  | 11 | 3  | 2  | 7  | 4  | 24 | na | 11          |
|        | 23 Aug 2023 | 3       | 1  | 11 | 0  | 1  | 7  | 7  | 2  | 4  | 2  | 1  | 0  | 6           |
|        | 25 Aug 2023 | 17      | 3  | 5  | 6  | 3  | 8  | 3  | 8  | 1  | 6  | 9  | 0  | 4           |
|        | 26 Aug 2023 | 15      | 7  | 4  | 2  | 4  | 13 | 1  | 2  | 7  | 12 | 8  | 0  | 7           |
|        | 30 Aug 2023 | 2       | 5  | 6  | 0  | 3  | 5  | 2  | 0  | 4  | 2  | 4  | 0  | 7           |

## APPENDIX 4: RETAINED AND RELEASED FISH (LODGE REPORTS AND NON-LODGE CREEL ESTIMATES) BY MONTH, SUBAREA, AND SPECIES.

### Estimated numbers of Chinook Salmon retained by lodge and non-lodge anglers, by month, 2023.

|                    | Subarea |       |       |     |       |       |       |      |       |       |       |       |       |
|--------------------|---------|-------|-------|-----|-------|-------|-------|------|-------|-------|-------|-------|-------|
| Month / Catch Type | 3I      | 3J    | 3K    | 4A  | 4B    | 4C    | 4D    | 4E   | 4F    | 4G    | 4H    | 4L    | TOTAL |
| May                |         |       |       |     |       |       |       |      |       |       |       |       |       |
| Creel Estimate     | 18.4    | 17.9  | 34.1  | 2.4 | 138.2 | 184.9 | 128.8 | 6.0  | 439.5 | 33.8  | 35.9  | 1.8   | 1,042 |
| Lodge Report       | 39.0    | 37.8  | 72.2  | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0     | 0     | 149   |
| MAY RETAINED       | 57.4    | 55.7  | 106.3 | 2.4 | 138.2 | 184.9 | 128.8 | 6.0  | 439.5 | 33.8  | 35.9  | 1.8   | 1,191 |
| SE                 | 7.6     | 7.5   | 32.2  | 1.3 | 19.4  | 27.4  | 25.8  | 4.3  | 30.9  | 12.6  | 12.5  | 0.5   | 65    |
| June               |         |       |       |     |       |       |       |      |       |       |       |       |       |
| Creel Estimate     | 162.8   | 150.0 | 100.1 | 0.3 | 70.2  | 291.8 | 390.3 | 5.0  | 994.9 | 256.0 | 193.0 | 113.8 | 2,728 |
| Lodge Report       | 178.2   | 164.2 | 109.6 | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0     | 0     | 452   |
| JUNE RETAINED      | 341.0   | 314.1 | 209.7 | 0.3 | 70.2  | 291.8 | 390.3 | 5.0  | 994.9 | 256.0 | 193.0 | 113.8 | 3,180 |
| SE                 | 37.9    | 27.5  | 25.8  | 0.2 | 14.3  | 49.8  | 45.1  | 3.8  | 64.4  | 85.2  | 40.1  | 31.6  | 147   |
| July               |         |       |       |     |       |       |       |      |       |       |       |       |       |
| Creel Estimate     | 82.1    | 51.5  | 131.3 | 0   | 60.1  | 155.6 | 73.3  | 11.6 | 225.9 | 20.0  | 110.1 | 14.0  | 935   |
| Lodge Report       | 68.1    | 42.8  | 109.0 | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0     | 0     | 220   |
| JULY RETAINED      | 150.2   | 94.3  | 240.4 | 0   | 60.1  | 155.6 | 73.3  | 11.6 | 225.9 | 20.0  | 110.1 | 14.0  | 1,155 |
| SE                 | 19.8    | 12.6  | 19.3  | 0   | 14.2  | 22.9  | 11.5  | 4.4  | 24.2  | 7.4   | 28.0  | 4.3   | 57    |
| August             |         |       |       |     |       |       |       |      |       |       |       |       |       |
| Creel Estimate     | 80.3    | 62.0  | 113.1 | 0.9 | 18.4  | 31.2  | 85.6  | 7.3  | 223.9 | 77.1  | 105.1 | 0     | 805   |
| Lodge Report       | 35.2    | 27.2  | 49.6  | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0     | 0     | 112   |
| AUGUST RETAINED    | 115.5   | 89.2  | 162.6 | 0.9 | 18.4  | 31.2  | 85.6  | 7.3  | 223.9 | 77.1  | 105.1 | 0     | 917   |
| SE                 | 25.3    | 20.6  | 23.8  | 0.6 | 4.6   | 11.5  | 17.1  | 6.1  | 30.1  | 19.9  | 24.4  | 0     | 63    |
| TOTAL              |         |       |       |     |       |       |       |      |       |       |       |       |       |
| Creel Estimate     | 344     | 281   | 379   | 4   | 287   | 664   | 678   | 30   | 1,884 | 387   | 444   | 130   | 5,510 |
| Lodge Report       | 321     | 272   | 340   | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0     | 0     | 933   |
| TOTAL RETAINED     | 664     | 553   | 719   | 4   | 287   | 664   | 678   | 30   | 1,884 | 387   | 444   | 130   | 6,443 |
| SE                 | 50      | 37    | 51    | 1   | 28    | 62    | 56    | 10   | 81    | 89    | 56    | 32    | 182   |

### Estimated numbers of Chinook Salmon released by lodge and non-lodge anglers, by month, 2023.

|                    | Subarea |       |       |     |       |       |       |      |        |       |       |       |       |
|--------------------|---------|-------|-------|-----|-------|-------|-------|------|--------|-------|-------|-------|-------|
| Month / Catch Type | 3I      | 3J    | 3K    | 4A  | 4B    | 4C    | 4D    | 4E   | 4F     | 4G    | 4H    | 4L    | TOTAL |
| May                |         |       |       |     |       |       |       |      |        |       |       |       |       |
| Creel Estimate     | 12.3    | 14.3  | 0     | 3.1 | 80.2  | 152.0 | 141.7 | 1.0  | 362.8  | 15.0  | 68.2  | 2.2   | 853   |
| Lodge Report       | 58.7    | 68.3  | 0     | 0   | 0     | 0     | 0     | 0    | 0      | 0     | 0     | 0     | 127   |
| MAY RELEASED       | 71.0    | 82.6  | 0     | 3.1 | 80.2  | 152.0 | 141.7 | 1.0  | 362.8  | 15.0  | 68.2  | 2.2   | 980   |
| SE                 | 8.9     | 12.6  | 0     | 1.9 | 21.2  | 29.8  | 33.2  | 0.7  | 35.8   | 7.5   | 26.8  | 0.8   | 69    |
| June               |         |       |       |     |       |       |       |      |        |       |       |       |       |
| Creel Estimate     | 69.8    | 139.2 | 229.7 | 0.3 | 39.8  | 368.6 | 261.5 | 24.0 | 1358.3 | 286.2 | 214.5 | 239.2 | 3,231 |
| Lodge Report       | 88.9    | 177.4 | 292.7 | 0   | 0     | 0     | 0     | 0    | 0      | 0     | 0     | 0     | 559   |
| JUNE RELEASED      | 158.7   | 316.7 | 522.3 | 0.3 | 39.8  | 368.6 | 261.5 | 24.0 | 1358.3 | 286.2 | 214.5 | 239.2 | 3,790 |
| SE                 | 15.9    | 26.7  | 78.5  | 0.2 | 9.5   | 66.6  | 45.6  | 15.2 | 122.0  | 135.1 | 53.6  | 60.1  | 232   |
| July               |         |       |       |     |       |       |       |      |        |       |       |       |       |
| Creel Estimate     | 133.3   | 177.5 | 255.9 | 0   | 123.4 | 589.5 | 151.7 | 3.9  | 616.6  | 219.6 | 330.3 | 12.0  | 2,614 |
| Lodge Report       | 188.5   | 250.9 | 361.6 | 0   | 0     | 0     | 0     | 0    | 0      | 0     | 0     | 0     | 801   |
| JULY RELEASED      | 321.8   | 428.4 | 617.5 | 0   | 123.4 | 589.5 | 151.7 | 3.9  | 616.6  | 219.6 | 330.3 | 12.0  | 3,415 |
| SE                 | 37.9    | 47.5  | 38.5  | 0   | 35.2  | 76.8  | 26.3  | 2.8  | 69.1   | 55.0  | 49.9  | 4.7   | 153   |
| August             |         |       |       |     |       |       |       |      |        |       |       |       |       |
| Creel Estimate     | 0       | 78.9  | 41.1  | 3.6 | 18.4  | 81.2  | 50.6  | 87.3 | 64.0   | 25.7  | 74.2  | 0     | 525   |
| Lodge Report       | 0       | 67.1  | 34.9  | 0   | 0     | 0     | 0     | 0    | 0      | 0     | 0     | 0     | 102   |
| AUGUST RELEASED    | 0       | 146.0 | 76.0  | 3.6 | 18.4  | 81.2  | 50.6  | 87.3 | 64.0   | 25.7  | 74.2  | 0     | 627   |
| SE                 | 0       | 29.5  | 12.5  | 1.6 | 5.1   | 27.1  | 18.1  | 42.1 | 25.1   | 10.2  | 23.7  | 0     | 72    |
| TOTAL              |         |       |       |     |       |       |       |      |        |       |       |       |       |
| Creel Estimate     | 215     | 410   | 527   | 7   | 262   | 1,191 | 606   | 116  | 2,402  | 547   | 687   | 253   | 7,223 |
| Lodge Report       | 336     | 564   | 689   | 0   | 0     | 0     | 0     | 0    | 0      | 0     | 0     | 0     | 1,589 |
| TOTAL RELEASED     | 551     | 974   | 1,216 | 7   | 262   | 1,191 | 606   | 116  | 2,402  | 547   | 687   | 253   | 8,812 |
| SE                 | 42      | 63    | 88    | 2   | 42    | 109   | 65    | 45   | 147    | 146   | 82    | 60    | 295   |

**Estimated numbers of Coho Salmon retained by lodge and non-lodge anglers, by month, 2023.**

|                    | Subarea |        |        |    |       |        |        |       |        |        |        |       |        |
|--------------------|---------|--------|--------|----|-------|--------|--------|-------|--------|--------|--------|-------|--------|
| Month / Catch Type | 3I      | 3J     | 3K     | 4A | 4B    | 4C     | 4D     | 4E    | 4F     | 4G     | 4H     | 4L    | TOTAL  |
| May                |         |        |        |    |       |        |        |       |        |        |        |       |        |
| Creel Estimate     | 0       | 0      | 0      | 0  | 0     | 0      | 0      | 0     | 1.7    | 0      | 0      | 0     | 2      |
| Lodge Report       | 0       | 0      | 0      | 0  | 0     | 0      | 0      | 0     | 0      | 0      | 0      | 0     | 0      |
| MAY RETAINED       | 0       | 0      | 0      | 0  | 0     | 0      | 0      | 0     | 1.7    | 0      | 0      | 0     | 2      |
| SE                 | 0       | 0      | 0      | 0  | 0     | 0      | 0      | 0     | 1.3    | 0      | 0      | 0     | 1      |
| June               |         |        |        |    |       |        |        |       |        |        |        |       |        |
| Creel Estimate     | 55.8    | 93.7   | 123.7  | 0  | 16.4  | 272.6  | 56.3   | 27.0  | 1265.9 | 105.4  | 85.8   | 315.1 | 2,418  |
| Lodge Report       | 28.1    | 47.2   | 62.3   | 0  | 0     | 0      | 0      | 0     | 0      | 0      | 0      | 0     | 138    |
| JUNE RETAINED      | 83.9    | 140.9  | 186.0  | 0  | 16.4  | 272.6  | 56.3   | 27.0  | 1265.9 | 105.4  | 85.8   | 315.1 | 2,555  |
| SE                 | 40.4    | 25.3   | 58.6   | 0  | 10.1  | 69.5   | 18.6   | 12.2  | 103.8  | 74.6   | 29.0   | 47.3  | 175    |
| July               |         |        |        |    |       |        |        |       |        |        |        |       |        |
| Creel Estimate     | 1102.6  | 552.6  | 1302.9 | 0  | 601.2 | 3131.9 | 1618.4 | 77.2  | 4673.4 | 958.4  | 1734.3 | 117.0 | 15,870 |
| Lodge Report       | 555.4   | 278.4  | 656.3  | 0  | 0     | 0      | 0      | 0     | 0      | 0      | 0      | 0     | 1,490  |
| JULY RETAINED      | 1658.0  | 831.0  | 1959.2 | 0  | 601.2 | 3131.9 | 1618.4 | 77.2  | 4673.4 | 958.4  | 1734.3 | 117.0 | 17,360 |
| SE                 | 150.2   | 77.0   | 97.9   | 0  | 74.7  | 205.6  | 85.6   | 41.5  | 161.7  | 154.6  | 150.4  | 21.5  | 410    |
| August             |         |        |        |    |       |        |        |       |        |        |        |       |        |
| Creel Estimate     | 2891.5  | 1043.2 | 3268.7 | 0  | 417.8 | 1385.9 | 1237.9 | 334.7 | 2161.9 | 1380.0 | 4824.1 | 34.0  | 18,980 |
| Lodge Report       | 1456.4  | 525.5  | 1646.5 | 0  | 0     | 0      | 0      | 0     | 0      | 0      | 0      | 0     | 3,628  |
| AUGUST RETAINED    | 4347.9  | 1568.7 | 4915.2 | 0  | 417.8 | 1385.9 | 1237.9 | 334.7 | 2161.9 | 1380.0 | 4824.1 | 34.0  | 22,608 |
| SE                 | 255.2   | 135.8  | 208.7  | 0  | 44.0  | 159.7  | 111.7  | 154.1 | 123.6  | 205.8  | 311.6  | 16.5  | 588    |
| TOTAL              |         |        |        |    |       |        |        |       |        |        |        |       |        |
| Creel Estimate     | 4,050   | 1,690  | 4,695  | 0  | 1,035 | 4,790  | 2,913  | 439   | 8,103  | 2,444  | 6,644  | 466   | 37,269 |
| Lodge Report       | 2,040   | 851    | 2,365  | 0  | 0     | 0      | 0      | 0     | 0      | 0      | 0      | 0     | 5,256  |
| TOTAL RETAINED     | 6,090   | 2,541  | 7,060  | 0  | 1,035 | 4,790  | 2,913  | 439   | 8,103  | 2,444  | 6,644  | 466   | 42,525 |
| SE                 | 299     | 158    | 238    | 0  | 87    | 270    | 142    | 160   | 228    | 268    | 347    | 55    | 738    |

**Estimated numbers of Coho Salmon released by lodge and non-lodge anglers, by month, 2023.**

|                    | Subarea |       |       |     |       |       |       |      |       |       |        |      |       |
|--------------------|---------|-------|-------|-----|-------|-------|-------|------|-------|-------|--------|------|-------|
| Month / Catch Type | 3I      | 3J    | 3K    | 4A  | 4B    | 4C    | 4D    | 4E   | 4F    | 4G    | 4H     | 4L   | TOTAL |
| May                |         |       |       |     |       |       |       |      |       |       |        |      |       |
| Creel Estimate     | 3.1     | 0     | 0     | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0      | 0    | 3     |
| Lodge Report       | 0       | 0     | 0     | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0      | 0    | 0     |
| MAY RELEASED       | 3.1     | 0     | 0     | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0      | 0    | 3     |
| SE                 | 2.9     | 0     | 0     | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0      | 0    | 3     |
| June               |         |       |       |     |       |       |       |      |       |       |        |      |       |
| Creel Estimate     | 0       | 5.4   | 0     | 0   | 9.4   | 15.4  | 0     | 0    | 199.6 | 0     | 14.3   | 11.7 | 256   |
| Lodge Report       | 0       | 0     | 0     | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0      | 0    | 0     |
| JUNE RELEASED      | 0       | 5.4   | 0     | 0   | 9.4   | 15.4  | 0     | 0    | 199.6 | 0     | 14.3   | 11.7 | 256   |
| SE                 | 0       | 3.8   | 0     | 0   | 6.6   | 7.4   | 0     | 0    | 39.6  | 0     | 11.0   | 9.8  | 44    |
| July               |         |       |       |     |       |       |       |      |       |       |        |      |       |
| Creel Estimate     | 677.0   | 134.6 | 589.2 | 1.6 | 180.4 | 393.0 | 561.7 | 0    | 614.4 | 294.5 | 350.0  | 4.0  | 3,800 |
| Lodge Report       | 0       | 0     | 0     | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0      | 0    | 0     |
| JULY RELEASED      | 677.0   | 134.6 | 589.2 | 1.6 | 180.4 | 393.0 | 561.7 | 0    | 614.4 | 294.5 | 350.0  | 4.0  | 3,800 |
| SE                 | 155.8   | 38.6  | 127.0 | 1.3 | 46.3  | 65.5  | 67.5  | 0    | 65.2  | 124.5 | 74.7   | 2.9  | 280   |
| August             |         |       |       |     |       |       |       |      |       |       |        |      |       |
| Creel Estimate     | 722.9   | 507.5 | 827.5 | 0.9 | 32.8  | 212.3 | 210.2 | 36.4 | 390.2 | 865.7 | 1663.7 | 0    | 5,470 |
| Lodge Report       | 0       | 0     | 0     | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0      | 0    | 0     |
| AUGUST RELEASED    | 722.9   | 507.5 | 827.5 | 0.9 | 32.8  | 212.3 | 210.2 | 36.4 | 390.2 | 865.7 | 1663.7 | 0    | 5,470 |
| SE                 | 186.6   | 139.4 | 120.2 | 0.6 | 9.4   | 59.0  | 68.5  | 30.5 | 69.0  | 246.5 | 338.3  | 0    | 508   |
| TOTAL              |         |       |       |     |       |       |       |      |       |       |        |      |       |
| Creel Estimate     | 1,403   | 647   | 1,417 | 3   | 222   | 621   | 772   | 36   | 1,204 | 1,160 | 2,028  | 16   | 9,529 |
| Lodge Report       | 0       | 0     | 0     | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0      | 0    | 0     |
| TOTAL RELEASED     | 1,403   | 647   | 1,417 | 3   | 222   | 621   | 772   | 36   | 1,204 | 1,160 | 2,028  | 16   | 9,529 |
| SE                 | 243     | 145   | 175   | 1   | 48    | 88    | 96    | 31   | 103   | 276   | 347    | 10   | 581   |

**Estimated numbers of Pink Salmon retained by lodge and non-lodge anglers, by month, 2023.**

|                    | Subarea |      |       |     |      |       |       |      |       |      |       |      |       |
|--------------------|---------|------|-------|-----|------|-------|-------|------|-------|------|-------|------|-------|
| Month / Catch Type | 3I      | 3J   | 3K    | 4A  | 4B   | 4C    | 4D    | 4E   | 4F    | 4G   | 4H    | 4L   | TOTAL |
| May                |         |      |       |     |      |       |       |      |       |      |       |      |       |
| Creel Estimate     | 0       | 0    | 0     | 0   | 0    | 0     | 0     | 0    | 0     | 0    | 7.2   | 0.2  | 7     |
| Lodge Report       | 0       | 0    | 0     | 0   | 0    | 0     | 0     | 0    | 0     | 0    | 0     | 0    | 0     |
| MAY RETAINED       | 0       | 0    | 0     | 0   | 0    | 0     | 0     | 0    | 0     | 0    | 7.2   | 0.2  | 7     |
| SE                 | 0       | 0    | 0     | 0   | 0    | 0     | 0     | 0    | 0     | 0    | 5.6   | 0.2  | 6     |
| June               |         |      |       |     |      |       |       |      |       |      |       |      |       |
| Creel Estimate     | 0       | 2.7  | 0     | 0   | 0    | 3.8   | 0     | 0    | 35.7  | 0    | 0     | 17.5 | 60    |
| Lodge Report       | 0       | 0.3  | 0     | 0   | 0    | 0     | 0     | 0    | 0     | 0    | 0     | 0    | 0     |
| JUNE RETAINED      | 0       | 2.9  | 0     | 0   | 0    | 3.8   | 0     | 0    | 35.7  | 0    | 0     | 17.5 | 60    |
| SE                 | 0       | 1.9  | 0     | 0   | 0    | 3.0   | 0     | 0    | 11.5  | 0    | 0     | 7.2  | 14    |
| July               |         |      |       |     |      |       |       |      |       |      |       |      |       |
| Creel Estimate     | 107.7   | 31.5 | 104.4 | 0   | 66.4 | 262.0 | 126.7 | 7.7  | 395.3 | 64.9 | 110.1 | 0    | 1,277 |
| Lodge Report       | 10.5    | 3.1  | 10.2  | 0   | 0    | 0     | 0     | 0    | 0     | 0    | 0     | 0    | 24    |
| JULY RETAINED      | 118.2   | 34.6 | 114.6 | 0   | 66.4 | 262.0 | 126.7 | 7.7  | 395.3 | 64.9 | 110.1 | 0    | 1,301 |
| SE                 | 47.7    | 12.3 | 40.1  | 0   | 12.3 | 40.7  | 21.2  | 3.8  | 44.1  | 22.9 | 32.5  | 0    | 99    |
| August             |         |      |       |     |      |       |       |      |       |      |       |      |       |
| Creel Estimate     | 80.3    | 62.0 | 30.8  | 0.9 | 51.2 | 287.2 | 85.6  | 43.7 | 278.2 | 51.4 | 191.7 | 0    | 1,163 |
| Lodge Report       | 7.9     | 6.1  | 3.0   | 0   | 0    | 0     | 0     | 0    | 0     | 0    | 0     | 0    | 17    |
| AUGUST RETAINED    | 88.2    | 68.1 | 33.9  | 0.9 | 51.2 | 287.2 | 85.6  | 43.7 | 278.2 | 51.4 | 191.7 | 0    | 1,180 |
| SE                 | 29.2    | 26.2 | 11.5  | 0.6 | 13.2 | 64.8  | 24.2  | 36.6 | 40.9  | 30.7 | 49.7  | 0    | 114   |
| TOTAL              |         |      |       |     |      |       |       |      |       |      |       |      |       |
| Creel Estimate     | 188     | 96   | 135   | 1   | 118  | 553   | 212   | 51   | 709   | 116  | 309   | 18   | 2,507 |
| Lodge Report       | 18      | 9    | 13    | 0   | 0    | 0     | 0     | 0    | 0     | 0    | 0     | 0    | 41    |
| TOTAL RETAINED     | 206     | 106  | 148   | 1   | 118  | 553   | 212   | 51   | 709   | 116  | 309   | 18   | 2,548 |
| SE                 | 56      | 29   | 42    | 1   | 18   | 77    | 32    | 37   | 61    | 38   | 60    | 7    | 152   |

**Estimated numbers of Pink Salmon released by lodge and non-lodge anglers, by month, 2023.**

|                    | Subarea |       |        |     |       |        |        |       |        |        |        |      |        |
|--------------------|---------|-------|--------|-----|-------|--------|--------|-------|--------|--------|--------|------|--------|
| Month / Catch Type | 3I      | 3J    | 3K     | 4A  | 4B    | 4C     | 4D     | 4E    | 4F     | 4G     | 4H     | 4L   | TOTAL  |
| May                |         |       |        |     |       |        |        |       |        |        |        |      |        |
| Creel Estimate     | 0       | 0     | 0      | 0   | 0     | 0      | 0      | 0     | 0      | 0      | 0      | 0    | 0      |
| Lodge Report       | 0       | 0     | 0      | 0   | 0     | 0      | 0      | 0     | 0      | 0      | 0      | 0    | 0      |
| MAY RELEASED       | 0       | 0     | 0      | 0   | 0     | 0      | 0      | 0     | 0      | 0      | 0      | 0    | 0      |
| SE                 | 0       | 0     | 0      | 0   | 0     | 0      | 0      | 0     | 0      | 0      | 0      | 0    | 0      |
| June               |         |       |        |     |       |        |        |       |        |        |        |      |        |
| Creel Estimate     | 4.7     | 5.4   | 0      | 0   | 0     | 15.4   | 0      | 0     | 0      | 0      | 0      | 17.5 | 43     |
| Lodge Report       | 0       | 0     | 0      | 0   | 0     | 0      | 0      | 0     | 0      | 0      | 0      | 0    | 0      |
| JUNE RELEASED      | 4.7     | 5.4   | 0      | 0   | 0     | 15.4   | 0      | 0     | 0      | 0      | 0      | 17.5 | 43     |
| SE                 | 3.4     | 3.8   | 0      | 0   | 0     | 8.5    | 0      | 0     | 0      | 0      | 0      | 14.7 | 18     |
| July               |         |       |        |     |       |        |        |       |        |        |        |      |        |
| Creel Estimate     | 702.6   | 489.6 | 700.3  | 0   | 496.8 | 917.0  | 775.0  | 38.6  | 1664.7 | 344.4  | 424.7  | 4.0  | 6,558  |
| Lodge Report       | 0       | 0     | 0      | 0   | 0     | 0      | 0      | 0     | 0      | 0      | 0      | 0    | 0      |
| JULY RELEASED      | 702.6   | 489.6 | 700.3  | 0   | 496.8 | 917.0  | 775.0  | 38.6  | 1664.7 | 344.4  | 424.7  | 4.0  | 6,558  |
| SE                 | 168.8   | 100.6 | 128.7  | 0   | 79.7  | 112.5  | 86.3   | 28.0  | 148.4  | 104.6  | 74.4   | 2.9  | 348    |
| August             |         |       |        |     |       |        |        |       |        |        |        |      |        |
| Creel Estimate     | 1506.0  | 507.5 | 1937.6 | 2.7 | 561.2 | 1123.7 | 1790.6 | 269.2 | 2900.6 | 1045.7 | 1843.0 | 2.0  | 13,490 |
| Lodge Report       | 0       | 0     | 0      | 0   | 0     | 0      | 0      | 0     | 0      | 0      | 0      | 0    | 0      |
| AUGUST RELEASED    | 1506.0  | 507.5 | 1937.6 | 2.7 | 561.2 | 1123.7 | 1790.6 | 269.2 | 2900.6 | 1045.7 | 1843.0 | 2.0  | 13,490 |
| SE                 | 329.4   | 136.8 | 249.7  | 1.4 | 157.1 | 239.8  | 241.7  | 138.1 | 756.9  | 248.4  | 328.9  | 1.4  | 1,045  |
| TOTAL              |         |       |        |     |       |        |        |       |        |        |        |      |        |
| Creel Estimate     | 2,213   | 1,003 | 2,638  | 3   | 1,058 | 2,056  | 2,566  | 308   | 4,565  | 1,390  | 2,268  | 24   | 20,091 |
| Lodge Report       | 0       | 0     | 0      | 0   | 0     | 0      | 0      | 0     | 0      | 0      | 0      | 0    | 0      |
| TOTAL RELEASED     | 2,213   | 1,003 | 2,638  | 3   | 1,058 | 2,056  | 2,566  | 308   | 4,565  | 1,390  | 2,268  | 24   | 20,091 |
| SE                 | 370     | 170   | 281    | 1   | 176   | 265    | 257    | 141   | 771    | 270    | 337    | 15   | 1,102  |



**Estimated numbers of Chum Salmon retained by lodge and non-lodge anglers, by month, 2023.**

|                    | Subarea |    |      |    |    |      |     |      |      |      |      |    |       |
|--------------------|---------|----|------|----|----|------|-----|------|------|------|------|----|-------|
| Month / Catch Type | 3I      | 3J | 3K   | 4A | 4B | 4C   | 4D  | 4E   | 4F   | 4G   | 4H   | 4L | TOTAL |
| May                |         |    |      |    |    |      |     |      |      |      |      |    |       |
| Creel Estimate     | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0    | 0    | 0    | 0    | 0  | 0     |
| Lodge Report       | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0    | 0    | 0    | 0    | 0  | 0     |
| MAY RETAINED       | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0    | 0    | 0    | 0    | 0  | 0     |
| SE                 | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0    | 0    | 0    | 0    | 0  | 0     |
| June               |         |    |      |    |    |      |     |      |      |      |      |    |       |
| Creel Estimate     | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0    | 8.9  | 15.1 | 0    | 0  | 24    |
| Lodge Report       | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0    | 0    | 0    | 0    | 0  | 0     |
| JUNE RETAINED      | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0    | 8.9  | 15.1 | 0    | 0  | 24    |
| SE                 | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0    | 6.3  | 12.4 | 0    | 0  | 14    |
| July               |         |    |      |    |    |      |     |      |      |      |      |    |       |
| Creel Estimate     | 0       | 0  | 0    | 0  | 0  | 57.3 | 3.3 | 0    | 11.3 | 0    | 0    | 0  | 72    |
| Lodge Report       | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0    | 0    | 0    | 0    | 0  | 0     |
| JULY RETAINED      | 0       | 0  | 0    | 0  | 0  | 57.3 | 3.3 | 0    | 11.3 | 0    | 0    | 0  | 72    |
| SE                 | 0       | 0  | 0    | 0  | 0  | 40.6 | 2.7 | 0    | 3.7  | 0    | 0    | 0  | 41    |
| August             |         |    |      |    |    |      |     |      |      |      |      |    |       |
| Creel Estimate     | 10.0    | 0  | 61.7 | 0  | 0  | 0    | 0   | 72.8 | 0    | 0    | 18.6 | 0  | 163   |
| Lodge Report       | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0    | 0    | 0    | 0    | 0  | 0     |
| AUGUST RETAINED    | 10.0    | 0  | 61.7 | 0  | 0  | 0    | 0   | 72.8 | 0    | 0    | 18.6 | 0  | 163   |
| SE                 | 7.2     | 0  | 44.7 | 0  | 0  | 0    | 0   | 61.0 | 0    | 0    | 9.8  | 0  | 77    |
| TOTAL              |         |    |      |    |    |      |     |      |      |      |      |    |       |
| Creel Estimate     | 10      | 0  | 62   | 0  | 0  | 57   | 3   | 73   | 20   | 15   | 19   | 0  | 259   |
| Lodge Report       | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0    | 0    | 0    | 0    | 0  | 0     |
| TOTAL RETAINED     | 10      | 0  | 62   | 0  | 0  | 57   | 3   | 73   | 20   | 15   | 19   | 0  | 259   |
| SE                 | 7       | 0  | 45   | 0  | 0  | 41   | 3   | 61   | 7    | 12   | 10   | 0  | 88    |

**Estimated numbers of Chum Salmon released by lodge and non-lodge anglers, by month, 2023.**

|                    | Subarea |    |      |    |    |      |     |    |      |    |      |    |       |
|--------------------|---------|----|------|----|----|------|-----|----|------|----|------|----|-------|
| Month / Catch Type | 3I      | 3J | 3K   | 4A | 4B | 4C   | 4D  | 4E | 4F   | 4G | 4H   | 4L | TOTAL |
| May                |         |    |      |    |    |      |     |    |      |    |      |    |       |
| Creel Estimate     | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0  | 0    | 0  | 0    | 0  | 0     |
| Lodge Report       | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0  | 0    | 0  | 0    | 0  | 0     |
| MAY RELEASED       | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0  | 0    | 0  | 0    | 0  | 0     |
| SE                 | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0  | 0    | 0  | 0    | 0  | 0     |
| June               |         |    |      |    |    |      |     |    |      |    |      |    |       |
| Creel Estimate     | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0  | 8.9  | 0  | 0    | 0  | 9     |
| Lodge Report       | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0  | 0    | 0  | 0    | 0  | 0     |
| JUNE RELEASED      | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0  | 8.9  | 0  | 0    | 0  | 9     |
| SE                 | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0  | 4.7  | 0  | 0    | 0  | 5     |
| July               |         |    |      |    |    |      |     |    |      |    |      |    |       |
| Creel Estimate     | 0       | 0  | 30.3 | 0  | 0  | 28.7 | 3.3 | 0  | 29.4 | 0  | 7.9  | 0  | 100   |
| Lodge Report       | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0  | 0    | 0  | 0    | 0  | 0     |
| JULY RELEASED      | 0       | 0  | 30.3 | 0  | 0  | 28.7 | 3.3 | 0  | 29.4 | 0  | 7.9  | 0  | 100   |
| SE                 | 0       | 0  | 15.1 | 0  | 0  | 15.6 | 2.7 | 0  | 16.7 | 0  | 5.6  | 0  | 28    |
| August             |         |    |      |    |    |      |     |    |      |    |      |    |       |
| Creel Estimate     | 0       | 0  | 15.4 | 0  | 0  | 49.9 | 0   | 0  | 0    | 0  | 18.6 | 0  | 84    |
| Lodge Report       | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0  | 0    | 0  | 0    | 0  | 0     |
| AUGUST RELEASED    | 0       | 0  | 15.4 | 0  | 0  | 49.9 | 0   | 0  | 0    | 0  | 18.6 | 0  | 84    |
| SE                 | 0       | 0  | 8.3  | 0  | 0  | 35.4 | 0   | 0  | 0    | 0  | 9.8  | 0  | 38    |
| TOTAL              |         |    |      |    |    |      |     |    |      |    |      |    |       |
| Creel Estimate     | 0       | 0  | 46   | 0  | 0  | 79   | 3   | 0  | 38   | 0  | 26   | 0  | 192   |
| Lodge Report       | 0       | 0  | 0    | 0  | 0  | 0    | 0   | 0  | 0    | 0  | 0    | 0  | 0     |
| TOTAL RELEASED     | 0       | 0  | 46   | 0  | 0  | 79   | 3   | 0  | 38   | 0  | 26   | 0  | 192   |
| SE                 | 0       | 0  | 17   | 0  | 0  | 39   | 3   | 0  | 17   | 0  | 11   | 0  | 47    |

**Estimated numbers of Sockeye Salmon retained by lodge and non-lodge anglers, by month, 2023.**

| Month / Catch Type | Subarea |    |      |    |     |     |    |    |     |    |    |    | TOTAL |
|--------------------|---------|----|------|----|-----|-----|----|----|-----|----|----|----|-------|
|                    | 3I      | 3J | 3K   | 4A | 4B  | 4C  | 4D | 4E | 4F  | 4G | 4H | 4L |       |
| May                |         |    |      |    |     |     |    |    |     |    |    |    |       |
| Creel Estimate     | 0       | 0  | 0    | 0  | 0   | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0     |
| Lodge Report       | 0       | 0  | 0    | 0  | 0   | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0     |
| MAY RETAINED       | 0       | 0  | 0    | 0  | 0   | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0     |
| SE                 | 0       | 0  | 0    | 0  | 0   | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0     |
| June               |         |    |      |    |     |     |    |    |     |    |    |    |       |
| Creel Estimate     | 0       | 0  | 0    | 0  | 0   | 0   | 0  | 0  | 3.0 | 0  | 0  | 0  | 3     |
| Lodge Report       | 0       | 0  | 0    | 0  | 0   | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0     |
| JUNE RETAINED      | 0       | 0  | 0    | 0  | 0   | 0   | 0  | 0  | 3.0 | 0  | 0  | 0  | 3     |
| SE                 | 0       | 0  | 0    | 0  | 0   | 0   | 0  | 0  | 2.1 | 0  | 0  | 0  | 2     |
| July               |         |    |      |    |     |     |    |    |     |    |    |    |       |
| Creel Estimate     | 0       | 0  | 0    | 0  | 6.3 | 4.1 | 0  | 0  | 2.3 | 0  | 0  | 0  | 13    |
| Lodge Report       | 0       | 0  | 0    | 0  | 0   | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0     |
| JULY RETAINED      | 0       | 0  | 0    | 0  | 6.3 | 4.1 | 0  | 0  | 2.3 | 0  | 0  | 0  | 13    |
| SE                 | 0       | 0  | 0    | 0  | 4.6 | 2.9 | 0  | 0  | 1.6 | 0  | 0  | 0  | 6     |
| August             |         |    |      |    |     |     |    |    |     |    |    |    |       |
| Creel Estimate     | 0       | 0  | 15.4 | 0  | 0   | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 15    |
| Lodge Report       | 0       | 0  | 0    | 0  | 0   | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0     |
| AUGUST RETAINED    | 0       | 0  | 15.4 | 0  | 0   | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 15    |
| SE                 | 0       | 0  | 8.3  | 0  | 0   | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 8     |
| TOTAL              |         |    |      |    |     |     |    |    |     |    |    |    |       |
| Creel Estimate     | 0       | 0  | 15   | 0  | 6   | 4   | 0  | 0  | 5   | 0  | 0  | 0  | 31    |
| Lodge Report       | 0       | 0  | 0    | 0  | 0   | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0     |
| TOTAL RETAINED     | 0       | 0  | 15   | 0  | 6   | 4   | 0  | 0  | 5   | 0  | 0  | 0  | 31    |
| SE                 | 0       | 0  | 8    | 0  | 5   | 3   | 0  | 0  | 3   | 0  | 0  | 0  | 10    |

**Estimated numbers of Sockeye Salmon released by lodge and non-lodge anglers, by month, 2023.**

| Month / Catch Type | Subarea |    |     |    |    |     |    |    |    |    |     |    | TOTAL |
|--------------------|---------|----|-----|----|----|-----|----|----|----|----|-----|----|-------|
|                    | 3I      | 3J | 3K  | 4A | 4B | 4C  | 4D | 4E | 4F | 4G | 4H  | 4L |       |
| May                |         |    |     |    |    |     |    |    |    |    |     |    |       |
| Creel Estimate     | 0       | 0  | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| Lodge Report       | 0       | 0  | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| MAY RELEASED       | 0       | 0  | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| SE                 | 0       | 0  | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| June               |         |    |     |    |    |     |    |    |    |    |     |    |       |
| Creel Estimate     | 0       | 0  | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| Lodge Report       | 0       | 0  | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| JUNE RELEASED      | 0       | 0  | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| SE                 | 0       | 0  | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| July               |         |    |     |    |    |     |    |    |    |    |     |    |       |
| Creel Estimate     | 0       | 0  | 3.4 | 0  | 0  | 4.1 | 0  | 0  | 0  | 0  | 7.9 | 0  | 15    |
| Lodge Report       | 0       | 0  | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| JULY RELEASED      | 0       | 0  | 3.4 | 0  | 0  | 4.1 | 0  | 0  | 0  | 0  | 7.9 | 0  | 15    |
| SE                 | 0       | 0  | 2.4 | 0  | 0  | 2.9 | 0  | 0  | 0  | 0  | 5.6 | 0  | 7     |
| August             |         |    |     |    |    |     |    |    |    |    |     |    |       |
| Creel Estimate     | 0       | 0  | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| Lodge Report       | 0       | 0  | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| AUGUST RELEASED    | 0       | 0  | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| SE                 | 0       | 0  | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| TOTAL              |         |    |     |    |    |     |    |    |    |    |     |    |       |
| Creel Estimate     | 0       | 0  | 3   | 0  | 0  | 4   | 0  | 0  | 0  | 0  | 8   | 0  | 15    |
| Lodge Report       | 0       | 0  | 0   | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 0   | 0  | 0     |
| TOTAL RELEASED     | 0       | 0  | 3   | 0  | 0  | 4   | 0  | 0  | 0  | 0  | 8   | 0  | 15    |
| SE                 | 0       | 0  | 2   | 0  | 0  | 3   | 0  | 0  | 0  | 0  | 6   | 0  | 7     |

**Estimated numbers of unknown salmon retained by lodge and non-lodge anglers, by month, 2023.**

|                    | Subarea |    |    |    |    |    |    |    |     |    |      |    |       |
|--------------------|---------|----|----|----|----|----|----|----|-----|----|------|----|-------|
| Month / Catch Type | 3I      | 3J | 3K | 4A | 4B | 4C | 4D | 4E | 4F  | 4G | 4H   | 4L | TOTAL |
| May                |         |    |    |    |    |    |    |    |     |    |      |    |       |
| Creel Estimate     | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0    | 0  | 0     |
| Lodge Report       | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0    | 0  | 0     |
| MAY RETAINED       | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0    | 0  | 0     |
| SE                 | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0    | 0  | 0     |
| June               |         |    |    |    |    |    |    |    |     |    |      |    |       |
| Creel Estimate     | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0    | 0  | 0     |
| Lodge Report       | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0    | 0  | 0     |
| JUNE RETAINED      | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0    | 0  | 0     |
| SE                 | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0    | 0  | 0     |
| July               |         |    |    |    |    |    |    |    |     |    |      |    |       |
| Creel Estimate     | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 2.3 | 0  | 0    | 0  | 2     |
| Lodge Report       | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0    | 0  | 0     |
| JULY RETAINED      | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 2.3 | 0  | 0    | 0  | 2     |
| SE                 | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1.6 | 0  | 0    | 0  | 2     |
| August             |         |    |    |    |    |    |    |    |     |    |      |    |       |
| Creel Estimate     | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 80.4 | 0  | 80    |
| Lodge Report       | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0    | 0  | 0     |
| AUGUST RETAINED    | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 80.4 | 0  | 80    |
| SE                 | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 57.4 | 0  | 57    |
| TOTAL              |         |    |    |    |    |    |    |    |     |    |      |    |       |
| Creel Estimate     | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 2   | 0  | 80   | 0  | 83    |
| Lodge Report       | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0    | 0  | 0     |
| TOTAL RETAINED     | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 2   | 0  | 80   | 0  | 83    |
| SE                 | 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 2   | 0  | 57   | 0  | 57    |

**Estimated numbers of unknown salmon released by lodge and non-lodge anglers, by month, 2023.**

|                    | Subarea |      |      |    |    |     |      |    |     |      |      |    |       |
|--------------------|---------|------|------|----|----|-----|------|----|-----|------|------|----|-------|
| Month / Catch Type | 3I      | 3J   | 3K   | 4A | 4B | 4C  | 4D   | 4E | 4F  | 4G   | 4H   | 4L | TOTAL |
| May                |         |      |      |    |    |     |      |    |     |      |      |    |       |
| Creel Estimate     | 0       | 0    | 0    | 0  | 0  | 0   | 0    | 0  | 0   | 0    | 0    | 0  | 0     |
| Lodge Report       | 0       | 0    | 0    | 0  | 0  | 0   | 0    | 0  | 0   | 0    | 0    | 0  | 0     |
| MAY RELEASED       | 0       | 0    | 0    | 0  | 0  | 0   | 0    | 0  | 0   | 0    | 0    | 0  | 0     |
| SE                 | 0       | 0    | 0    | 0  | 0  | 0   | 0    | 0  | 0   | 0    | 0    | 0  | 0     |
| June               |         |      |      |    |    |     |      |    |     |      |      |    |       |
| Creel Estimate     | 0       | 0    | 0    | 0  | 0  | 0   | 0    | 0  | 0   | 0    | 0    | 0  | 0     |
| Lodge Report       | 0       | 0    | 0    | 0  | 0  | 0   | 0    | 0  | 0   | 0    | 0    | 0  | 0     |
| JUNE RELEASED      | 0       | 0    | 0    | 0  | 0  | 0   | 0    | 0  | 0   | 0    | 0    | 0  | 0     |
| SE                 | 0       | 0    | 0    | 0  | 0  | 0   | 0    | 0  | 0   | 0    | 0    | 0  | 0     |
| July               |         |      |      |    |    |     |      |    |     |      |      |    |       |
| Creel Estimate     | 0       | 0    | 0    | 0  | 0  | 8.2 | 0    | 0  | 4.5 | 10.0 | 11.8 | 0  | 34    |
| Lodge Report       | 0       | 0    | 0    | 0  | 0  | 0   | 0    | 0  | 0   | 0    | 0    | 0  | 0     |
| JULY RELEASED      | 0       | 0    | 0    | 0  | 0  | 8.2 | 0    | 0  | 4.5 | 10.0 | 11.8 | 0  | 34    |
| SE                 | 0       | 0    | 0    | 0  | 0  | 5.8 | 0    | 0  | 3.3 | 7.7  | 8.4  | 0  | 13    |
| August             |         |      |      |    |    |     |      |    |     |      |      |    |       |
| Creel Estimate     | 0       | 39.5 | 77.1 | 0  | 0  | 0   | 11.7 | 0  | 0   | 0    | 0    | 0  | 128   |
| Lodge Report       | 0       | 0    | 0    | 0  | 0  | 0   | 0    | 0  | 0   | 0    | 0    | 0  | 0     |
| AUGUST RELEASED    | 0       | 39.5 | 77.1 | 0  | 0  | 0   | 11.7 | 0  | 0   | 0    | 0    | 0  | 128   |
| SE                 | 0       | 27.9 | 55.9 | 0  | 0  | 0   | 8.7  | 0  | 0   | 0    | 0    | 0  | 63    |
| TOTAL              |         |      |      |    |    |     |      |    |     |      |      |    |       |
| Creel Estimate     | 0       | 39   | 77   | 0  | 0  | 8   | 12   | 0  | 5   | 10   | 12   | 0  | 163   |
| Lodge Report       | 0       | 0    | 0    | 0  | 0  | 0   | 0    | 0  | 0   | 0    | 0    | 0  | 0     |
| TOTAL RELEASED     | 0       | 39   | 77   | 0  | 0  | 8   | 12   | 0  | 5   | 10   | 12   | 0  | 163   |
| SE                 | 0       | 28   | 56   | 0  | 0  | 6   | 9    | 0  | 3   | 8    | 8    | 0  | 64    |

**Estimated numbers of Pacific Halibut retained by lodge and non-lodge anglers, by month, 2023.**

|                    | Subarea |        |       |     |       |       |       |       |        |       |       |       |        |
|--------------------|---------|--------|-------|-----|-------|-------|-------|-------|--------|-------|-------|-------|--------|
| Month / Catch Type | 3I      | 3J     | 3K    | 4A  | 4B    | 4C    | 4D    | 4E    | 4F     | 4G    | 4H    | 4L    | TOTAL  |
| May                |         |        |       |     |       |       |       |       |        |       |       |       |        |
| Creel Estimate     | 39.9    | 100.1  | 34.1  | 0.6 | 49.0  | 116.5 | 124.6 | 29.0  | 230.2  | 67.6  | 75.4  | 4.6   | 872    |
| Lodge Report       | 14.2    | 35.6   | 12.2  | 0   | 0     | 0     | 0     | 0     | 0      | 0     | 0     | 0     | 62     |
| MAY RETAINED       | 54.1    | 135.7  | 46.3  | 0.6 | 49.0  | 116.5 | 124.6 | 29.0  | 230.2  | 67.6  | 75.4  | 4.6   | 934    |
| SE                 | 14.6    | 23.6   | 22.6  | 0.5 | 14.7  | 18.9  | 32.1  | 7.3   | 24.8   | 16.0  | 21.1  | 0.8   | 65     |
| June               |         |        |       |     |       |       |       |       |        |       |       |       |        |
| Creel Estimate     | 69.8    | 112.5  | 47.1  | 0   | 44.5  | 326.4 | 84.5  | 17.0  | 744.7  | 512.1 | 153.7 | 180.9 | 2,293  |
| Lodge Report       | 104.3   | 168.2  | 70.5  | 0   | 0     | 0     | 0     | 0     | 0      | 0     | 0     | 0     | 343    |
| JUNE RETAINED      | 174.1   | 280.7  | 117.6 | 0   | 44.5  | 326.4 | 84.5  | 17.0  | 744.7  | 512.1 | 153.7 | 180.9 | 2,636  |
| SE                 | 23.8    | 18.3   | 20.4  | 0   | 8.4   | 43.0  | 21.2  | 4.0   | 51.0   | 85.0  | 26.4  | 17.4  | 121    |
| July               |         |        |       |     |       |       |       |       |        |       |       |       |        |
| Creel Estimate     | 343.6   | 604.2  | 138.0 | 0   | 129.7 | 761.5 | 206.7 | 96.5  | 1034.5 | 454.2 | 515.2 | 122.0 | 4,406  |
| Lodge Report       | 152.2   | 267.6  | 61.1  | 0   | 0     | 0     | 0     | 0     | 0      | 0     | 0     | 0     | 481    |
| JULY RETAINED      | 495.8   | 871.8  | 199.2 | 0   | 129.7 | 761.5 | 206.7 | 96.5  | 1034.5 | 454.2 | 515.2 | 122.0 | 4,887  |
| SE                 | 63.8    | 53.7   | 40.0  | 0   | 27.7  | 76.5  | 31.7  | 23.5  | 63.5   | 65.3  | 56.3  | 10.5  | 168    |
| August             |         |        |       |     |       |       |       |       |        |       |       |       |        |
| Creel Estimate     | 612.4   | 806.4  | 241.6 | 0   | 73.7  | 586.8 | 218.0 | 167.4 | 614.0  | 342.9 | 884.4 | 43.0  | 4,591  |
| Lodge Report       | 168.9   | 222.4  | 66.6  | 0   | 0     | 0     | 0     | 0     | 0      | 0     | 0     | 0     | 458    |
| AUGUST RETAINED    | 781.4   | 1028.8 | 308.2 | 0   | 73.7  | 586.8 | 218.0 | 167.4 | 614.0  | 342.9 | 884.4 | 43.0  | 5,049  |
| SE                 | 101.4   | 94.5   | 56.2  | 0   | 15.4  | 80.7  | 43.2  | 53.5  | 59.6   | 69.2  | 98.1  | 7.0   | 228    |
| TOTAL              |         |        |       |     |       |       |       |       |        |       |       |       |        |
| Creel Estimate     | 1,066   | 1,623  | 461   | 1   | 297   | 1,791 | 634   | 310   | 2,623  | 1,377 | 1,629 | 350   | 12,161 |
| Lodge Report       | 440     | 694    | 210   | 0   | 0     | 0     | 0     | 0     | 0      | 0     | 0     | 0     | 1,344  |
| TOTAL RETAINED     | 1,505   | 2,317  | 671   | 1   | 297   | 1,791 | 634   | 310   | 2,623  | 1,377 | 1,629 | 350   | 13,505 |
| SE                 | 123     | 113    | 75    | 1   | 36    | 121   | 66    | 59    | 104    | 129   | 118   | 21    | 315    |

**Estimated numbers of Pacific Halibut released by lodge and non-lodge anglers, by month, 2023.**

|                    | Subarea |       |       |    |      |       |       |       |       |       |       |       |       |
|--------------------|---------|-------|-------|----|------|-------|-------|-------|-------|-------|-------|-------|-------|
| Month / Catch Type | 3I      | 3J    | 3K    | 4A | 4B   | 4C    | 4D    | 4E    | 4F    | 4G    | 4H    | 4L    | TOTAL |
| May                |         |       |       |    |      |       |       |       |       |       |       |       |       |
| Creel Estimate     | 15.3    | 96.5  | 0     | 0  | 8.9  | 30.4  | 34.4  | 4.0   | 144.8 | 71.4  | 18.0  | 2.7   | 426   |
| Lodge Report       | 0       | 0     | 0     | 0  | 0    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| MAY RELEASED       | 15.3    | 96.5  | 0     | 0  | 8.9  | 30.4  | 34.4  | 4.0   | 144.8 | 71.4  | 18.0  | 2.7   | 426   |
| SE                 | 7.7     | 40.0  | 0     | 0  | 6.4  | 10.7  | 15.6  | 2.2   | 24.8  | 25.4  | 8.0   | 0.8   | 58    |
| June               |         |       |       |    |      |       |       |       |       |       |       |       |       |
| Creel Estimate     | 23.3    | 160.7 | 82.4  | 0  | 11.7 | 92.2  | 60.3  | 100.0 | 399.1 | 225.9 | 128.7 | 332.6 | 1,617 |
| Lodge Report       | 0       | 0     | 0     | 0  | 0    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| JUNE RELEASED      | 23.3    | 160.7 | 82.4  | 0  | 11.7 | 92.2  | 60.3  | 100.0 | 399.1 | 225.9 | 128.7 | 332.6 | 1,617 |
| SE                 | 13.7    | 55.3  | 50.9  | 0  | 4.9  | 27.0  | 29.7  | 76.9  | 69.2  | 124.3 | 45.0  | 72.3  | 202   |
| July               |         |       |       |    |      |       |       |       |       |       |       |       |       |
| Creel Estimate     | 138.5   | 463.9 | 30.3  | 0  | 60.1 | 110.5 | 58.3  | 73.3  | 370.4 | 144.8 | 357.9 | 118.0 | 1,926 |
| Lodge Report       | 0       | 0     | 0     | 0  | 0    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| JULY RELEASED      | 138.5   | 463.9 | 30.3  | 0  | 60.1 | 110.5 | 58.3  | 73.3  | 370.4 | 144.8 | 357.9 | 118.0 | 1,926 |
| SE                 | 56.8    | 78.7  | 13.9  | 0  | 21.2 | 22.7  | 17.6  | 32.6  | 45.6  | 36.1  | 76.9  | 25.7  | 148   |
| August             |         |       |       |    |      |       |       |       |       |       |       |       |       |
| Creel Estimate     | 20.1    | 394.7 | 159.3 | 0  | 16.4 | 174.8 | 120.7 | 50.9  | 342.2 | 111.4 | 358.7 | 14.0  | 1,763 |
| Lodge Report       | 0       | 0     | 0     | 0  | 0    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| AUGUST RELEASED    | 20.1    | 394.7 | 159.3 | 0  | 16.4 | 174.8 | 120.7 | 50.9  | 342.2 | 111.4 | 358.7 | 14.0  | 1,763 |
| SE                 | 14.3    | 67.0  | 42.8  | 0  | 5.3  | 40.3  | 42.8  | 25.1  | 74.2  | 40.7  | 86.6  | 4.5   | 159   |
| TOTAL              |         |       |       |    |      |       |       |       |       |       |       |       |       |
| Creel Estimate     | 197     | 1,116 | 272   | 0  | 97   | 408   | 274   | 228   | 1,257 | 553   | 863   | 467   | 5,732 |
| Lodge Report       | 0       | 0     | 0     | 0  | 0    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| TOTAL RELEASED     | 197     | 1,116 | 272   | 0  | 97   | 408   | 274   | 228   | 1,257 | 553   | 863   | 467   | 5,732 |
| SE                 | 61      | 124   | 68    | 0  | 23   | 55    | 57    | 87    | 114   | 138   | 124   | 77    | 302   |

**Estimated numbers of Lingcod retained by lodge and non-lodge anglers, by month, 2023.**

|                    | Subarea |       |      |     |      |       |       |      |       |       |       |       |       |
|--------------------|---------|-------|------|-----|------|-------|-------|------|-------|-------|-------|-------|-------|
| Month / Catch Type | 3I      | 3J    | 3K   | 4A  | 4B   | 4C    | 4D    | 4E   | 4F    | 4G    | 4H    | 4L    | TOTAL |
| May                |         |       |      |     |      |       |       |      |       |       |       |       |       |
| Creel Estimate     | 12.3    | 85.8  | 0    | 0.6 | 53.5 | 12.7  | 55.8  | 62.0 | 303.5 | 52.6  | 39.5  | 2.8   | 681   |
| Lodge Report       | 15.2    | 105.8 | 0    | 0   | 0    | 0     | 0     | 0    | 0     | 0     | 0     | 0     | 121   |
| MAY RETAINED       | 27.4    | 191.6 | 0    | 0.6 | 53.5 | 12.7  | 55.8  | 62.0 | 303.5 | 52.6  | 39.5  | 2.8   | 802   |
| SE                 | 8.9     | 30.9  | 0    | 0.5 | 17.9 | 6.2   | 19.4  | 11.0 | 40.3  | 20.8  | 18.7  | 0.8   | 66    |
| June               |         |       |      |     |      |       |       |      |       |       |       |       |       |
| Creel Estimate     | 41.9    | 168.7 | 11.8 | 0.3 | 9.4  | 142.1 | 40.2  | 14.0 | 592.8 | 677.8 | 96.5  | 110.9 | 1,906 |
| Lodge Report       | 83.0    | 334.6 | 23.4 | 0   | 0    | 0     | 0     | 0    | 0     | 0     | 0     | 0     | 441   |
| JUNE RETAINED      | 124.9   | 503.3 | 35.1 | 0.3 | 9.4  | 142.1 | 40.2  | 14.0 | 592.8 | 677.8 | 96.5  | 110.9 | 2,347 |
| SE                 | 22.0    | 41.2  | 8.5  | 0.2 | 4.7  | 50.2  | 16.1  | 9.0  | 67.0  | 185.6 | 24.8  | 32.7  | 214   |
| July               |         |       |      |     |      |       |       |      |       |       |       |       |       |
| Creel Estimate     | 148.7   | 352.2 | 43.8 | 0   | 3.2  | 257.9 | 138.3 | 34.7 | 702.5 | 279.5 | 338.2 | 54.0  | 2,353 |
| Lodge Report       | 173.1   | 409.9 | 50.9 | 0   | 0    | 0     | 0     | 0    | 0     | 0     | 0     | 0     | 634   |
| JULY RETAINED      | 321.8   | 762.1 | 94.7 | 0   | 3.2  | 257.9 | 138.3 | 34.7 | 702.5 | 279.5 | 338.2 | 54.0  | 2,987 |
| SE                 | 41.6    | 56.0  | 22.2 | 0   | 2.3  | 59.5  | 27.8  | 13.9 | 69.8  | 69.4  | 66.3  | 10.9  | 155   |
| August             |         |       |      |     |      |       |       |      |       |       |       |       |       |
| Creel Estimate     | 341.4   | 355.3 | 56.5 | 0   | 38.9 | 118.6 | 93.4  | 21.8 | 335.8 | 231.4 | 674.1 | 10.0  | 2,277 |
| Lodge Report       | 213.0   | 221.7 | 35.3 | 0   | 0    | 0     | 0     | 0    | 0     | 0     | 0     | 0     | 470   |
| AUGUST RETAINED    | 554.4   | 577.0 | 91.8 | 0   | 38.9 | 118.6 | 93.4  | 21.8 | 335.8 | 231.4 | 674.1 | 10.0  | 2,747 |
| SE                 | 88.0    | 78.8  | 28.4 | 0   | 22.5 | 34.8  | 32.2  | 9.8  | 52.3  | 57.9  | 126.2 | 3.4   | 199   |
| TOTAL              |         |       |      |     |      |       |       |      |       |       |       |       |       |
| Creel Estimate     | 544     | 962   | 112  | 1   | 105  | 531   | 328   | 133  | 1,935 | 1,241 | 1,148 | 178   | 7,218 |
| Lodge Report       | 484     | 1,072 | 110  | 0   | 0    | 0     | 0     | 0    | 0     | 0     | 0     | 0     | 1,666 |
| TOTAL RETAINED     | 1,029   | 2,034 | 222  | 1   | 105  | 531   | 328   | 133  | 1,935 | 1,241 | 1,148 | 178   | 8,884 |
| SE                 | 100     | 110   | 37   | 1   | 29   | 86    | 49    | 22   | 117   | 208   | 146   | 35    | 337   |

**Estimated numbers of Lingcod released by lodge and non-lodge anglers, by month, 2023.**

|                    | Subarea |       |     |    |     |      |      |     |      |      |      |     |       |
|--------------------|---------|-------|-----|----|-----|------|------|-----|------|------|------|-----|-------|
| Month / Catch Type | 3I      | 3J    | 3K  | 4A | 4B  | 4C   | 4D   | 4E  | 4F   | 4G   | 4H   | 4L  | TOTAL |
| May                |         |       |     |    |     |      |      |     |      |      |      |     |       |
| Creel Estimate     | 9.2     | 71.5  | 0   | 0  | 4.5 | 7.6  | 8.6  | 2.0 | 40.1 | 18.8 | 3.6  | 0.6 | 166   |
| Lodge Report       | 0       | 0     | 0   | 0  | 0   | 0    | 0    | 0   | 0    | 0    | 0    | 0   | 0     |
| MAY RELEASED       | 9.2     | 71.5  | 0   | 0  | 4.5 | 7.6  | 8.6  | 2.0 | 40.1 | 18.8 | 3.6  | 0.6 | 166   |
| SE                 | 8.7     | 39.5  | 0   | 0  | 3.2 | 3.2  | 6.5  | 1.4 | 11.9 | 11.1 | 2.8  | 0.3 | 44    |
| June               |         |       |     |    |     |      |      |     |      |      |      |     |       |
| Creel Estimate     | 0       | 26.8  | 0   | 0  | 2.3 | 11.5 | 0    | 0   | 35.7 | 30.1 | 0    | 0   | 107   |
| Lodge Report       | 0       | 0     | 0   | 0  | 0   | 0    | 0    | 0   | 0    | 0    | 0    | 0   | 0     |
| JUNE RELEASED      | 0       | 26.8  | 0   | 0  | 2.3 | 11.5 | 0    | 0   | 35.7 | 30.1 | 0    | 0   | 107   |
| SE                 | 0       | 17.1  | 0   | 0  | 1.7 | 6.7  | 0    | 0   | 10.6 | 17.0 | 0    | 0   | 27    |
| July               |         |       |     |    |     |      |      |     |      |      |      |     |       |
| Creel Estimate     | 0       | 100.2 | 3.4 | 0  | 0   | 4.1  | 26.7 | 0   | 79.1 | 5.0  | 74.7 | 3.0 | 296   |
| Lodge Report       | 0       | 0     | 0   | 0  | 0   | 0    | 0    | 0   | 0    | 0    | 0    | 0   | 0     |
| JULY RELEASED      | 0       | 100.2 | 3.4 | 0  | 0   | 4.1  | 26.7 | 0   | 79.1 | 5.0  | 74.7 | 3.0 | 296   |
| SE                 | 0       | 38.2  | 2.4 | 0  | 0   | 2.9  | 16.5 | 0   | 20.9 | 3.8  | 29.4 | 2.2 | 55    |
| August             |         |       |     |    |     |      |      |     |      |      |      |     |       |
| Creel Estimate     | 0       | 73.3  | 0   | 0  | 2.0 | 0    | 0    | 0   | 41.6 | 0    | 30.9 | 0   | 148   |
| Lodge Report       | 0       | 0     | 0   | 0  | 0   | 0    | 0    | 0   | 0    | 0    | 0    | 0   | 0     |
| AUGUST RELEASED    | 0       | 73.3  | 0   | 0  | 2.0 | 0    | 0    | 0   | 41.6 | 0    | 30.9 | 0   | 148   |
| SE                 | 0       | 32.5  | 0   | 0  | 1.5 | 0    | 0    | 0   | 15.1 | 0    | 18.2 | 0   | 40    |
| TOTAL              |         |       |     |    |     |      |      |     |      |      |      |     |       |
| Creel Estimate     | 9       | 272   | 3   | 0  | 9   | 23   | 35   | 2   | 196  | 54   | 109  | 4   | 717   |
| Lodge Report       | 0       | 0     | 0   | 0  | 0   | 0    | 0    | 0   | 0    | 0    | 0    | 0   | 0     |
| TOTAL RELEASED     | 9       | 272   | 3   | 0  | 9   | 23   | 35   | 2   | 196  | 54   | 109  | 4   | 717   |
| SE                 | 9       | 66    | 2   | 0  | 4   | 8    | 18   | 1   | 30   | 21   | 35   | 2   | 86    |

**Estimated numbers of other groundfish retained by lodge and non-lodge anglers, by month, 2023. Included are Big and Longnose skates, Pacific Spiny Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, greenlings, Spotted Ratfish, Arrow-tooth and Starry flounders, Pacific Sanddab, Pacific Rock and English soles, and other miscellaneous groundfishes.**

|                    | Subarea |      |      |    |      |      |      |     |      |       |       |      |       |
|--------------------|---------|------|------|----|------|------|------|-----|------|-------|-------|------|-------|
| Month / Catch Type | 3I      | 3J   | 3K   | 4A | 4B   | 4C   | 4D   | 4E  | 4F   | 4G    | 4H    | 4L   | TOTAL |
| May                |         |      |      |    |      |      |      |     |      |       |       |      |       |
| Creel Estimate     | 27.6    | 10.7 | 0    | 0  | 11.1 | 12.7 | 0    | 2.0 | 19.2 | 15.0  | 10.8  | 0.7  | 110   |
| Lodge Report       | 0       | 0    | 0    | 0  | 0    | 0    | 0    | 0   | 0    | 0     | 0     | 0    | 0     |
| MAY RETAINED       | 27.6    | 10.7 | 0    | 0  | 11.1 | 12.7 | 0    | 2.0 | 19.2 | 15.0  | 10.8  | 0.7  | 110   |
| SE                 | 18.6    | 9.5  | 0    | 0  | 5.7  | 5.6  | 0    | 1.4 | 5.5  | 11.0  | 6.1   | 0.3  | 26    |
| June               |         |      |      |    |      |      |      |     |      |       |       |      |       |
| Creel Estimate     | 0       | 8.0  | 11.8 | 0  | 2.3  | 23.0 | 32.2 | 0   | 29.8 | 105.4 | 71.5  | 17.5 | 302   |
| Lodge Report       | 0       | 0    | 0    | 0  | 0    | 0    | 0    | 0   | 0    | 0     | 0     | 0    | 0     |
| JUNE RETAINED      | 0       | 8.0  | 11.8 | 0  | 2.3  | 23.0 | 32.2 | 0   | 29.8 | 105.4 | 71.5  | 17.5 | 302   |
| SE                 | 0       | 4.2  | 5.8  | 0  | 1.7  | 12.8 | 10.7 | 0   | 9.9  | 75.2  | 33.2  | 6.8  | 85    |
| July               |         |      |      |    |      |      |      |     |      |       |       |      |       |
| Creel Estimate     | 20.5    | 37.2 | 6.7  | 0  | 22.1 | 61.4 | 16.7 | 0   | 83.6 | 79.9  | 169.1 | 7.0  | 504   |
| Lodge Report       | 0       | 0    | 0    | 0  | 0    | 0    | 0    | 0   | 0    | 0     | 0     | 0    | 0     |
| JULY RETAINED      | 20.5    | 37.2 | 6.7  | 0  | 22.1 | 61.4 | 16.7 | 0   | 83.6 | 79.9  | 169.1 | 7.0  | 504   |
| SE                 | 7.3     | 12.4 | 3.3  | 0  | 10.1 | 18.3 | 6.0  | 0   | 21.4 | 40.2  | 39.5  | 3.6  | 66    |
| August             |         |      |      |    |      |      |      |     |      |       |       |      |       |
| Creel Estimate     | 50.2    | 0    | 20.6 | 0  | 12.3 | 18.7 | 3.9  | 0   | 73.6 | 120.0 | 129.9 | 0    | 429   |
| Lodge Report       | 0       | 0    | 0    | 0  | 0    | 0    | 0    | 0   | 0    | 0     | 0     | 0    | 0     |
| AUGUST RETAINED    | 50.2    | 0    | 20.6 | 0  | 12.3 | 18.7 | 3.9  | 0   | 73.6 | 120.0 | 129.9 | 0    | 429   |
| SE                 | 20.9    | 0    | 9.0  | 0  | 4.1  | 9.8  | 2.9  | 0   | 30.8 | 47.3  | 32.5  | 0    | 70    |
| TOTAL              |         |      |      |    |      |      |      |     |      |       |       |      |       |
| Creel Estimate     | 98      | 56   | 39   | 0  | 48   | 116  | 53   | 2   | 206  | 320   | 381   | 25   | 1,345 |
| Lodge Report       | 0       | 0    | 0    | 0  | 0    | 0    | 0    | 0   | 0    | 0     | 0     | 0    | 0     |
| TOTAL RETAINED     | 98      | 56   | 39   | 0  | 48   | 116  | 53   | 2   | 206  | 320   | 381   | 25   | 1,345 |
| SE                 | 29      | 16   | 11   | 0  | 12   | 25   | 13   | 1   | 39   | 98    | 61    | 8    | 131   |

**Estimated numbers of other groundfish released by lodge and non-lodge anglers, by month, 2023. Included are Big and Longnose skates, Pacific Spiny Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, greenlings, Spotted Ratfish, Arrow-tooth and Starry flounders, Pacific Sanddab, Pacific Rock and English soles, and other miscellaneous groundfishes.**

|                    | Subarea |     |      |     |      |       |      |      |      |       |       |     |       |
|--------------------|---------|-----|------|-----|------|-------|------|------|------|-------|-------|-----|-------|
| Month / Catch Type | 3I      | 3J  | 3K   | 4A  | 4B   | 4C    | 4D   | 4E   | 4F   | 4G    | 4H    | 4L  | TOTAL |
| May                |         |     |      |     |      |       |      |      |      |       |       |     |       |
| Creel Estimate     | 6.1     | 7.1 | 0    | 3.1 | 11.1 | 5.1   | 12.9 | 0    | 29.7 | 3.8   | 10.8  | 0.7 | 90    |
| Lodge Report       | 0       | 0   | 0    | 0   | 0    | 0     | 0    | 0    | 0    | 0     | 0     | 0   | 0     |
| MAY RELEASED       | 6.1     | 7.1 | 0    | 3.1 | 11.1 | 5.1   | 12.9 | 0    | 29.7 | 3.8   | 10.8  | 0.7 | 90    |
| SE                 | 4.1     | 6.3 | 0    | 1.7 | 4.8  | 3.7   | 5.7  | 0    | 9.5  | 2.7   | 4.8   | 0.2 | 16    |
| June               |         |     |      |     |      |       |      |      |      |       |       |     |       |
| Creel Estimate     | 0       | 0   | 11.8 | 0   | 9.4  | 11.5  | 16.1 | 0    | 26.8 | 0     | 7.1   | 0   | 83    |
| Lodge Report       | 0       | 0   | 0    | 0   | 0    | 0     | 0    | 0    | 0    | 0     | 0     | 0   | 0     |
| JUNE RELEASED      | 0       | 0   | 11.8 | 0   | 9.4  | 11.5  | 16.1 | 0    | 26.8 | 0     | 7.1   | 0   | 83    |
| SE                 | 0       | 0   | 8.5  | 0   | 4.1  | 5.2   | 9.2  | 0    | 7.6  | 0     | 3.9   | 0   | 85    |
| July               |         |     |      |     |      |       |      |      |      |       |       |     |       |
| Creel Estimate     | 20.5    | 5.7 | 16.8 | 0   | 25.3 | 69.6  | 31.7 | 19.3 | 54.2 | 89.8  | 19.7  | 2.0 | 355   |
| Lodge Report       | 0       | 0   | 0    | 0   | 0    | 0     | 0    | 0    | 0    | 0     | 0     | 0   | 0     |
| JULY RELEASED      | 20.5    | 5.7 | 16.8 | 0   | 25.3 | 69.6  | 31.7 | 19.3 | 54.2 | 89.8  | 19.7  | 2.0 | 355   |
| SE                 | 9.1     | 4.1 | 6.3  | 0   | 12.3 | 18.1  | 14.1 | 11.5 | 10.7 | 48.4  | 8.4   | 1.0 | 59    |
| August             |         |     |      |     |      |       |      |      |      |       |       |     |       |
| Creel Estimate     | 90.4    | 0   | 15.4 | 0   | 24.6 | 199.8 | 3.9  | 0    | 44.8 | 120.0 | 259.8 | 4.0 | 763   |
| Lodge Report       | 0       | 0   | 0    | 0   | 0    | 0     | 0    | 0    | 0    | 0     | 0     | 0   | 0     |
| AUGUST RELEASED    | 90.4    | 0   | 15.4 | 0   | 24.6 | 199.8 | 3.9  | 0    | 44.8 | 120.0 | 259.8 | 4.0 | 763   |
| SE                 | 37.2    | 0   | 8.3  | 0   | 8.7  | 54.5  | 2.9  | 0    | 13.6 | 73.3  | 95.6  | 2.0 | 139   |
| TOTAL              |         |     |      |     |      |       |      |      |      |       |       |     |       |
| Creel Estimate     | 117     | 13  | 44   | 3   | 70   | 286   | 65   | 19   | 155  | 214   | 297   | 7   | 1,290 |
| Lodge Report       | 0       | 0   | 0    | 0   | 0    | 0     | 0    | 0    | 0    | 0     | 0     | 0   | 0     |
| TOTAL RELEASED     | 117     | 13  | 44   | 3   | 70   | 286   | 65   | 19   | 155  | 214   | 297   | 7   | 1,290 |
| SE                 | 39      | 8   | 13   | 2   | 16   | 58    | 18   | 12   | 21   | 88    | 96    | 2   | 152   |

**Estimated numbers of other groundfish retained by lodge and non-lodge anglers, by taxon, 2023.**

| Month / Catch Type | Taxon                   |              |         |                 |            |                   |                |                    |                      |                  |                   |           |                    |                    |                    | TOTAL |
|--------------------|-------------------------|--------------|---------|-----------------|------------|-------------------|----------------|--------------------|----------------------|------------------|-------------------|-----------|--------------------|--------------------|--------------------|-------|
|                    | Arrow-tooth<br>Flounder | Big<br>Skate | Cabezon | English<br>Sole | Greenlings | Longnose<br>Skate | Pacific<br>Cod | Pacific<br>Sanddab | Pacific Rock<br>Sole | Spiny<br>Dogfish | Red Irish<br>Lord | Sablefish | Spotted<br>Ratfish | Starry<br>Flounder | Unknown /<br>Other |       |
| May                |                         |              |         |                 |            |                   |                |                    |                      |                  |                   |           |                    |                    |                    |       |
| Creel Estimate     | 0                       | 5.1          | 0       | 0               | 5.2        | 0                 | 88.4           | 0                  | 6.7                  | 0                | 0                 | 0         | 0                  | 0                  | 4.5                | 110   |
| Lodge Report       | 0                       | 0            | 0       | 0               | 0          | 0                 | 0              | 0                  | 0                    | 0                | 0                 | 0         | 0                  | 0                  | 0                  | 0     |
| MAY RETAINED       | 0                       | 5.1          | 0       | 0               | 5.2        | 0                 | 88.4           | 0                  | 6.7                  | 0                | 0                 | 0         | 0                  | 0                  | 4.5                | 110   |
| SE                 | 0                       | 3.7          | 0       | 0               | 3.9        | 0                 | 25.0           | 0                  | 4.8                  | 0                | 0                 | 0         | 0                  | 0                  | 3.2                | 26    |
| June               |                         |              |         |                 |            |                   |                |                    |                      |                  |                   |           |                    |                    |                    |       |
| Creel Estimate     | 3.0                     | 0            | 8.9     | 11.9            | 18.2       | 0                 | 236.5          | 0                  | 0                    | 15.1             | 0                 | 0         | 0                  | 0                  | 8.0                | 302   |
| Lodge Report       | 0                       | 0            | 0       | 0               | 0          | 0                 | 0              | 0                  | 0                    | 0                | 0                 | 0         | 0                  | 0                  | 0                  | 0     |
| JUNE RETAINED      | 3.0                     | 0            | 8.9     | 11.9            | 18.2       | 0                 | 236.5          | 0                  | 0                    | 15.1             | 0                 | 0         | 0                  | 0                  | 8.0                | 302   |
| SE                 | 2.1                     | 0            | 3.9     | 8.4             | 6.2        | 0                 | 83.2           | 0                  | 0                    | 12.4             | 0                 | 0         | 0                  | 0                  | 5.8                | 85    |
| July               |                         |              |         |                 |            |                   |                |                    |                      |                  |                   |           |                    |                    |                    |       |
| Creel Estimate     | 23.8                    | 0            | 2.3     | 0               | 15.6       | 0                 | 416.7          | 0                  | 5.6                  | 9.6              | 0                 | 23.6      | 0                  | 0                  | 6.9                | 504   |
| Lodge Report       | 0                       | 0            | 0       | 0               | 0          | 0                 | 0              | 0                  | 0                    | 0                | 0                 | 0         | 0                  | 0                  | 0                  | 0     |
| JULY RETAINED      | 23.8                    | 0            | 2.3     | 0               | 15.6       | 0                 | 416.7          | 0                  | 5.6                  | 9.6              | 0                 | 23.6      | 0                  | 0                  | 6.9                | 504   |
| SE                 | 11.8                    | 0            | 1.6     | 0               | 5.2        | 0                 | 62.0           | 0                  | 3.1                  | 4.4              | 0                 | 16.8      | 0                  | 0                  | 3.5                | 66    |
| August             |                         |              |         |                 |            |                   |                |                    |                      |                  |                   |           |                    |                    |                    |       |
| Creel Estimate     | 5.2                     | 3.9          | 25.7    | 0               | 26.5       | 0                 | 349.0          | 0                  | 0                    | 0                | 0                 | 12.4      | 0                  | 0                  | 6.4                | 429   |
| Lodge Report       | 0                       | 0            | 0       | 0               | 0          | 0                 | 0              | 0                  | 0                    | 0                | 0                 | 0         | 0                  | 0                  | 0                  | 0     |
| AUGUST RETAINED    | 5.2                     | 3.9          | 25.7    | 0               | 26.5       | 0                 | 349.0          | 0                  | 0                    | 0                | 0                 | 12.4      | 0                  | 0                  | 6.4                | 429   |
| SE                 | 2.7                     | 2.9          | 13.4    | 0               | 13.1       | 0                 | 66.6           | 0                  | 0                    | 0                | 0                 | 8.8       | 0                  | 0                  | 3.2                | 70    |
| TOTAL              |                         |              |         |                 |            |                   |                |                    |                      |                  |                   |           |                    |                    |                    |       |
| Creel Estimate     | 32                      | 9            | 37      | 12              | 66         | 0                 | 1,091          | 0                  | 12                   | 25               | 0                 | 36        | 0                  | 0                  | 26                 | 1,345 |
| Lodge Report       | 0                       | 0            | 0       | 0               | 0          | 0                 | 0              | 0                  | 0                    | 0                | 0                 | 0         | 0                  | 0                  | 0                  | 0     |
| TOTAL RETAINED     | 32                      | 9            | 37      | 12              | 66         | 0                 | 1,091          | 0                  | 12                   | 25               | 0                 | 36        | 0                  | 0                  | 26                 | 1,345 |
| SE                 | 12                      | 5            | 14      | 8               | 16         | 0                 | 126            | 0                  | 6                    | 13               | 0                 | 19        | 0                  | 0                  | 8                  | 131   |

**Estimated numbers of other groundfish released by lodge and non-lodge anglers, by taxon, 2023.**

| Month / Catch Type | Taxon                   |              |         |                 |            |                   |                |                    |                      |                  |                   |           |                    |                    |                    | TOTAL |
|--------------------|-------------------------|--------------|---------|-----------------|------------|-------------------|----------------|--------------------|----------------------|------------------|-------------------|-----------|--------------------|--------------------|--------------------|-------|
|                    | Arrow-tooth<br>Flounder | Big<br>Skate | Cabezon | English<br>Sole | Greenlings | Longnose<br>Skate | Pacific<br>Cod | Pacific<br>Sanddab | Pacific Rock<br>Sole | Spiny<br>Dogfish | Red Irish<br>Lord | Sablefish | Spotted<br>Ratfish | Starry<br>Flounder | Unknown /<br>Other |       |
| May                |                         |              |         |                 |            |                   |                |                    |                      |                  |                   |           |                    |                    |                    |       |
| Creel Estimate     | 9.6                     | 28.0         | 0.6     | 0               | 11.0       | 0                 | 16.1           | 0                  | 0                    | 8.9              | 0                 | 0         | 13.0               | 0                  | 3.1                | 90    |
| Lodge Report       | 0                       | 0            | 0       | 0               | 0          | 0                 | 0              | 0                  | 0                    | 0                | 0                 | 0         | 0                  | 0                  | 0                  | 0     |
| MAY RELEASED       | 9.6                     | 28.0         | 0.6     | 0               | 11.0       | 0                 | 16.1           | 0                  | 0                    | 8.9              | 0                 | 0         | 13.0               | 0                  | 3.1                | 90    |
| SE                 | 4.1                     | 9.2          | 0.5     | 0               | 4.8        | 0                 | 8.3            | 0                  | 0                    | 3.8              | 0                 | 0         | 5.7                | 0                  | 2.9                | 16    |
| June               |                         |              |         |                 |            |                   |                |                    |                      |                  |                   |           |                    |                    |                    |       |
| Creel Estimate     | 6.6                     | 9.3          | 0       | 0               | 7.7        | 0                 | 19.2           | 0                  | 0                    | 23.3             | 0                 | 3.0       | 13.6               | 0                  | 0                  | 83    |
| Lodge Report       | 0                       | 0            | 0       | 0               | 0          | 0                 | 0              | 0                  | 0                    | 0                | 0                 | 0         | 0                  | 0                  | 0                  | 0     |
| JUNE RELEASED      | 6.6                     | 9.3          | 0       | 0               | 7.7        | 0                 | 19.2           | 0                  | 0                    | 23.3             | 0                 | 3.0       | 13.6               | 0                  | 0                  | 83    |
| SE                 | 3.5                     | 4.0          | 0       | 0               | 3.9        | 0                 | 9.4            | 0                  | 0                    | 10.0             | 0                 | 2.1       | 6.0                | 0                  | 0                  | 17    |
| July               |                         |              |         |                 |            |                   |                |                    |                      |                  |                   |           |                    |                    |                    |       |
| Creel Estimate     | 23.1                    | 20.1         | 15.3    | 0               | 5.1        | 4.5               | 137.3          | 0                  | 16.7                 | 39.6             | 0                 | 15.8      | 43.2               | 0                  | 34.0               | 355   |
| Lodge Report       | 0                       | 0            | 0       | 0               | 0          | 0                 | 0              | 0                  | 0                    | 0                | 0                 | 0         | 0                  | 0                  | 0                  | 0     |
| JULY RELEASED      | 23.1                    | 20.1         | 15.3    | 0               | 5.1        | 4.5               | 137.3          | 0                  | 16.7                 | 39.6             | 0                 | 15.8      | 43.2               | 0                  | 34.0               | 355   |
| SE                 | 7.8                     | 6.5          | 6.9     | 0               | 3.0        | 3.3               | 51.1           | 0                  | 7.3                  | 11.8             | 0                 | 11.6      | 15.8               | 0                  | 10.7               | 59    |
| August             |                         |              |         |                 |            |                   |                |                    |                      |                  |                   |           |                    |                    |                    |       |
| Creel Estimate     | 82.0                    | 44.8         | 0       | 0               | 31.2       | 0                 | 8.6            | 0                  | 0                    | 400.8            | 0                 | 36.3      | 19.7               | 0                  | 139.2              | 763   |
| Lodge Report       | 0                       | 0            | 0       | 0               | 0          | 0                 | 0              | 0                  | 0                    | 0                | 0                 | 0         | 0                  | 0                  | 0                  | 0     |
| AUGUST RELEASED    | 82.0                    | 44.8         | 0       | 0               | 31.2       | 0                 | 8.6            | 0                  | 0                    | 400.8            | 0                 | 36.3      | 19.7               | 0                  | 139.2              | 763   |
| SE                 | 37.5                    | 12.7         | 0       | 0               | 18.2       | 0                 | 6.1            | 0                  | 0                    | 104.5            | 0                 | 21.9      | 9.4                | 0                  | 76.0               | 139   |
| TOTAL              |                         |              |         |                 |            |                   |                |                    |                      |                  |                   |           |                    |                    |                    |       |
| Creel Estimate     | 121                     | 102          | 16      | 0               | 55         | 5                 | 181            | 0                  | 17                   | 473              | 0                 | 55        | 90                 | 0                  | 176                | 1,290 |
| Lodge Report       | 0                       | 0            | 0       | 0               | 0          | 0                 | 0              | 0                  | 0                    | 0                | 0                 | 0         | 0                  | 0                  | 0                  | 0     |
| TOTAL RELEASED     | 121                     | 102          | 16      | 0               | 55         | 5                 | 181            | 0                  | 17                   | 473              | 0                 | 55        | 90                 | 0                  | 176                | 1,290 |
| SE                 | 39                      | 17           | 7       | 0               | 19         | 3                 | 53             | 0                  | 7                    | 106              | 0                 | 25        | 20                 | 0                  | 77                 | 152   |

**Estimated numbers of rockfish retained by lodge and non-lodge anglers, by month, 2023. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.**

|                    | Subarea |       |      |    |      |       |       |      |       |       |       |      |       |
|--------------------|---------|-------|------|----|------|-------|-------|------|-------|-------|-------|------|-------|
| Month / Catch Type | 3I      | 3J    | 3K   | 4A | 4B   | 4C    | 4D    | 4E   | 4F    | 4G    | 4H    | 4L   | TOTAL |
| May                |         |       |      |    |      |       |       |      |       |       |       |      |       |
| Creel Estimate     | 15.3    | 82.2  | 0    | 0  | 71.3 | 38.0  | 73.0  | 21.0 | 129.1 | 18.8  | 46.7  | 1.9  | 497   |
| Lodge Report       | 38.8    | 45.2  | 0    | 0  | 0    | 0     | 0     | 0    | 0     | 0     | 0     | 0    | 84    |
| MAY RETAINED       | 54.2    | 127.4 | 0    | 0  | 71.3 | 38.0  | 73.0  | 21.0 | 129.1 | 18.8  | 46.7  | 1.9  | 581   |
| SE                 | 8.7     | 36.7  | 0    | 0  | 14.6 | 13.0  | 21.0  | 6.1  | 20.8  | 11.3  | 16.5  | 0.6  | 56    |
| June               |         |       |      |    |      |       |       |      |       |       |       |      |       |
| Creel Estimate     | 4.7     | 83.0  | 11.8 | 0  | 18.7 | 119.0 | 76.4  | 13.0 | 440.8 | 210.9 | 78.6  | 67.1 | 1,124 |
| Lodge Report       | 0       | 184.0 | 0    | 0  | 0    | 0     | 0     | 0    | 0     | 0     | 0     | 0    | 184   |
| JUNE RETAINED      | 4.7     | 267.0 | 11.8 | 0  | 18.7 | 119.0 | 76.4  | 13.0 | 440.8 | 210.9 | 78.6  | 67.1 | 1,308 |
| SE                 | 3.4     | 20.8  | 8.5  | 0  | 7.1  | 25.5  | 18.8  | 9.3  | 60.5  | 57.2  | 30.5  | 20.4 | 100   |
| July               |         |       |      |    |      |       |       |      |       |       |       |      |       |
| Creel Estimate     | 118.0   | 177.5 | 37.0 | 0  | 82.3 | 212.9 | 105.0 | 27.0 | 415.6 | 89.8  | 106.2 | 51.0 | 1,422 |
| Lodge Report       | 73.8    | 92.7  | 60.5 | 0  | 0    | 0     | 0     | 0    | 0     | 0     | 0     | 0    | 227   |
| JULY RETAINED      | 191.7   | 270.2 | 97.6 | 0  | 82.3 | 212.9 | 105.0 | 27.0 | 415.6 | 89.8  | 106.2 | 51.0 | 1,649 |
| SE                 | 58.7    | 33.4  | 15.6 | 0  | 20.5 | 42.7  | 22.9  | 19.6 | 54.0  | 25.7  | 28.3  | 14.5 | 112   |
| August             |         |       |      |    |      |       |       |      |       |       |       |      |       |
| Creel Estimate     | 301.2   | 304.5 | 56.5 | 0  | 79.9 | 274.7 | 136.2 | 65.5 | 249.4 | 145.7 | 494.8 | 24.0 | 2,132 |
| Lodge Report       | 234     | 17    | 0    | 0  | 0    | 0     | 0     | 0    | 0     | 0     | 0     | 0    | 251   |
| AUGUST RETAINED    | 534.8   | 321.9 | 56.5 | 0  | 79.9 | 274.7 | 136.2 | 65.5 | 249.4 | 145.7 | 494.8 | 24.0 | 2,383 |
| SE                 | 83.4    | 75.1  | 16.9 | 0  | 18.9 | 52.4  | 32.8  | 26.0 | 43.4  | 41.9  | 100.4 | 6.5  | 177   |
| TOTAL              |         |       |      |    |      |       |       |      |       |       |       |      |       |
| Creel Estimate     | 439     | 647   | 105  | 0  | 252  | 645   | 391   | 127  | 1,235 | 465   | 726   | 144  | 5,176 |
| Lodge Report       | 346     | 339   | 61   | 0  | 0    | 0     | 0     | 0    | 0     | 0     | 0     | 0    | 746   |
| TOTAL RETAINED     | 785     | 987   | 166  | 0  | 252  | 645   | 391   | 127  | 1,235 | 465   | 726   | 144  | 5,922 |
| SE                 | 102     | 92    | 24   | 0  | 32   | 73    | 49    | 34   | 94    | 76    | 110   | 26   | 239   |

**Estimated numbers of rockfish released by lodge and non-lodge anglers, by month, 2023. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.**

|                    | Subarea |       |      |     |       |       |       |      |       |       |       |      |       |
|--------------------|---------|-------|------|-----|-------|-------|-------|------|-------|-------|-------|------|-------|
| Month / Catch Type | 3I      | 3J    | 3K   | 4A  | 4B    | 4C    | 4D    | 4E   | 4F    | 4G    | 4H    | 4L   | TOTAL |
| May                |         |       |      |     |       |       |       |      |       |       |       |      |       |
| Creel Estimate     | 24.6    | 157.2 | 0    | 4.9 | 122.6 | 134.2 | 193.3 | 31.0 | 254.7 | 67.6  | 111.3 | 4.9  | 1,106 |
| Lodge Report       | 0       | 0     | 0    | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0     | 0    | 0     |
| MAY RELEASED       | 24.6    | 157.2 | 0    | 4.9 | 122.6 | 134.2 | 193.3 | 31.0 | 254.7 | 67.6  | 111.3 | 4.9  | 1,106 |
| SE                 | 13.7    | 74.3  | 0    | 1.9 | 19.3  | 33.3  | 50.3  | 7.4  | 34.7  | 29.2  | 29.7  | 1.1  | 113   |
| June               |         |       |      |     |       |       |       |      |       |       |       |      |       |
| Creel Estimate     | 0       | 69.6  | 5.9  | 0   | 30.4  | 122.9 | 160.9 | 0    | 375.3 | 150.6 | 39.3  | 14.6 | 970   |
| Lodge Report       | 0       | 0     | 0    | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0     | 0    | 0     |
| JUNE RELEASED      | 0       | 69.6  | 5.9  | 0   | 30.4  | 122.9 | 160.9 | 0    | 375.3 | 150.6 | 39.3  | 14.6 | 970   |
| SE                 | 0       | 27.2  | 4.2  | 0   | 10.6  | 28.2  | 42.7  | 0    | 71.6  | 58.8  | 14.0  | 8.6  | 111   |
| July               |         |       |      |     |       |       |       |      |       |       |       |      |       |
| Creel Estimate     | 123.1   | 220.5 | 10.1 | 0   | 88.6  | 237.5 | 120.0 | 23.2 | 280.1 | 104.8 | 212.4 | 13.0 | 1,433 |
| Lodge Report       | 0       | 0     | 0    | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0     | 0    | 0     |
| JULY RELEASED      | 123.1   | 220.5 | 10.1 | 0   | 88.6  | 237.5 | 120.0 | 23.2 | 280.1 | 104.8 | 212.4 | 13.0 | 1,433 |
| SE                 | 30.6    | 42.4  | 4.1  | 0   | 19.9  | 46.2  | 34.0  | 12.5 | 37.5  | 30.3  | 61.3  | 5.4  | 113   |
| August             |         |       |      |     |       |       |       |      |       |       |       |      |       |
| Creel Estimate     | 471.9   | 535.7 | 56.5 | 0   | 73.7  | 262.2 | 159.6 | 7.3  | 297.4 | 102.9 | 383.5 | 3.0  | 2,354 |
| Lodge Report       | 0       | 0     | 0    | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0     | 0    | 0     |
| AUGUST RELEASED    | 471.9   | 535.7 | 56.5 | 0   | 73.7  | 262.2 | 159.6 | 7.3  | 297.4 | 102.9 | 383.5 | 3.0  | 2,354 |
| SE                 | 136.4   | 146.6 | 21.9 | 0   | 19.7  | 114.6 | 45.6  | 6.1  | 41.9  | 51.4  | 86.7  | 1.6  | 261   |
| TOTAL              |         |       |      |     |       |       |       |      |       |       |       |      |       |
| Creel Estimate     | 620     | 983   | 73   | 5   | 315   | 757   | 634   | 61   | 1,207 | 426   | 746   | 35   | 5,863 |
| Lodge Report       | 0       | 0     | 0    | 0   | 0     | 0     | 0     | 0    | 0     | 0     | 0     | 0    | 0     |
| TOTAL RELEASED     | 620     | 983   | 73   | 5   | 315   | 757   | 634   | 61   | 1,207 | 426   | 746   | 35   | 5,863 |
| SE                 | 140     | 172   | 23   | 2   | 36    | 131   | 87    | 16   | 97    | 89    | 111   | 10   | 325   |



**Estimated numbers of rockfish retained by lodge and non-lodge anglers, by taxon, 2023.**

| Month / Catch Type | Taxon          |                   |                 |                |                 |                |                    |                  |                    |                |                |                    |                     |                  | TOTAL |
|--------------------|----------------|-------------------|-----------------|----------------|-----------------|----------------|--------------------|------------------|--------------------|----------------|----------------|--------------------|---------------------|------------------|-------|
|                    | Black Rockfish | Boccacio Rockfish | Canary Rockfish | China Rockfish | Copper Rockfish | Dusky Rockfish | Quillback Rockfish | Redband Rockfish | Redstripe Rockfish | Tiger Rockfish | Widow Rockfish | Yelloweye Rockfish | Yellowtail Rockfish | Unknown rockfish |       |
| May                |                |                   |                 |                |                 |                |                    |                  |                    |                |                |                    |                     |                  |       |
| Creel Estimate     | 198.0          | 0                 | 1.7             | 17.2           | 26.4            | 0              | 136.1              | 0                | 0                  | 7.2            | 11.1           | 0                  | 22.9                | 76.6             | 497   |
| Lodge Report       | 0              | 0                 | 0               | 0              | 0               | 0              | 0                  | 0                | 0                  | 0              | 0              | 0                  | 0                   | 84.0             | 84    |
| MAY RETAINED       | 198.0          | 0                 | 1.7             | 17.2           | 26.4            | 0              | 136.1              | 0                | 0                  | 7.2            | 11.1           | 0                  | 22.9                | 160.6            | 581   |
| SE                 | 42.2           | 0                 | 1.3             | 5.1            | 11.9            | 0              | 25.5               | 0                | 0                  | 3.3            | 8.3            | 0                  | 8.6                 | 19.0             | 56    |
| June               |                |                   |                 |                |                 |                |                    |                  |                    |                |                |                    |                     |                  |       |
| Creel Estimate     | 406.8          | 0                 | 0               | 12.6           | 82.4            | 0              | 311.0              | 15.1             | 0                  | 54.4           | 5.8            | 15.1               | 19.7                | 201.2            | 1,124 |
| Lodge Report       | 0              | 0                 | 0               | 0              | 0               | 0              | 0                  | 0                | 0                  | 0              | 0              | 0                  | 0                   | 184.0            | 184   |
| JUNE RETAINED      | 406.8          | 0                 | 0               | 12.6           | 82.4            | 0              | 311.0              | 15.1             | 0                  | 54.4           | 5.8            | 15.1               | 19.7                | 385.2            | 1,308 |
| SE                 | 60.9           | 0                 | 0               | 4.7            | 29.2            | 0              | 48.5               | 12.4             | 0                  | 26.0           | 4.9            | 12.4               | 7.5                 | 44.0             | 100   |
| July               |                |                   |                 |                |                 |                |                    |                  |                    |                |                |                    |                     |                  |       |
| Creel Estimate     | 560.3          | 14.2              | 0               | 23.2           | 67.7            | 0              | 311.3              | 0                | 0                  | 24.0           | 3.2            | 13.6               | 82.6                | 322.4            | 1,422 |
| Lodge Report       | 0              | 0                 | 0               | 0              | 0               | 0              | 0                  | 0                | 0                  | 0              | 0              | 0                  | 0                   | 227.0            | 227   |
| JULY RETAINED      | 560.3          | 14.2              | 0               | 23.2           | 67.7            | 0              | 311.3              | 0                | 0                  | 24.0           | 3.2            | 13.6               | 82.6                | 549.4            | 1,649 |
| SE                 | 85.6           | 7.7               | 0               | 6.9            | 23.4            | 0              | 39.2               | 0                | 0                  | 6.9            | 2.3            | 9.9                | 27.7                | 45.9             | 112   |
| August             |                |                   |                 |                |                 |                |                    |                  |                    |                |                |                    |                     |                  |       |
| Creel Estimate     | 897.2          | 41.3              | 6.2             | 63.1           | 66.3            | 0              | 582.7              | 0                | 0                  | 48.3           | 17.3           | 11.3               | 57.5                | 341.4            | 2,132 |
| Lodge Report       | 0              | 0                 | 0               | 0              | 0               | 0              | 0                  | 0                | 0                  | 0              | 0              | 1                  | 0                   | 250              | 251   |
| AUGUST RETAINED    | 897.2          | 41.3              | 6.2             | 63.1           | 66.3            | 0              | 582.7              | 0                | 0                  | 48.3           | 17.3           | 12.3               | 57.5                | 591.4            | 2,383 |
| SE                 | 129.1          | 17.9              | 4.4             | 14.9           | 17.1            | 0              | 84.7               | 0                | 0                  | 12.4           | 6.6            | 8.0                | 16.1                | 79.3             | 177   |
| TOTAL              |                |                   |                 |                |                 |                |                    |                  |                    |                |                |                    |                     |                  |       |
| Creel Estimate     | 2,062          | 55                | 8               | 116            | 243             | 0              | 1,341              | 15               | 0                  | 134            | 37             | 40                 | 183                 | 942              | 5,176 |
| Lodge Report       | 0              | 0                 | 0               | 0              | 0               | 0              | 0                  | 0                | 0                  | 0              | 0              | 1                  | 0                   | 745              | 746   |
| TOTAL RETAINED     | 2,062          | 55                | 8               | 116            | 243             | 0              | 1,341              | 15               | 0                  | 134            | 37             | 41                 | 183                 | 1,687            | 5,922 |
| SE                 | 172            | 19                | 5               | 18             | 43              | 0              | 108                | 12               | 0                  | 30             | 12             | 18                 | 34                  | 103              | 239   |

**Estimated numbers of rockfish released by lodge and non-lodge anglers, by taxon, 2023.**

| Month / Catch Type | Taxon          |                   |                 |                |                 |                |                    |                  |                    |                |                |                    |                     |                  | TOTAL |
|--------------------|----------------|-------------------|-----------------|----------------|-----------------|----------------|--------------------|------------------|--------------------|----------------|----------------|--------------------|---------------------|------------------|-------|
|                    | Black Rockfish | Boccacio Rockfish | Canary Rockfish | China Rockfish | Copper Rockfish | Dusky Rockfish | Quillback Rockfish | Redband Rockfish | Redstripe Rockfish | Tiger Rockfish | Widow Rockfish | Yelloweye Rockfish | Yellowtail Rockfish | Unknown rockfish |       |
| May                |                |                   |                 |                |                 |                |                    |                  |                    |                |                |                    |                     |                  |       |
| Creel Estimate     | 134.2          | 0                 | 4.5             | 32.9           | 14.9            | 0              | 102.4              | 2.0              | 21.1               | 6.8            | 0              | 115.3              | 18.9                | 653.2            | 1,106 |
| Lodge Report       | 0              | 0                 | 0               | 0              | 0               | 0              | 0                  | 0                | 0                  | 0              | 0              | 0                  | 0                   | 0                | 0     |
| MAY RELEASED       | 134.2          | 0                 | 4.5             | 32.9           | 14.9            | 0              | 102.4              | 2.0              | 21.1               | 6.8            | 0              | 115.3              | 18.9                | 653.2            | 1,106 |
| SE                 | 35.1           | 0                 | 3.2             | 12.6           | 7.1             | 0              | 25.9               | 1.4              | 13.1               | 4.0            | 0              | 34.9               | 13.2                | 94.9             | 113   |
| June               |                |                   |                 |                |                 |                |                    |                  |                    |                |                |                    |                     |                  |       |
| Creel Estimate     | 58.1           | 0                 | 0               | 15.9           | 3.8             | 0              | 43.5               | 0                | 6.0                | 0              | 19.2           | 131.5              | 24.0                | 667.6            | 970   |
| Lodge Report       | 0              | 0                 | 0               | 0              | 0               | 0              | 0                  | 0                | 0                  | 0              | 0              | 0                  | 0                   | 0                | 0     |
| JUNE RELEASED      | 58.1           | 0                 | 0               | 15.9           | 3.8             | 0              | 43.5               | 0                | 6.0                | 0              | 19.2           | 131.5              | 24.0                | 667.6            | 970   |
| SE                 | 18.0           | 0                 | 0               | 5.9            | 3.0             | 0              | 13.7               | 0                | 4.2                | 0              | 15.1           | 41.9               | 10.6                | 98.3             | 111   |
| July               |                |                   |                 |                |                 |                |                    |                  |                    |                |                |                    |                     |                  |       |
| Creel Estimate     | 83.9           | 4.1               | 0               | 53.8           | 5.0             | 8.2            | 136.9              | 0                | 5.1                | 16.2           | 0              | 330.0              | 3.3                 | 786.6            | 1,433 |
| Lodge Report       | 0              | 0                 | 0               | 0              | 0               | 0              | 0                  | 0                | 0                  | 0              | 0              | 0                  | 0                   | 0                | 0     |
| JULY RELEASED      | 83.9           | 4.1               | 0               | 53.8           | 5.0             | 8.2            | 136.9              | 0                | 5.1                | 16.2           | 0              | 330.0              | 3.3                 | 786.6            | 1,433 |
| SE                 | 25.1           | 2.9               | 0               | 16.6           | 3.8             | 5.8            | 45.3               | 0                | 3.7                | 8.1            | 0              | 52.1               | 2.7                 | 83.0             | 113   |
| August             |                |                   |                 |                |                 |                |                    |                  |                    |                |                |                    |                     |                  |       |
| Creel Estimate     | 546.9          | 0                 | 0               | 105.2          | 14.4            | 0              | 323.2              | 0                | 0                  | 31.4           | 0              | 238.6              | 0                   | 1094.0           | 2,354 |
| Lodge Report       | 0              | 0                 | 0               | 0              | 0               | 0              | 0                  | 0                | 0                  | 0              | 0              | 0                  | 0                   | 0                | 0     |
| AUGUST RELEASED    | 546.9          | 0                 | 0               | 105.2          | 14.4            | 0              | 323.2              | 0                | 0                  | 31.4           | 0              | 238.6              | 0                   | 1094.0           | 2,354 |
| SE                 | 173.5          | 0                 | 0               | 53.0           | 9.0             | 0              | 68.1               | 0                | 0                  | 16.4           | 0              | 51.1               | 0                   | 166.1            | 261   |
| TOTAL              |                |                   |                 |                |                 |                |                    |                  |                    |                |                |                    |                     |                  |       |
| Creel Estimate     | 823            | 4                 | 4               | 208            | 38              | 8              | 606                | 2                | 32                 | 54             | 19             | 815                | 46                  | 3,201            | 5,863 |
| Lodge Report       | 0              | 0                 | 0               | 0              | 0               | 0              | 0                  | 0                | 0                  | 0              | 0              | 0                  | 0                   | 0                | 0     |
| TOTAL RELEASED     | 823            | 4                 | 4               | 208            | 38              | 8              | 606                | 2                | 32                 | 54             | 19             | 815                | 46                  | 3,201            | 5,863 |
| SE                 | 180            | 3                 | 3               | 57             | 12              | 6              | 87                 | 1                | 14                 | 19             | 15             | 91                 | 17                  | 231              | 325   |

## APPENDIX 5: ADIPOSE FIN-CLIP INCIDENCE

Incidence of fin clipping observed in biosamples, by species, month, and subarea, in 2023. Excludes any interviews for which observers were not able to confirm adipose status, or for which biosampling was not a complete sample of fish kept.

| Species        |                 | Subarea |    |    |    |    |    |     |     |    |     |    |    |     |    |       |
|----------------|-----------------|---------|----|----|----|----|----|-----|-----|----|-----|----|----|-----|----|-------|
| Month          | Adipose         | 3I      | 3J | 3K | 3? | 4A | 4B | 4C  | 4D  | 4E | 4F  | 4G | 4H | 4L  | 4? | TOTAL |
| Chinook Salmon |                 |         |    |    |    |    |    |     |     |    |     |    |    |     |    |       |
| May            | Clipped         | 0       | 0  | 0  | 0  | 0  | 2  | 5   | 0   | 0  | 5   | 0  | 0  | 0   | 0  | 12    |
|                | Not Clipped     | 1       | 0  | 3  | 0  | 2  | 22 | 22  | 7   | 0  | 50  | 2  | 3  | 0   | 0  | 112   |
|                | Percent Clipped | 0%      | -  | 0% | -  | 0% | 8% | 19% | 0%  | -  | 9%  | 0% | 0% | -   | -  | 10%   |
| June           | Clipped         | 1       | 0  | 0  | 0  | 0  | 0  | 0   | 1   | 0  | 4   | 0  | 0  | 2   | 0  | 8     |
|                | Not Clipped     | 7       | 5  | 8  | 0  | 0  | 10 | 11  | 25  | 0  | 34  | 2  | 2  | 7   | 0  | 111   |
|                | Percent Clipped | 13%     | 0% | 0% | -  | -  | 0% | 0%  | 4%  | -  | 11% | 0% | 0% | 22% | -  | 7%    |
| July           | Clipped         | 0       | 0  | 0  | 0  | 0  | 0  | 0   | 1   | 0  | 1   | 0  | 0  | 0   | 0  | 2     |
|                | Not Clipped     | 0       | 1  | 3  | 0  | 0  | 1  | 0   | 9   | 1  | 1   | 1  | 0  | 0   | 0  | 17    |
|                | Percent Clipped | -       | 0% | 0% | -  | -  | 0% | -   | 10% | 0% | 50% | 0% | -  | -   | -  | 11%   |
| August         | Clipped         | 0       | 0  | 0  | 0  | 0  | 0  | 0   | 0   | 0  | 0   | 0  | 0  | 0   | 0  | 0     |
|                | Not Clipped     | 0       | 0  | 0  | 0  | 1  | 1  | 0   | 3   | 0  | 4   | 0  | 0  | 0   | 0  | 9     |
|                | Percent Clipped | -       | -  | -  | -  | 0% | 0% | -   | 0%  | -  | 0%  | -  | -  | -   | -  | 0%    |
| Total          | Clipped         | 1       | 0  | 0  | 0  | 0  | 2  | 5   | 2   | 0  | 10  | 0  | 0  | 2   | 0  | 22    |
|                | Not Clipped     | 8       | 6  | 14 | 0  | 3  | 34 | 33  | 44  | 1  | 89  | 5  | 5  | 7   | 0  | 249   |
|                | Percent Clipped | 11%     | 0% | 0% | -  | 0% | 6% | 13% | 4%  | 0% | 10% | 0% | 0% | 22% | -  | 8.1%  |
| Coho Salmon    |                 |         |    |    |    |    |    |     |     |    |     |    |    |     |    |       |
| May            | Clipped         | 0       | 0  | 0  | 0  | 0  | 0  | 0   | 0   | 0  | 0   | 0  | 0  | 0   | 0  | 0     |
|                | Not Clipped     | 0       | 0  | 0  | 0  | 0  | 0  | 0   | 0   | 0  | 0   | 0  | 0  | 0   | 0  | 0     |
|                | Percent Clipped | -       | -  | -  | -  | -  | -  | -   | -   | -  | -   | -  | -  | -   | -  | -     |
| June           | Clipped         | 0       | 0  | 0  | 0  | 0  | 0  | 0   | 0   | 0  | 1   | 0  | 0  | 0   | 0  | 1     |
|                | Not Clipped     | 0       | 2  | 8  | 0  | 0  | 0  | 5   | 2   | 0  | 31  | 6  | 4  | 16  | 0  | 74    |
|                | Percent Clipped | -       | 0% | 0% | -  | -  | -  | 0%  | 0%  | -  | 3%  | 0% | 0% | 0%  | -  | 1.3%  |
| July           | Clipped         | 0       | 0  | 1  | 0  | 0  | 0  | 0   | 0   | 0  | 0   | 0  | 0  | 0   | 0  | 1     |
|                | Not Clipped     | 1       | 10 | 21 | 0  | 0  | 42 | 51  | 114 | 2  | 191 | 1  | 22 | 0   | 0  | 455   |
|                | Percent Clipped | 0%      | 0% | 5% | -  | -  | 0% | 0%  | 0%  | 0% | 0%  | 0% | 0% | -   | -  | 0.2%  |
| August         | Clipped         | 0       | 0  | 1  | 0  | 0  | 0  | 0   | 0   | 0  | 0   | 0  | 0  | 0   | 0  | 1     |
|                | Not Clipped     | 4       | 0  | 11 | 0  | 0  | 40 | 9   | 58  | 1  | 47  | 11 | 50 | 0   | 0  | 231   |
|                | Percent Clipped | 0%      | -  | 8% | -  | -  | 0% | 0%  | 0%  | 0% | 0%  | 0% | 0% | -   | -  | 0.4%  |
| Total          | Clipped         | 0       | 0  | 2  | 0  | 0  | 0  | 0   | 0   | 0  | 1   | 0  | 0  | 0   | 0  | 3     |
|                | Not Clipped     | 5       | 12 | 40 | 0  | 0  | 82 | 65  | 174 | 3  | 269 | 18 | 76 | 16  | 0  | 760   |
|                | Percent Clipped | 0%      | 0% | 5% | -  | -  | 0% | 0%  | 0%  | 0% | 0%  | 0% | 0% | 0%  | -  | 0.4%  |